Study Topics in Phylogenomics and Bioinformatics of Metazoans (Draft in Development)

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Preface

This book brings together study material on phylogenomics and bioinformatics of metazoans, ranging from the theoretical-philosophical foundations of phylogenetic systematics to contemporary issues in artificial intelligence, Brazilian legislation, and computational infrastructure. The work is organized into eleven chapters that cover the main topics necessary for understanding and practicing large-scale phylogenomic reconstruction.

The text emphasizes both the theoretical principles and practical aspects of implementation, including the mathematical formulation of probabilistic methods, the diversity of available molecular data, sequencing strategies, challenges imposed by repetitive elements, and computational infrastructure requirements. The final chapters address the emerging role of artificial intelligence in biomedical science and the legal and ethical aspects of genomic data use, with emphasis on Brazilian legislation and international frameworks.

Each chapter contains its own bibliographic references, which allows for independent consultation. The chapter order follows a logical progression from fundamentals to applications, but each chapter was written to be reasonably self-contained. A study guide is presented at the end of each chapter.

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Chapter 1

Theoretical-Philosophical Foundations of Phylogenetics

1.1 Introduction

Inferring the evolutionary history of organisms requires organized hereditary observations capable of discriminating between competing phylogenetic hypotheses. The transformation of raw biological observations into such evidence is not a trivial operation: it demands a principled understanding of what constitutes a character, how characters support the hypothesis of common ancestry, and how apparent conflicts between characters are interpreted and resolved (Wheeler et al. 2006).

These problems have a long and contested history. Lankester (1870) introduced the term *homoplasy* to distinguish features that resemble superficially but do not share a common origin. Hennig (1966) formalized the distinction between ancestral and derived character states that are shared, providing the logical foundation of cladistic methodology. Pinna (1991) and Patterson (1982) subsequently addressed the epistemological status of homology as a hypothesis. Wheeler et al. (2006) and Wheeler (2012) synthesized these contributions into an integrated computational framework, capable of dealing with the scale and complexity of modern genomic data.

Characters are, in this framework, theory-laden objects in the sense of Popper (1959): the prior understanding that the investigator has of the biology of organisms informs the decision to recognize a feature as a character. Phylogenetic systematics is, therefore, an idiographic science — not nomothetic — that seeks to discover the series of historical transformation events that explain hereditary variation among lineages (Grant and Kluge 2004b).

The advent of high-throughput sequencing (HTS) has fundamentally altered the scale at which phylogenetic data are collected: hundreds or thousands of loci can now be sequenced for dozens to hundreds of taxa in a single experiment. This abundance of data does not simplify the conceptual problems of character delimitation, homology assessment, and homoplasy management; rather, it amplifies them. The practical urgency of these questions is illustrated by large-scale analyses such as that of Jacob Machado et al. (2021c), which analyzed 2,006 complete genomes of the subfamily *Orthocoronavirinae* using total evidence parsimony, producing the most comprehensive phylogenetic hypothesis for this group at the time and elucidating the origin and host history of SARS-CoV-2. Jacob Machado et al. (2021d) reviewed the trajectory of genomic epidemiology from its origins in molecular epidemiology through the COVID-19 pandemic, demonstrating how the dramatic reduction in sequencing costs enabled phylogenomic surveillance at a global scale.

Recent developments in software have further extended the scope of phylogenetic inference. PhyG (Wheeler et al. 2024), successor to POY5, extends phylogenetic search to general graphs, including softwired and hardwired networks, and offers substantial improvements in optimization quality and execution speed. Wheeler and Varón (2025) proposed phylogenetic minimum description length (PMDL) as a unified optimality criterion grounded in algorithmic information theory. The present chapter examines these conceptual foundations and their practical implementation in contemporary phylogenetic analysis workflows.

1.2 The Structure of Phylogenetic Trees

1.2.1 Trees, Cladograms, and Topologies

In phylogenetic systematics, the terms **phylogenetic tree**, **cladogram**, and **topology** are frequently used interchangeably (Wheeler 2012). The phylogenetic tree is the primary tool for describing evolutionary relationships among organisms. Formally, trees are connected acyclic graphs that represent nested hierarchical relationships among groups. In YBYRÁ and in other phylogenetic analysis programs, trees are typically represented in parenthetic notation, following the Newick format or the Hennig86 format (also called TREAD) (Machado 2015).

1.2.2 Elementary Components: Nodes, Branches, Terminals, and Root

The structural elements of a phylogenetic tree are the following:

Terminals The terminals — also called *operational taxonomic units* (OTUs) or leaves — are the objects that occur at the extremities of the tree. Terminals frequently represent taxa, but can represent any comparable entity (e.g., sequences, populations, fossil species).

Nodes Nodes are the intersection points in a tree, identifying each component. Internal nodes represent hypothetical ancestors and are called *hypothetical taxonomic units* (HTUs); terminal nodes correspond to OTUs. In a rooted tree, the root is the only node with degree two.

Branches Branches are the edges that connect two adjacent nodes. Each branch can be associated with a length representing the number of transformations or the evolutionary duration of the interval.

Root The root determines the polarity of character transformations in the cladogram (Schuh and Brower 2009). Rooting converts an unrooted tree into a rooted tree, establishing the direction of evolutionary transformations.

1.2.3 Sister Groups, Splits, and Strict Consensus

In a binary tree, **sister groups** are two groups that share a single exclusive common ancestor (Schuh and Brower 2009). A **split** is the bipartition of a set of terminals induced by the removal of an edge from the tree. For example, the tree `(A,(C,(D,(E,F))))` can be split into the subtrees `(E,F)` and `(A,(C,D))` by removing the branch connecting them. A **strict consensus tree** contains exclusively the nodes present in *all* trees from a set of equally optimal results; any inconsistency results in the collapse of the corresponding node (Schuh and Brower 2009).

1.2.4 File Formats for Phylogenetic Trees

Two file formats are widely used to represent trees in parenthetic notation:

Newick Adopted by programs such as GARLI, MrBayes, PAUP*, PHYLIP, PROTML, and TREE-PUZZLE. Terminals are separated by commas and sister group relationships by nested parentheses. Supports branch lengths and internal node labels: `((A:5,B:7)Ancestral:10,C:3);`

TREAD (Hennig86) Represents trees preceded by the header `tread`. Multiple trees are separated by asterisks, and the last ends with a semicolon. It is common to include `proc-;` at the end of the file for use in TNT (Goloboff et al. 2008).

1.3 Taxa, Clades, and Phylogenetic Classification

1.3.1 Monophyletic, Paraphyletic, and Polyphyletic Groups

A **taxon** is any group of organisms recognized at any level of the systematic hierarchy (Schuh and Brower 2009). Phylogenetic systematics distinguishes three types of grouping (Hennig 1966; Schuh and Brower 2009):

Monophyletic group (clade) Composed of a single common ancestor and *all* of its descendants. Only monophyletic groups are recognized as natural and valid units of cladistic classification (Hennig 1966).

Paraphyletic group One that is missing exactly one monophyletic group to become monophyletic (e.g., "Reptilia" in the classical sense, which excludes Aves).

Polyphyletic group One that is missing two or more monophyletic groups to become monophyletic; typically reflects evolutionary convergence (Schuh and Brower 2009).

The monophyly of any grouping is a proposition to be evaluated by the congruence of characters, not a methodological assumption (Nixon and Carpenter 1993).

1.4 The Concept of Character

1.4.1 The Historical Nature of Character

A character, in the phylogenetic sense, is "a historically independent transformation series" (Wheeler et al. 2006, p. 333). Characters are not simple observations, but theoretically organized statements about hereditary variation among taxa, carrying notions of relevance, comparability, and correspondence (Wheeler 2012). They are theory-laden objects in the sense of Popper (1959): the prior understanding of the investigator informs the decision to recognize a feature as a character.

Grant and Kluge (2004b) argue that this is the only character concept consistent with the epistemological requirements of phylogenetic systematics. As transformation series, characters are historical individuals, analogous in ontological status to species and clades. The independence relevant to phylogenetic inference is **historical or transformational independence**, not functional or developmental correlation (Grant and Kluge 2004b).

Character states are the alternative forms that a transformation series can assume. For binary characters, states typically designate presence or absence of a structure; for sequence data, states are the nucleotide or amino acid residues at a given position.

1.4.2 Types of Character

Wheeler et al. (2006) recognize five fundamental types of character:

Additive Assume linear ordering of states; the cost of transformation is the additive distance between states (Farris 1970).

Non-additive Assign equal costs to all transformations, without nested homology among states (Fitch 1971).

Matrix (Sankoff) Allow arbitrary transformation costs specified by a cost matrix (Sankoff 1975; Sankoff and Rousseau 1975).

Sequence Treat contiguous chains of nucleotides or amino acids as the unit of comparison, with transformations involving substitution, insertion, or deletion.

Chromosomal Optimize simultaneously the variation at the nucleotide level and at the locus level; applicable to mitochondrial, bacterial, and viral genomes (Wheeler et al. 2006).

PhyG (Wheeler et al. 2024) extends this representation further to effectively unlimited alphabets, allowing the analysis of linguistic data, developmental data, and other types of non-standard comparative data.

1.4.3 Phenotypic and Genotypic Characters

Phenotypic characters are structural or functional characteristics that express the genotype in conjunction with environmental influences, including morphological, behavioral, and amino acid sequence data. Genotypic characters derive directly from DNA or RNA sequences, eliminating concerns about heritability; their main challenge is that nucleotide homology cannot be tested in isolation, but requires a congruence test with other characters (Wheeler et al. 2006). The principle of **total evidence** (Kluge 1989) maintains that all available evidence — phenotypic and genotypic — has equal initial value for discriminating among phylogenetic hypotheses and should be analyzed simultaneously.

1.5 Homology: Definition, Epistemology, and Debate

1.5.1 The Operational Definition and Homology as Null Hypothesis

In the formulation of Wheeler et al. (2006, p. 10), "features are homologous when their origins can be traced to a single transformation in a branch of the cladogram leading to the most recent common ancestor." Three consequences follow from this definition: (1) homology is **cladogram-dependent** — there is no notion of homology without reference to a cladogram, nor choice among cladograms without statements of homology; (2) homology is the **null hypothesis** in the construction of monophyletic groups during tree search; (3) every statement of homology is simultaneously a phylogenetic hypothesis, subject to falsification by character incongruence.

Grant and Kluge (2004b) argue, independently, that homology refers to a **historical relation of identity** — the identity of being derived from the same single transformation event — and as such is conceptually prior to and distinct from the cladistic test of character congruence.

1.5.2 The Homology–Synapomorphy Debate

Nixon and Carpenter (2012) argue that the equation of homology with synapomorphy, proposed by Patterson (1982), is historically and conceptually incorrect. Hennig (1966) clearly intended that homology encompass both plesiomorphy and synapomorphy, since both represent features shared by taxa as a result of common ancestry. A symplesiomorphy is homologous at the most inclusive level at which it constitutes a synapomorphy; at the level of the ingroup, it is homologous but phylogenetically non-informative (Nixon and Carpenter 2012).

1.5.3 The Primary–Secondary Homology Debate

Pinna (1991) proposed a two-stage model: **primary homology** refers to initial propositions based on similarity, generated before analysis; **secondary homology** refers to propositions that survive the test of character congruence. Wheeler et al. (2006) reject this separation as "neither a methodological necessity nor an epistemological one" (Wheeler et al. 2006, p. 10). Grant and Kluge (2004b) argue that the distinction confuses the *concept* of homology with the *discovery operations* used to detect instances of that concept; since both are distinct, no intermediate epistemological category is necessary. The congruence test can be applied directly to the unrestricted set of homology hypotheses — which is the core of the **dynamic homology** framework's claim (Wheeler 2001a,b).

1.5.4 Static and Dynamic Homology

Static homology encodes homology hypotheses in a fixed matrix before phylogenetic analysis. In molecular systematics, multiple sequence alignment is the canonical method: each column is treated as a character and each residue within the column as a homologous state (positional homology). The critical limitation is that the optimal alignment is topology-dependent; an alignment produced under a guide tree may differ from the optimal alignment for the optimal cladogram (Wheeler et al. 2006).

Dynamic homology treats homology as an optimization problem solved simultaneously with tree search, employing the same optimality criterion for both steps; the globally optimal solution is the combination of minimum cladogram cost plus homology scheme (Wheeler et al. 2006).

An important practical development is **implied alignments**, introduced by Wheeler (2003a) as visual representations of synapomorphies in a given cladogram. Each implied alignment is unique to a topology and a set of specific transformation parameters; using it as the basis for a subsequent independent analysis would reintroduce the problems of static homology (Wheeler et al. 2006).

1.5.5 Types of Transformations

Character states are classified according to polarity and distribution in the tree (Hennig 1966; Schuh and Brower 2009):

Apomorphy Derived character state, originating from an ancestral state in a transformation event.

Plesiomorphy Ancestral character state. Plesiomorphies are legitimate homologies, but are not informative for discriminating relationship hypotheses (Nixon and Carpenter 2012).

Synapomorphy Shared occurrence of an apomorphic state in a monophyletic group (Grant and Kluge 2004a); fundamental evidence of monophyly.

Symplesiomorphy Ancestral state shared by a non-monophyletic group of terminals; does not support natural groupings.

Autapomorphy Apomorphy exclusive to a single terminal or taxon; diagnostic for individual terminals, but not informative for relationships among them.

1.6 Homoplasy: Definition, Measurement, and Interpretation

1.6.1 Definition and Epistemological Status

Homoplasy occurs "when a character state is shared by two or more taxa without being the result of common ancestry, but rather of parallelism or convergence" (Wheeler et al. 2006, p. 336). The term was introduced by Lankester (1870). In operational terms, homoplasy is any non-minimal transformation of a character in a cladogram — a definition explicitly tree-specific: a homoplastic feature in one cladogram may be optimized without extra steps in another (Wheeler 2012).

Farris (1983) and Kluge (2009) argue that homoplasy functions as a **falsifier** of the tree hypothesis in which it occurs. The preferred cladogram is the one least falsified by homoplasy, i.e., the one that minimizes *ad hoc* hypotheses of convergence and parallelism. This is the criterion of parsimony stated in epistemological terms. An ***ad hoc* hypothesis** is an assumption invoked to eliminate inconvenient observations; in cladistics, the *a priori* resort to homoplasy — explaining similarity as non-homologous without topological justification — is the paradigmatic example (Grant and Kluge 2004a).

1.6.2 Measurement of Homoplasy

The **consistency index** (CI) for a character is the ratio of the minimum possible number of steps to the number observed in a given tree (Kluge and Farris 1969). A CI of 1.0 indicates absence of homoplasy. The **retention index** (RI; Farris 1989) measures the fraction of apparent synapomorphy effectively retained as synapomorphy in the tree; it is less sensitive to the number of taxa than the CI.

Goloboff (1991) introduced the distinction between homoplasy and **decisiveness** — the information content of a matrix with respect to choice of trees, proportional to the variance in tree length over all possible cladograms. Lower values of CI or RI do not necessarily imply lower decisiveness: a dataset can have high homoplasy in the most parsimonious cladogram and still be highly decisive if alternative trees are substantially less parsimonious. This result warns against confusing the level of homoplasy with the strength of evidence for a cladogram (Goloboff 1991).

1.6.3 InDels and Homoplasy

Insertions and deletions (InDels) exhibit patterns of homoplasy distinct from nucleotide substitutions. Redelings et al. (2024) observe that InDel rates vary substantially among taxonomic groups and genomic regions, and that a deletion bias is observed in most studied lineages. Because InDels are relatively rare events compared to point mutations, and because the probability of an identical InDel occurring independently in two unrelated lineages at the same position is very low, shared InDels constitute particularly strong synapomorphic evidence (Redelings et al. 2024).

1.7 Outgroup Comparison, Polarity, and Rooting

1.7.1 Outgroup, Ingroup, and the Identity of Three Problems

The **ingroup** is the putative monophyletic group whose internal relationships constitute the primary focus of the study; ingroup monophyly is a hypothesis to be tested, not an assumption to be imposed (Nixon and Carpenter 1993). The **outgroup** is any taxon external to the ingroup included to root the resulting network and establish character polarity (Grant 2019; Nixon and Carpenter 1993).

Nixon and Carpenter (1993) clarified that outgroup comparison, determination of character polarity, and rooting all reduce to a single operation: polarity need not — and should not — be decided before phylogenetic analysis; it is a consequence of rooting, not a prerequisite. The "outgroup algorithm" of Maddison et al. (1984, discussed in Nixon and Carpenter 1993), which applies parsimony independently to ingroup and outgroup, constitutes a theoretically indefensible relaxation of parsimony (Nixon and Carpenter 1993). The solution is unrestricted simultaneous analysis of all terminals in a single unrooted network, which is then rooted between the ingroup and outgroup.

1.7.2 Lundberg Rooting

Lundberg rooting consists of constructing a hypothetical ancestor to root the network. Nixon and Carpenter (1993) concluded that this procedure is consistent with cladistic parsimony only when the hypothetical ancestor is derived from independent ontogenetic or paleontological evidence, not from outgroup comparison; using it otherwise is circular.

1.7.3 Successive Outgroup Expansion

Grant (2019) proposed the procedure of **successive outgroup expansion**: the outgroup sample is expanded iteratively, with priority for terminals more closely related to the ingroup and for terminals in regions of the external topology weakly supported, until the topology and homology hypotheses of the ingroup remain stable over at least two consecutive rounds of expansion. This procedure provides a defensible stopping rule without implying that results are beyond refutation (Grant 2019).

1.7.4 Simultaneous Analysis and Total Evidence

Simultaneous analysis refers to the joint analysis of multiple datasets in opposition to separate analysis followed by consensus (Nixon and Carpenter 1996). The term **total evidence** is often synonymous (Kluge 1989), although some authors reserve "total evidence" for analyses that incorporate *all* available data (Kluge 1989; Nixon and Carpenter 1996). Both approaches oppose partitioned analysis followed by consensus, which can obscure informative conflict between partitions (Grant and Kluge 2003).

1.8 Dynamic Homology, InDels, and High-Throughput Sequencing

1.8.1 The Problem of InDels in Static Alignment

Insertions and deletions (InDels) constitute the second most important source of genomic variation, after point mutations (Redelings et al. 2024). Most alignment algorithms are not phylogeny-aware

and tend to infer alignments in which characters are independently deleted many times when considered in a tree, resulting in overestimation of convergent loss events and loss of the genuine phylogenetic signal carried by shared InDels (Redelings et al. 2024).

The empirical consequences of coding gaps as missing data in maximum likelihood (ML) analysis were studied systematically by Jacob Machado et al. (2021a). This study demonstrated that ML scores correlate negatively with gap-opening costs, and that variations in the number and distribution of gaps produce substantial and largely unpredictable effects on tree topology. Using Monte Carlo simulations that replaced nucleotides with gaps while holding nucleotide homology fixed, Jacob Machado et al. (2021a) confirmed that, under the standard convention of gaps as unknown nucleotides, the ML criterion rewards the alignment with more gaps, converging on the **trivial identity alignment** (TIA) as optimal when the alignment space is fully explored. This is a fundamental inconsistency: aligning sequences with MAFFT and then evaluating alignment quality with the ML score applies distinct optimality criteria at different stages of the same analysis.

To quantify the number and distribution of gaps in any alignment, Jacob Machado et al. (2021a) introduced four indices: the **Gap Contiguity Index** (GCI), the **Nucleotide Contiguity Index** (NCI), the **Shared Gap Index** (SGI), and the **Topological Gap Index** (TGI), which provide a quantitative basis for characterizing alignment decisions independently of any particular tree.

1.8.2 Sequence Optimization Algorithms under Dynamic Homology

Dynamic homology, as implemented in direct analysis (Wheeler 1996), treats homology as an optimization problem solved simultaneously with tree search, employing the same optimality criterion for both steps. Wheeler et al. (2006) describe four heuristic procedures for solving the problem of HTU sequence determination:

Direct Optimization (Wheeler 1996) Evaluates simultaneously the homologies of nucleic acid sequences and cladogram cost without requiring pre-analysis alignment. InDels are treated as informative events via an edit cost matrix.

Iterative Pass Optimization (Wheeler 2003b) Combines direct optimization of three sequences with iterative improvement. HTU sequences are optimized based on descendant and ancestral sequences simultaneously, until total cost stabilizes.

Fixed-States Optimization (Wheeler 1999) Treats each sequence as a single multistate character whose state is one of the observed terminal sequences. HTU sequences are drawn from this predetermined pool via dynamic programming.

Search-Based Optimization (Wheeler 2003c) Extends fixed-states optimization by allowing the user to provide a larger set of candidate solutions for ancestral states.

PhyG (Wheeler et al. 2024) enhances all these procedures through character-specific graph traversals (Varón and Wheeler 2012), typically generating shorter trees than POY5, with score improvements reaching 7% on empirical datasets.

1.8.3 Likelihood under Dynamic Homology

ML analysis can be performed under the dynamic homology framework without requiring pre-analysis alignment (Wheeler et al. 2006). In this implementation, a five-state general time reversible

(GTR) model is used, with states A, C, G, T, and gap — treating gap as a fifth character state in the Markov process, unlike standard implementations.

PhyG (Wheeler et al. 2024) additionally implements No-Common-Mechanism (NCM) likelihood (Tuffley and Steel 1997), which does not assume a shared model among sites and is equivalent to parsimony for treeless networks.

1.8.4 Parallel Computation

The scale of HTS data demands distributed computational resources. The problem of cladogram search is amenable to parallelization because many independent heuristic replicates can be executed concurrently on separate processors (Wheeler et al. 2006). PhyG (Wheeler et al. 2024) adopts a different model: it uses lightweight operating system threads instead of MPI, eliminating the need for external libraries and enabling deployment on common hardware and high-performance clusters without additional configuration.

1.9 Phylogenetic Networks

Traditional phylogenetic analysis is restricted to trees — directed acyclic graphs in which each non-root node has exactly one parent. This structure represents only vertical inheritance and strictly bifurcating relationships. Processes such as horizontal gene transfer, hybridization, and introgression produce reticulate evolutionary histories that trees cannot represent without invoking *ad hoc* hypotheses of homoplasy (Wheeler et al. 2024).

PhyG (Wheeler et al. 2024) extends phylogenetic search to softwired and hardwired networks:

Softwired network Each character or data block follows a single display tree; different blocks may follow distinct paths through the network, capturing heterogeneous histories produced by horizontal transfer.

Hardwired network Each reticulation vertex inherits from two parents simultaneously, and the cost is determined by transformation costs in both parental branches. Hardwired network costs are always at least as high as those of any display tree derivable from the graph (Wheeler et al. 2024).

Individual data blocks can be assigned to separate reticulate paths via the `reblock` command, allowing the user to specify which loci share a common history (Wheeler et al. 2024).

1.10 Optimality Criteria

1.10.1 Overview and Incommensurability

Phylogenetic systematics requires an optimality criterion: a metric by which we compare candidate trees and select the one or ones that best explain the data. The three dominant criteria — parsimony, maximum likelihood, and Bayesian posterior probability — are **incommensurable**: a parsimony score, a log-likelihood, and a posterior probability cannot be directly compared among methods (Wheeler and Varón 2025). This prevents principled model selection or comparison of alternative graph types under existing frameworks.

1.10.2 Parsimony

Parsimony selects the cladogram that minimizes total cost — the total number of transformation events postulated (Farris 1983; Wheeler et al. 2006). Minimizing homoplasy is not merely computational convenience, but an epistemological principle grounded in Occam's razor: each instance of homoplasy represents an independent evolutionary event that must be explained without reference to shared ancestry.

1.10.3 Maximum Likelihood

Maximum likelihood selects the tree that maximizes $\Pr(D \mid T, \theta)$, the probability of the data D given the tree T and a set of model parameters θ. It requires explicit specification of a substitution model.

1.10.4 Bayesian Inference

Bayesian inference calculates the posterior probability distribution $P(T \mid D)$ over the space of topologies, under a specified *a priori* distribution, combining information from the data with prior beliefs formally expressed.

1.10.5 Phylogenetic Minimum Description Length (PMDL)

Wheeler and Varón (2025) proposed PMDL as a unified optimality criterion grounded in algorithmic information theory (Kolmogorov 1968; Li and Vitányi 1997). PMDL defines the complexity of a phylogenetic hypothesis H as:

$$K(H) = K(M) + K(D \mid M) \tag{1.1}$$

where $K(M)$ is the algorithmic complexity of the model M (encompassing both the structure of the graph and the character transformation model) and $K(D \mid M)$ is the complexity of the data D given the model — equivalent to the likelihood of the data under the model (Wheeler and Varón 2025). The preferred hypothesis is the one that minimizes $K(H)$.

This formulation has several desirable properties. First, PMDL generates natural character weighting schemes without external specification: each data type contributes its own complexities via the same framework. Second, it includes graph complexity, allowing competition among trees, networks, and forests on a common numerical scale. Third, it converges on parsimony for low-complexity models, on likelihood for large datasets, and on Bayesian posterior probability when priors are specified, unifying the three dominant frameworks (Wheeler and Varón 2025). PMDL is implemented in PhyG (Wheeler et al. 2024).

1.11 Character Weighting

1.11.1 Implied Weighting

Goloboff (1993) proposed implied weighting as a method for simultaneously estimating character weights and searching for the optimal cladogram. The fitness of a character is defined as $F = K/(K + S)$, where K is a concavity constant set by the user and S is the number of extra steps (homoplasy) for that character in a given tree (Wheeler et al. 2006). The analysis maximizes total

fitness during tree search. The constant K controls the rate at which the function decreases as homoplasy increases: low values heavily penalize highly homoplastic characters. The sensitivity to this choice should be examined as part of any analysis employing implied weighting (Goloboff 1993, 2013).

PMDL (Wheeler and Varón 2025) provides an alternative: by generating weights intrinsically from the algorithmic complexity of the model and data, it does not require the user to specify a concavity constant.

1.11.2 Successive Approximation Weighting

Farris (1969) proposed successive approximation weighting as an iterative procedure: characters are first analyzed with equal weights; the resulting tree is used to calculate character fit statistics (e.g., CI); characters are reweighted and the analysis is repeated until stabilization. This method differs from implied weighting in that it performs weighting between analytical cycles, not continuously during search.

1.12 Clade Support

Identifying the optimal cladogram is only one component of a phylogenetic analysis. Assessing confidence in the recovered groups is equally important. Grant and Kluge (2008) argue that support values should reflect the strength of evidence for a group, not merely the frequency with which it is recovered under data perturbation.

1.12.1 Bremer Support

The **Bremer support index** (Bremer 1988) measures the difference in cost between the optimal cladogram containing a given clade and the optimal cladogram that does not contain it. A higher value indicates that more evidence supports the clade. Under likelihood, it is expressed as difference in log-likelihood scores.

1.12.2 Jackknife

The **jackknife** (Farris et al. 1996) is a resampling method without replacement: a subset of characters is randomly deleted in each replicate, and the frequency with which each clade appears is reported. The jackknife is considered more theoretically defensible than the bootstrap under parsimony because it does not resample characters with replacement, which distorts character independence (Farris et al. 1996).

1.12.3 Bootstrap

The **bootstrap** (Felsenstein 1985) resamples characters with replacement; each replicate is a new matrix of the same size in which some characters are absent and others duplicated. Bootstrap frequencies are widely reported but criticized on theoretical grounds (Farris et al. 1996). Jacob Machado et al. (2021b) investigated the empirical relationship between bootstrap support values and likelihood ratio tests, finding a consistent positive relationship, although the precise correspondence depends on model assumptions and data properties.

1.13 Sensitivity Analysis and Topological Comparison Metrics

1.13.1 Sensitivity Analysis

In phylogenetic systematics, **sensitivity analysis** assesses the extent to which obtained hypotheses are affected by analytical variables such as search strategies, optimality criteria, alignment methods, and transformation cost schemes (Machado 2015; Wheeler 1995). Morrison and Ellis (1997) demonstrated that phylogenetic results can be more sensitive to these parameter values than to the choice of analytical method.

The results of a sensitivity analysis can be summarized in several ways (Wheeler et al. 2006): (1) percentage of parameter sets under which each node is recovered (Giribet et al. 2001); (2) "Navajo rug" plots, in which the parameter space is a two-dimensional grid with monophyly of each clade indicated by color coding; (3) consensus of the clades recovered under all parameter sets. Optimal parameter sets can be identified by minimizing partition incongruence between data subsets (Wheeler et al. 2006). The **Meta-Retention Index** (MRI), calculated as the RI over all characters, was proposed as a partition-free metric for this purpose (Wheeler et al. 2006).

1.13.2 Robinson-Foulds Distance

Originally proposed by Robinson and Foulds (1981), the **Robinson-Foulds** (RF) distance is calculated by the number of elementary operations necessary to transform one tree into another, equivalent to the number of splits exclusive to each tree. A known limitation is that shifting a single OTU can alter multiple nodes, causing the metric to overestimate the differences between two topologies.

1.13.3 Match Split Distance

The **match split** (MS) distance for binary unrooted phylogenetic trees is defined by Bogdanowicz and Giaro (2012) as the cost of the best global assignment between the internal edges of the two compared trees. The MS distance is more sensitive than RF and, at the same time, more resistant to the displacement of a small number of terminals (Bogdanowicz and Giaro 2012), making it especially suitable for identifying rogue taxa.

1.14 Open Perspectives and Debates

1.14.1 Static versus Dynamic Homology

The most fundamental debate in molecular phylogenetics concerns the role of multiple sequence alignment. Proponents of static homology argue that alignment is a necessary step in which positional homology hypotheses are explicitly stated and evaluated independently of the tree. Proponents of dynamic homology argue that fixing alignments before search introduces topology-dependent bias and prevents identification of the globally optimal cladogram-plus-homology-scheme (Wheeler et al. 2006). Redelings et al. (2024) provide additional evidence from the InDel perspective: most alignment algorithms are not phylogeny-aware, introduce guide-tree bias, and lead to overestimation of independent loss events.

1.14.2 Treatment of InDels

Redelings et al. (2024) identify three dominant approaches for InDel treatment: (1) exclusion of columns containing gaps; (2) treatment of gaps as missing data; and (3) InDel coding — which codes each InDel event as a binary presence/absence character (Simmons et al. 2007). Probabilistic models of InDels, such as those based on the TKF framework (Thorne et al. 1991), offer a more principled treatment, but have been implemented in relatively limited contexts (Redelings et al. 2024). The absence of a widely accepted gold standard for InDel treatment remains a significant limitation of contemporary phylogenetic practice.

1.14.3 Incommensurability between Optimality Criteria

A parsimony score, a log-likelihood, and a Bayesian posterior probability cannot be directly compared among methods (Wheeler and Varón 2025). PMDL addresses this limitation by expressing all components of the phylogenetic hypothesis in a common unit of bits. Whether PMDL will achieve wide empirical adoption depends on continued improvements in the upper-bound approximations used to estimate Kolmogorov complexity and the availability of user-friendly implementations (Wheeler and Varón 2025).

1.14.4 Gene Trees, Species Tree, and Horizontal Transfer

Phylogenomic analyses frequently recover incongruences among gene trees, which can arise from deep coalescence, hybridization, incomplete lineage sorting, or horizontal gene transfer (HGT). Wheeler (2012) warns against the tendency to attribute all incongruence to HGT without independent evidence, arguing that this is *ad hoc*. The explicit modeling of reticulate events in PhyG (Wheeler et al. 2024) offers a rigorous alternative: if a network with specified reticulation nodes provides a significantly lower PMDL score than the best tree, the network hypothesis is supported by the data, not merely assumed.

1.14.5 Point-and-Click Systematics

The democratization of bioinformatic software has made phylogenetic analysis accessible to investigators with limited training in the theoretical foundations of the discipline. Grant et al. (2003) documented the risks of this development, showing how uncritical application of default parameters, inadequate treatment of InDels, and use of theoretically unjustified analytical choices produce results that superficially resemble rigorous phylogenetic analysis while lacking its logical foundation. The epistemological coherence demanded by the cladistic framework requires that analytical procedures be transparent, repeatable, and theoretically justified (Wheeler et al. 2006).

1.15 Conclusion

The concepts of character, homology, and homoplasy form an interconnected theoretical triad that underlies all phylogenetic systematics. Characters are transformation series laden with theory. Homology is a tree-dependent hypothesis of shared ancestry that serves as the null hypothesis in cladogram construction. Homoplasy is a tree-specific non-minimal transformation that acts as a falsifier of the cladogram in which it is observed.

Outgroup comparison, character polarity, and rooting are the same problem, solved by unrestricted simultaneous analysis of all terminals (Nixon and Carpenter 1993). Ingroup monophyly is a hypothesis subject to empirical test, not a prior assumption, and the successive outgroup expansion procedure (Grant 2019) provides an objective and repeatable criterion for limiting the outgroup sample.

High-throughput sequencing did not alter the conceptual foundations of phylogenetics; it amplified the consequences of how these foundations are implemented. Treating InDels as missing data is an empirically demonstrated source of unpredictable topological error and score inflation (Jacob Machado et al. 2021a). The dynamic homology framework, by integrating sequence alignment and tree search into a single optimization problem, provides the most theoretically coherent approach. PMDL (Wheeler and Varón 2025) provides a unified criterion that can dissolve the tripartition parsimony-likelihood-Bayesian by expressing all forms of phylogenetic inference in a common language of information theory.

Chapter Review Outline

1. Characters are **historical individuals** laden with theory (transformation series; Hennig 1966); phylogenetics = **idiographic** science, not nomothetic (Grant and Kluge 2004b; Popper 1959).
2. Five types of character: additive (Farris 1970), non-additive (Fitch 1971), Sankoff (Sankoff 1975), sequence and chromosomal (Wheeler et al. 2006); phenotypic *vs.* genotypic; total evidence (Kluge 1989).
3. Homology ≠ synapomorphy (Nixon and Carpenter 2012); homology = null hypothesis in group construction; historical relation of identity (Grant and Kluge 2004b).
4. Primary/secondary debate (de Pinna 1991): rejected (Grant and Kluge 2004b; Wheeler et al. 2006); no intermediate category necessary under dynamic homology (Wheeler 2001a,b).
5. Static homology (fixed alignment) vs. dynamic (simultaneous optimization); implied alignments: topology and parameter specific (Wheeler 2003a).
6. InDels: 2nd largest source of genomic variation; gaps as missing data → TIA and unpredictable topologies (Jacob Machado et al. 2021a); indices GCI, NCI, SGI, TGI (Jacob Machado et al. 2021a).
7. Homoplasy as **falsifier** of tree (Farris 1983; Kluge 2009); CI (Kluge and Farris 1969), RI (Farris 1989), decisiveness ≠ homoplasy (Goloboff 1991).
8. Polarity + outgroup + rooting = **same problem**; unrestricted simultaneous analysis (Nixon and Carpenter 1993). Lundberg rooting valid only with independent evidence (Nixon and Carpenter 1993).
9. Successive outgroup expansion (Grant 2019); ingroup monophyly = hypothesis to test.
10. Simultaneous analysis / total evidence (Kluge 1989; Nixon and Carpenter 1996) vs. partitioned (Grant and Kluge 2003).
11. 4 optimization algorithms: DO (Wheeler 1996), iterative passes (Wheeler 2003b), fixed states (Wheeler 1999), search-based (Wheeler 2003c); PhyG exceeds POY5 (Varón and Wheeler 2012; Wheeler et al. 2024).
12. Likelihood under DH: GTR + gap as 5th state; NCM ≡ parsimony for trees (Tuffley and Steel 1997).
13. Implied weighting: $F = K/(K + S)$ (Goloboff 1993, 2013); successive approximations (Farris 1969).
14. Support: Bremer (Bremer 1988), jackknife (Farris et al. 1996), bootstrap (Felsenstein 1985); bootstrap/LRT relationship (Jacob Machado et al. 2021b).
15. 4 criteria: parsimony (Farris 1983); ML; BI; PMDL: $K(H) = K(M) + K(D \mid M)$ (Wheeler and Varón 2025). First three are **incommensurable**.
16. Softwired and hardwired networks in PhyG (Wheeler et al. 2024); hardwired cost ≥ best display tree.
17. Sensitivity: Navajo rugs, MRI, partition incongruence (Morrison and Ellis 1997; Wheeler 1995; Wheeler et al. 2006).
18. Point-and-click systematics: risks of uncritical defaults (Grant et al. 2003). HGT ≠ default explanation for incongruence (Wheeler 2012).
19. Empirical applications: Orthocoronavirinae (Jacob Machado et al. 2021c); genomic epidemiology (Jacob Machado et al. 2021d).

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Chapter 2

Probabilistic Inference Methods for Phylogenomic Reconstruction

2.1 Introduction

The reconstruction of phylogenetic relationships from genomic data sits at the intersection of evolutionary biology, statistics, and computational science. Over the past three decades, the dominant analytical paradigm in phylogenetics has shifted from parsimony-based methods to probabilistic approaches—maximum likelihood (ML) and Bayesian inference (BI). This shift has been driven by the explosion of molecular sequence data, the development of increasingly sophisticated models of sequence evolution, and the availability of computational power that would have seemed fantastic to systematists of the 1990s.

As Wheeler (2012) reminds us, "there is not a single answer to 'which is the best method?' any more than there is a single answer to 'which is the best tree?' A criterion must be specified in both cases" (Wheeler 2012). The choice of a probabilistic framework is itself a hypothesis about how inference should proceed—and it is one that has been vigorously contested on theoretical, philosophical, and practical grounds.

2.2 Theoretical Foundations of Probability in Phylogenetics

2.2.1 The Likelihood Function

The likelihood of a tree T given data D is proportional to the probability of the data given the tree (Edwards 1972):

$$l(T|D) \propto \Pr(D|T)$$

A systematic ML method selects the tree T that maximizes $\Pr(D|T)$. This maximization requires specifying a set of *nuisance parameters* θ, defined as all quantities necessary to calculate $\Pr(D|T)$ beyond the data and tree topology (Wheeler 2012). The three most consequential nuisance parameters are: (1) the character transformation model; (2) branch parameters (branch lengths); and (3) the distribution of rates of change across characters.

2.2.2 The General Markov Model

The most general reversible description of character change for n states is specified by an instantaneous rate matrix R and a stationary frequency vector $\boldsymbol{\pi}$. The symmetry conditions imposed on R are:

$$\forall i,\ R_{ii} = 0; \quad \forall i, j,\ R_{ij} = R_{ji}; \quad \sum_{i=1}^{n}\sum_{j=1}^{n} \pi_i \cdot \pi_j \cdot R_{ij} = 1$$

Here is how we would read this equation, step-by-step:

- **Part 1:** "For all i, ar-sub-ii equals zero;"
- **Part 2:** "For all i and j, ar-sub-ij equals ar-sub-ji;"
- **Part 3:** "The double summation, with i and j from 1 to n, of pi-sub-i times pi-sub-j times ar-sub-ij equals 1."

See some pronunciation and context observations:

Symbol	Description and Pronunciation
$\forall$	Read as "for all" or "for any".
R_{ii}, R_{ij}	Pronounced as the letter and subscripts ("ar-ii" or "ar-ij").
π_i	The sound of π in English is always "pi", never "pie".
$\sum\sum$	Called the "double summation".
n	Read simply as the letter "en".

These matrices are combined in the rate matrix Q:

$$Q_{ij} = \begin{cases} R_{i,j} \cdot \pi_j & \text{if } i \neq j \\ -\sum_{m=1}^{n} R_{i,m} \cdot \pi_m & \text{if } i = j \end{cases}$$

The probability of change between states i and j in time t is:

$$P_{i,j}(t) = \sum_{m=1}^{n} e^{\lambda_m t} \cdot U_{m,i} \cdot U^{-1}_{j,m}$$

where λ_m (lambda-sub-m) are the eigenvalues of Q, U is the matrix of eigenvectors, and U^{-1} is its inverse.

2.3 Classification of Maximum Likelihood Methods

2.3.1 The Concept of Nuisance Parameters

Wheeler (2012) presents a hierarchical classification of likelihood methods used in systematics. The key idea is that each method differs in *how it treats nuisance parameters and ancestral state assignments*. This is a fundamental point frequently neglected: different formulations of what exactly

is being maximized—averages over states (AML), best states at nodes (MPL), or best complete evolutionary paths (EPL)—lead to genuinely different methods with distinct properties.

2.3.2 Integrated Maximum Likelihood (IML)

If we knew the distribution of nuisance parameters, $\Phi(\theta|T)$, we could integrate them out of the parameter space Θ:

$$p(D|T) = \int p(D|T,\theta)d\Phi(\theta|T)$$

Here is how we would read this equation, step-by-step:

- **Left Side:** "The probability pee (p) of dee (D) given tee (T) equals..."
- **Right Side:** "The integral of the probability pee (p) of dee (D) given tee (T) and theta (θ), with respect to the measure phi (Φ) of theta (θ) given tee (T)."
- **Fluent Reading:** "The probability of D given T is the integral of the likelihood of D given T and theta (θ), integrated over the prior distribution (or measure) of theta (θ) given T."

Some pronunciation and context observations:

Symbol	Description and Pronunciation
$p(A\|B)$	The vertical bar "|" is always read as **"given"**.
$\int$	Read as **"integral"**.
θ	Greek letter **"theta"**.
Φ	Greek letter **"phi"** (uppercase).
$d\Phi$	The "d" indicates the differential, read as **"with respect to"** or **"in the measure of"**.

The tree that maximizes this integrated probability is the **Integrated Maximum Likelihood** (IML) tree. This is the theoretically most satisfactory approach because it accounts for our uncertainty in the nuisance parameters rather than choosing a single "best" set.

2.3.3 Maximum A Posteriori (MAP)

If we also have a **prior distribution over trees**, $p(T)$, we can ask: which tree has the highest posterior probability? Székely and Steel (1999) showed that the method with the greatest expectation of returning the true tree selects the tree maximizing $p(T) \times \Pr(D|T)$. This is the **Maximum A Posteriori** (MAP) tree—the Bayesian estimate.

Here is the important connection: when all tree priors are **equal** (a flat, non-informative prior), the MAP tree is identical to the IML tree. As Wheeler notes, "the use of non-uniform tree priors (such as empirical or Yule) breaks this identity" (Wheeler 2012).

2.3.4 Relative Maximum Likelihood (RML) and Its Three Variants

In practice, we rarely know the distribution of nuisance parameters. Rather than integrating them, we **optimize**: we choose θ to maximize $p(D|T, \theta)$. This is called **Relative Maximum Likelihood** (RML), and it is "in general, the methodology used in empirical analyses" (Wheeler 2012).

Average Maximum Likelihood (AML)

This is the most commonly used approach. It **sums over all possible vertex state assignments**, weighted by their probabilities. When people refer to "ML" in typical phylogenetic analyses, they generally mean AML—and Wheeler notes that "for the remainder of this discussion, when we speak of ML methods, we will be referring to AML" (Wheeler 2012).

In AML, ancestral states are nuisance parameters. We do not really care which state an ancestor had; we care about the likelihood of the *tree and the data*. The statistically correct way to remove unknown quantities from a probability calculation is to **marginalize** (that is, sum or integrate) over them, not optimize them.

Most Parsimonious Likelihood (MPL)

Also suggested by Barry and Hartigan (1987), MPL **assigns specific vertex states** (along with other parameters) such that the overall likelihood of the tree is maximized. The name suggests a connection to parsimony, and indeed the idea of selecting "better" ancestral states sounds similar. However, there is a detail: in MPL, the **branch probabilities are the same across all characters**, so "MPL will not (in general) choose the same tree as parsimony" (Wheeler 2012).

MPL differs from AML in that it optimizes over ancestral states. It chooses the single best assignment. In statistics, replacing an integral with an optimization is known as taking a "plug-in" or point estimator, and it tends to **overestimate the evidence** for a particular tree.

Evolutionary Path Likelihood (EPL)

Originally proposed by Farris (1973), EPL goes beyond MPL. Instead of only specifying ancestral vertex states, EPL specifies **the entire sequence of intermediate character states between vertices** such that the overall likelihood of the tree is maximized. Think of it as maximizing the likelihood along each single "path" of evolution in the tree.

A remarkable result: the tree that maximizes EPL **is precisely the most parsimonious tree**. Wheeler states this directly:

> "The tree that maximizes this form of likelihood is precisely the most parsimonious tree. This result holds for a broad and robust set of assumptions (there is no requirement for low or homogeneous rates of character change, for example)" (Wheeler 2012)

2.3.5 The Paradox: Felsenstein vs. Farris

The relationship between parsimony and likelihood creates what appears to be a contradiction. Felsenstein (1973, 1978) argued that ML methods are statistically consistent and showed that parsimony can be inconsistent under certain conditions (the "Felsenstein Zone," where long branches attract each other). But Farris (1973a) showed that parsimony *is* an ML method under the EPL formulation. If ML is consistent and parsimony is ML, then parsimony should be consistent too—but

Felsenstein showed it is not always so. Wheeler (2012, p. 218) resolves this apparent paradox by recognizing that Farris and Felsenstein were discussing **different forms of likelihood**: Felsenstein's ML refers to MAL (averaging over ancestral states with a common model across characters), while Farris's ML refers to EPL (maximizing over the entire path of evolutionary changes). These are genuinely different optimization problems that can yield different optimal trees. Felsenstein's consistency results apply to MAL, not to EPL, and so the contradiction dissolves.

2.3.6 Additional Connections: Goldman (1990) and Tuffley & Steel (1997)

The link between parsimony and likelihood was further explored by Goldman (1990), who showed that when edge weights are held constant over the tree, likelihood and parsimony converge (Goldman 1990). Tuffley and Steel (1997) introduced the **No-Common-Mechanism (NCM) model**, where each character has a potentially unique rate that may vary among edges (Tuffley and Steel 1997). Under this model and a Neyman-type framework, the overall likelihood becomes a direct function of parsimony length. However, as Wheeler (2012, p. 224) cautions, this equivalence holds strictly only "when the number of states of each character (r_i) is a constant over the data set." When r_i varies across characters, the NCM likelihood introduces a character-specific weighting factor, so that "NCM can be viewed as a likelihood-based character weighting scheme in parsimony analyses" (Wheeler 2012).

It is important to distinguish EPL from the NCM model: EPL is a variant in *how ancestral states are handled*, sitting within Wheeler's classification as a flavor of Maximum Relative Likelihood. The NCM model, by contrast, is a variant in the *evolutionary model itself*. EPL yields an exact equivalence with parsimony under very general conditions (no requirement of low or homogeneous rates). NCM yields equivalence with parsimony only conditionally—when r_i is constant across characters. Despite these differences, both results converge on the same conclusion: that parsimony has a principled foundation within the likelihood framework.

2.4 The Bayesian Perspective

2.4.1 Bayes' Theorem and Posterior Inference

Bayesian inference in phylogenetics reformulates tree inference using Bayes' Theorem. The posterior probability of a topology T given data D, substitution parameters θ, and branch weights t_T is:

$$p(T|D,\theta,t_T) = \frac{p(T)\cdot p(D|T,\theta,t_T)}{\sum_{T_j\in\tau} p(T_j)\cdot p(D|T_j,\theta,t_T)}$$

Here is how we would read this equation, step-by-step:

- **Left Side:** "The probability pee (p) of tee (T) given dee (D), theta (θ) and tee sub-tee (t_T) equals..."
- **Numerator:** "The prior probability of tee (T) times the likelihood of dee (D) given tee (T), theta (θ) and tee sub-tee (t_T)..."
- **Denominator:** "...divided by the summation, for all topologies tee-j (T_j) belonging to the set tau (τ), of the probability of tee-sub-j (T_j) times the likelihood of dee (D) given tee-sub-j (T_j), theta (θ) and tee sub-tee (t_T)."

- **Fluent Reading:** "The posterior of tree T is the product of its prior and its likelihood, normalized by the summation of the probabilities of all possible topologies in tree space τ."

Some pronunciation and context observations:

Symbol	Description and Pronunciation
$p(T)$	Often called the **"prior"** of the topology.
t_T	Read as **"tee sub-tee"**, represents the branch lengths of tree T.
$\sum_{T_j \in \tau}$	Read as **"summation for each tee-j belonging to tau"**.
τ	Greek letter **"tau"**, here representing the set of all possible topologies.
$\frac{A}{B}$	Aloud, we say the numerator and use the expression **"over"** or **"divided by"** for the denominator (in this case, A over B or A divided by B).

Integration over nuisance parameters removes the conditionality on them:

$$p(T|D) = \frac{\iint p(\theta), p(T|\theta), p(t_T|\theta, T), p(D|\theta, T, t_T)\ dt_T, d\theta}{\sum_{T_j \in \tau} \iint p(\theta), p(T_j|\theta), p(t_{T_j}|\theta, T_j), p(D|\theta, T_j, t_{T_j})\ dt_{T_j}, d\theta}$$

Here is how we would read this equation, step-by-step:

- **Left Side:** "The posterior probability of tree (T) given only the data (D)."

- **Numerator:** "The double integral over parameters theta (θ) and branch weights (t_T) of the joint probability of all model components."

- **Denominator:** "The summation, for all possible trees (T_j), of the same double integral performed in the numerator."

- **Fluent Reading:** "The probability of topology T is its likelihood marginalized over all possible values of parameters and branch lengths, normalized by the sum of the marginal likelihoods of all other topologies."

Some pronunciation and context observations:

Symbol	Description and Pronunciation
$\iint$	Read as **"double integral"**. Represents integration over multiple parameters simultaneously.
$dt_T, d\theta$	Indicate the variables of integration. We read as **"with respect to tee-sub-tee and theta"**.
Marginalization	The process of using integrals to "remove" nuisance parameters (θ, t_T) and focus only on what matters (T).
$p(T\|D)$	Note that θ and t_T disappeared from the left side; this happens because they were **"integrated"** out of the equation.

2.4.2 Bayesian Convergence and the Bernstein–von Mises Theorem

A fundamental theoretical guarantee often cited in Bayesian analysis is that, given sufficient data, the posterior distribution will converge to the "truth". This is formalized by the **Bernstein–von Mises Theorem** (BvM). The BvM Theorem states that under specific regularity conditions, as the amount of data (in phylogenetics, the sequence length n) approaches infinity, the posterior distribution converges to a multivariate normal distribution centered at the true parameter values (or the maximum likelihood estimator), with variance shrinking to zero (Vaart 1998).

However, it is critical to recognize that this convergence guarantee is both **asymptotic** and **conditional**:

- **Asymptotic.** Applies strictly only in the limit as $n \to \infty$. For any finite alignment (which is all we have), the guarantee is an approximation whose quality depends on how large n is relative to the problem's complexity.
- **Conditional.** The standard formulation of the BvM Theorem requires that the true data-generating mechanism (i.e., the actual biological process that produced the sequences) lies perfectly within the space of predefined models being explored. If the true model M_{true} is not an element of the model space $\mathcal{M}$, the fundamental guarantees of the standard BvM Theorem do not apply directly.

2.4.3 Discrete Topologies and BHV Tree Space

An additional subtlety frequently neglected: the classical BvM Theorem applies to continuous parameters in Euclidean space. Tree topologies, however, are **discrete combinatorial objects**. The phylogenetic parameter space is not a smooth manifold. It is, instead, a collection of orthants (one per topology) glued together along their boundaries, forming the Billera–Holmes–Vogtmann (BHV) tree space (Billera et al. 2001).

The BvM Theorem can be applied to continuous parameters (branch lengths, substitution model parameters) *conditional on a fixed topology*, but the convergence of posterior probability over topologies is a separate and more difficult problem. For the discrete topology itself, convergence behavior depends on additional conditions, including whether the true topology is identifiable under the assumed model, and whether the data provides sufficient signal to distinguish among competing topologies.

Explaining It Simply: The Problem with Trees

Imagine you are trying to map the path of a road. In ordinary geometry, you can move smoothly in any direction. But in phylogenetics, changing the structure of a tree (the "topology") is like jumping from one universe to another.
Each possible tree is like a separate room. You can move freely within the room (adjusting branch sizes), but to change who is "related" to whom, you need to cross a narrow door that connects these rooms. The **BHV tree space** is the mathematical name for that "maze of rooms" glued together. The challenge is that the normal rules of probability work well within each room, but become much more complicated when we try to decide which room the data should really be in.

Practical Consequences for Research

- **The Convergence Problem (MCMC):** Because tree space is a "maze of rooms", algorithms (like those in MrBayes or BEAST) often get "stuck" in one room (a topology) that looks good, but cannot find the door to an even better room. This is why we require millions of generations and the use of multiple cold and hot chains (MCMCMC).
- **Non-Euclidean Uncertainty:** You cannot simply take the "average" of two different trees to get an average tree. In practice, we use the *Majority Rule Consensus* or *Maximum Clade Credibility Trees* because arithmetic averaging does not make sense in discrete spaces.
- **Phylogenetic Signal vs. Noise:** If the "truth" lies exactly on the boundary between two rooms (e.g., a very fast radiation where the internal branch is nearly zero), the model may oscillate infinitely between topologies, generating low support (low posterior values) not from error, but from the very geometry of space itself.

2.5 Model Misspecification

2.5.1 Convergence to the Best Wrong Answer

In phylogenetics, we can state with confidence that the model is *always* misspecified to some degree. Biological sequence evolution is a profoundly complex process driven by heterogeneous mutation rates, natural selection operating at multiple levels, incomplete lineage sorting, horizontal gene transfer, changing population dynamics, and context-dependent substitution processes. Standard Markov models of sequence evolution (*e.g.*, GTR+Γ) are mathematically tractable abstractions.

When the model is misspecified ($M_{\text{true}} \notin \mathcal{M}$), we must ask: to what does the Bayesian posterior distribution converge as sequence length increases?

Under model misspecification, the posterior distribution does not converge to absolute truth. Instead, as $n \to \infty$, the posterior probability concentrates on the parameter values within the model space explored that minimize the Kullback–Leibler (KL) divergence from the true data-generating distribution. Formally, convergence is to:

$$M^* = \arg\min_{M \in \mathcal{M}} D_{KL}(P_{\text{true}} \| P_M)$$

Here is how we would read this equation, step-by-step:

- **Left Side:** "M-star (M^*) equals the argument of the minimum..."
- **Right Side:** "...over the set of models M ($\mathcal{M}$), of the Kullback–Leibler Divergence (D_{KL}) between the true distribution (P_{true}) and the model distribution (P_M)."
- **Fluent Reading:** "The ideal model M^* is the one, within our set of possible models, that minimizes the information loss (KL Divergence) relative to the reality of the data."

Some pronunciation and context observations:

Symbol	Description and Pronunciation
M^*	The asterisk indicates the **optimal** value or solution of the problem.
$\arg\min$	We do not want the minimum value of the distance, but rather **which model** generates that minimum.
$\mathcal{M}$	Calligraphic letter representing the **search space** (all the models you are testing).
$D_{KL}(P\|Q)$	The **Kullback–Leibler Divergence**. Measures the "distance" (or extra surprise) between two distributions.

This M^* is sometimes called the "pseudo-true" parameter value: the best approximation to truth available within the wrong model family.

2.5.2 Topological Consequence

For **continuous** parameters (branch lengths, substitution rates), convergence to the KL minimizer may produce values that are "close enough" to be biologically useful, even if not exactly correct. But for tree topology (a **discrete** parameter), the consequences of misspecification are more prominent.

A misspecified model may cause the posterior to concentrate on a topology that **is not the true tree**, and this concentration may occur with arbitrarily high posterior probability as $n \to \infty$. This is the mechanism underlying the well-documented phenomenon of posterior probability inflation: Bayesian posteriors may assign probability approaching 1.0 to incorrect clades when the model is wrong (Cummings et al. 2003; Suzuki et al. 2002; Yang 2007).

In phylogenomics, where massive datasets produce steep likelihood surfaces, this failure mode becomes especially dangerous: more data does not correct the error but instead increases the confidence with which the wrong answer is delivered (Kumar et al. 2012).

2.6 Finite Data and Inescapable Priors

Although asymptotic theorems are valuable for understanding theoretical properties of estimators, **empirical systematics deals exclusively with finite data**. The sequence length n of any physical alignment is always a finite integer.

Because **we operate permanently in a finite data regime**, the asymptotic state where perfect likelihood dominates the prior is never strictly reached.

This equation explains the asymptotic state we spoke of above:

$$\lim_{t\to\infty} P_t = \pi$$

Here is how we would read this definition, step-by-step:

- **Expression:** "The limit of pee-sub-tee (P_t), as time tee (t) approaches infinity, equals pi (π)."
- **Technical Meaning:** "The distribution of the system converges to its stationary (or asymptotic) state π, regardless of where the system started."
- **Fluent Reading:** "As time passes indefinitely, the configuration of the system stabilizes in a stable equilibrium distribution."

Some pronunciation and context observations:

Symbol	Description and Pronunciation
$\lim_{t\to\infty}$	Read as **"limit with tee approaching infinity"**. Represents long-term behavior.
P_t	The probability distribution at instant t.
π	In phylogenetics, represents the **equilibrium frequencies** of bases (A, C, G, T) or amino acids.
$\to$	The arrow is read as **"approaches"** or **"goes to"**.

Now, having explained the asymptotic state we referred to earlier, we can return to the proportionality of Bayes' Theorem:

$$P(\tau, \theta|X) \propto P(X|\tau, \theta)\, P(\tau, \theta)$$

Here is how we would read this equation, step-by-step:

- **Left Side:** "The posterior probability of tau (τ) and theta (θ) given xi (X) is proportional to..."
- **Right Side:** "...the likelihood of xi (X) given tau (τ) and theta (θ), times the prior probability of tau (τ) and theta (θ)."
- **Fluent Reading:** "The posterior is proportional to the product of the likelihood and the prior."

Some pronunciation and context observations:

Symbol	Description and Pronunciation
$\propto$	Read as **"proportional to"**. Replaces the equals sign when we omit the denominator.
τ	Greek letter **"tau"**. In phylogenetics, almost always represents the **tree topology**.
θ	Greek letter **"theta"**. Represents the **model parameters** (such as substitution rates).
X	Represents the **observed data** (the DNA or amino acid sequences).

The posterior is always a mathematical compromise between the information contained in the data (the likelihood) and the information specified before the data were observed (the prior). As n grows, the likelihood surface becomes steeper and exerts much more leverage, flattening the relative impact of the prior. However, because n is finite, the likelihood never achieves infinite precision.

Consequently, **the prior distribution is strictly *always* part of the final answer**.

2.6.1 Persistent Influence of the Prior

The influence of the prior (or "*prior*") is not uniform across all parameters. For substitution model parameters (*e.g.*, rate matrix entries and base frequencies), moderate amounts of data typically dominate reasonable priors, and the practical effect of the prior becomes negligible. These are parameters for which data are highly informative, the parameter space is low-dimensional, and different reasonable priors converge on similar posteriors.

However, for other parameters listed below, *the prior* can persist tenaciously:

- **Tree topology.** The prior over topologies is a probability distribution over a combinatorially vast discrete space. For $n = 50$ taxa, there are approximately 2.8×10^{76} unrooted bifurcating trees. A uniform prior assigns infinitesimal probability to each topology, and different structural priors (uniform vs. Yule vs. birth-death) assign different probabilities to different tree shapes. In regions of the tree where the data lack strong signal—rapid radiations, short internodes, saturated substitutions—the posterior over competing topologies will depend strongly on whatever prior was specified.

- **Branch lengths close to zero.** Short internodes, which characterize rapid radiations and the "anomaly zone" of phylogenomics, are precisely the branches where data provide the minimum signal and priors have the maximum influence. Brown *et al.* (2010) demonstrated that the standard exponential branch-length prior in MrBayes can produce inflated posterior probabilities and unreliable branch-length estimates, not from a software bug, but because the prior was inadvertently informative in ways that interacted poorly with the data. Rannala *et al.* (2012) confirmed that branch-length priors can have substantial effects on tree posterior probabilities even with moderately large datasets.

- **Divergence times.** In molecular clock analyses, the prior on branch rates (*e.g.*, lognormal, exponential, or uniform) can shift estimated divergence dates by tens of millions of years.

Wheeler (2012) suggests that "given our lack of knowledge, sampling, lineage, environmental, and locus-specific (among a myriad of other known and unknown) effects, the uniform distribution would seem as appropriate as lognormal or exponential". However, a uniform prior on rates is itself a strong statement, assigning equal prior probability to biologically absurd rates and plausible ones. Every prior says *something*, and "I don't know" is not a probability distribution.

2.7 Responsible Practices for Working with Priors and Models

If priors always matter and models are always wrong, the question is not whether these limitations can be eliminated (they cannot). Instead, the question becomes how to work responsibly within them.

2.7.1 Sensitivity Analysis

The most important practical answer is to vary both priors and models and examine how results change. If a clade is supported under one prior but not another, or under one model but not another, the data are not strong enough to support that clade independent of the method. This logic parallels Wheeler's (2012, Section 10.11) recommendation for sensitivity analysis under different cost regimes in parsimony. Conclusions robust to analytical assumptions are more reliable than those that are not.

2.7.2 Model Adequacy Testing

Rather than assuming the chosen model is "close enough", researchers should explicitly test whether the model fits the data, using tools such as posterior predictive simulations and model adequacy assessments. If the model fails its own goodness-of-fit test, the posterior it produces is conditional on an inadequate assumption, and no amount of data will rescue it.

The Problem of Model Adequacy

- **Selection vs. Adequacy:** Selecting a model is like choosing the shoe that fits best among three small options. Adequacy is measuring whether your foot really fits in any of them. If all are small, the "best" will still hurt your foot.
- **Posterior Predictive Simulations (PPS):** This is a "sanity check". The software uses results from your analysis to generate synthetic (fictional) data. If data simulated by the model are completely different from your real DNA data, the model did not "understand" the biological process.
- **The Danger of Data Quantity:** There is a myth that having large amounts of data (phylogenomics) corrects bad models. Quite the opposite! If the model is wrong, more data only gives it more confidence to arrive at the **wrong answer** with high posterior probability.

2.7.3 Honest Reporting of Uncertainty

A posterior probability of 1.0 for a clade means "this clade is overwhelmingly supported **given** the model and priors." It does not mean that "this clade is certainly correct." Reporting clade support alongside concordance factors that quantify what proportion of the genome really supports a clade will provide a more honest picture of the evidence than posterior probabilities alone.

Posterior 1.0 vs. Biological Reality

- **The Illusion of 1.0:** In large datasets (phylogenomics), the posterior probability often "inflates" to 1.0. This happens because the Bayesian model accumulates so much mathematical evidence that it becomes "mathematically certain", even though the model itself oversimplifies biology.
- **Conflicting Signal:** The genome is not a single unit. Different genes may have different histories (due to incomplete lineage sorting or hybridization). A posterior of 1.0 may hide the fact that, for example, 40% of your gene data support a different tree.
- **Concordance Factors (CF):** Unlike the posterior, CFs (such as the *Gene Concordance Factor*) tell you: "Of all the genes we examined, how many actually show this clade?". If the posterior is 1.0 but the CF is 0.2, you know the support is mathematically strong, but there is massive underlying biological conflict.

2.8 Parsimony Reconsidered: Premises for an Alternative Criterion

The preceding sections have demonstrated that probabilistic methods rest on assumptions—correct model specification, infinite data, well-behaved priors—that are never fully met in practice. This raises a natural question: can the case for parsimony be made on its own terms, not merely as a special case of likelihood, but as a criterion that is better suited to the realities of phylogenetic inference?

2.8.1 Premises

Several premises underpin this argument. First, all models are incorrect, and model incorrectness is not merely imprecise—it can be actively misleading. As Wheeler (2012, p. 279) states, the consistency of ML requires "that the model used for reconstruction is the same as that which generated the tree in the first place. In general, likelihood will be inconsistent otherwise." Chang (1996) showed that even a case as simple as using one model when the true scenario was generated by two destroys consistency (Chang 1996). Second, statistical consistency is an asymptotic guarantee that is routinely violated in practice: empirical sequence data are not independent, not identically distributed, and finite (Wheeler 2012). Third, ML itself suffers from systematic failures: Pol and Siddall (2001) demonstrated that likelihood is prone to long-branch attraction regardless of model correctness, and the "Farris Zone" (Siddall, 1998)—a failure mode not found in parsimony analyses—shows that parsimony can outperform likelihood under specific conditions (Pol and Siddall 2001; Siddall 1998). Fourth, each character's evolutionary history is unique, shaped by unique selective pressures, population dynamics, and mutational contexts; treating characters as having uncorrelated evolutionary rates (as in the NCM model) is biologically more realistic than imposing

a single parametric distribution shared across all sites and lineages (Tuffley and Steel 1997).

2.8.2 Practical Advantages of Parsimony

From these premises, several practical advantages follow. Parsimony is **non-parametric and robust**: it does not assume a specific model of character evolution, and therefore cannot be misled by model misspecification in the way that ML and Bayesian methods can. Parsimony **avoids the nuisance parameter problem**: it does not require specifying branch lengths, rate distributions, or base frequencies, all of which introduce additional dimensions of uncertainty and potential error. Parsimony is **computationally efficient**: because it does not require matrix exponentiation or numerical optimization of continuous parameters, tree scoring under parsimony is substantially faster, enabling more thorough searches of tree space. Finally, parsimony has **maximal explanatory power** in the cladistic sense: it minimizes the number of *ad hoc* hypotheses of homoplasy required to explain the data, thereby preferring the tree that explains the most character distributions as historical events rather than convergences (Farris 1983).

2.9 Conclusion

Probabilistic phylogenetic inference is a mathematically rigorous and practically powerful tool, but its results must be interpreted with a clear understanding of its mathematical limits. The convergence of posteriors to the "truth" is an asymptotic guarantee applicable to continuous parameters conditional on the unrealistic assumption of perfect model specification. This guarantee does not extend directly to discrete tree topologies. In biological reality, models are always approximations, meaning that under misspecification, inference converges to the best approximation available within the wrong model family. In the case of tree topology, this may mean convergence to an incorrect tree with high confidence.

Furthermore, the finite nature of any genomic data ensures that prior assumptions remain permanently an active ingredient in the posterior distributions we calculate, with their influence persisting most strongly precisely where we need reliability: on short branches, in rapid radiations, and in divergence time estimates. As our collective understanding of Bayesian methods evolves over time, they may or may not remain popular and considered a flexible and informative tool. However, they must be used with the intellectual honesty they demand: testing model adequacy, conducting sensitivity analyses, and reporting uncertainty in ways that distinguish what the data say from what the assumptions contribute.

Chapter Review Outline

1. Likelihood: $l(T|D) \propto \Pr(D|T)$; nuisance parameters θ: substitution model, branch lengths, rate distribution (Edwards 1972; Wheeler 2012).
2. Four forms of likelihood (Wheeler, 2012): **AML** (sum over ancestral states—the standard "ML"), **MPL** (optimize states at nodes), **EPL** (Farris, 1973—specifies complete evolutionary path; $\text{EPL}_{\max} \equiv$ parsimony) and **IML** (integrates over θ; = MAP with flat prior).
3. Felsenstein–Farris paradox: resolved by recognizing Felsenstein's "ML" = AML (averaging over ancestral states), Farris's "ML" = EPL (maximizing complete evolutionary paths); these are different optimization criteria.
4. No-Common-Mechanism (NCM) model (Tuffley & Steel, 1997): each character has unique rate per edge; under constant r_i, NCM likelihood $\equiv$ parsimony; differs from EPL (which is a variant in ancestral-state handling, not in the model itself).
5. Premises for parsimony: model misspecification destroys ML consistency (Chang, 1996); ML also inconsistent in "Farris Zone" (Siddall, 1998); parsimony is non-parametric, avoids nuisance parameters, computationally efficient, and maximizes explanatory power.
6. Bernstein–von Mises Theorem (BvM): posterior $\to$ normal centered at truth when $n \to \infty$ — but **asymptotic** and **conditional** on true model $\in \mathcal{M}$.
7. Topologies are discrete objects; BHV tree space (Billera–Holmes–Vogtmann, 2001) is not smooth manifold; BvM for topologies is separate problem.
8. Misspecification ($M_{\text{true}} \notin \mathcal{M}$): posterior converges to KL minimizer $M^* = \arg\min D_{KL}(P_{\text{true}} \| P_M)$, not to truth.
9. Consequence: more data $\Rightarrow$ greater certainty in the **wrong answer** (posterior probability inflation) (Kumar et al. 2012; Suzuki et al. 2002).
10. Prior is **always** active ingredient with finite data; influence persists in: topologies (combinatorial space $\sim 2{,}8 \times 10^{76}$ for 50 taxa), short branches (rapid radiations) and divergence times.
11. Standard exponential prior on branch lengths in MrBayes can inflate posteriors (Brown et al., 2010; Rannala et al., 2012).
12. Responsible practices: sensitivity analysis (vary priors and models), model adequacy testing (posterior predictive simulations), honest reporting of uncertainty (concordance factors).

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Chapter 3

Substitution Models in Phylogenetic Reconstruction

3.1 Introduction

Phylogenetic reconstruction from molecular sequence data requires an explicit probabilistic framework for modeling how characters change along a tree. This framework is incorporated in substitution models, which specify the instantaneous rates of change between character states and the stationary frequencies of those states. Substitution models underpin two main optimality criteria in modern systematics: maximum likelihood (ML) and Bayesian posterior probability. Both criteria share a common probabilistic core but differ in how they treat unknown parameters and how they summarize inference.

This chapter synthesizes the theoretical principles, implementation strategies, interpretive limits, and recent developments associated with substitution models in phylogenetic analysis. The primary sources are chapters 11 and 12 of Wheeler (2012), which address likelihood and Bayesian optimality criteria respectively.

3.2 Theoretical Principles

3.2.1 The Likelihood Function in Systematics

Note that this section was already explained in the previous chapter, where additional details are provided to help understand the equations. Here, some information is repeated to aid study of the material at the expense of page count.

The foundation of model-based phylogenetics is the likelihood function. For a tree T and data D, the likelihood is proportional to the probability of the data given the tree and a set of parameters (Edwards 1972):

$$l(T|D) \propto \Pr(D|T)$$

A systematic ML method selects the tree T that maximizes $\Pr(D|T)$. This maximization requires specifying a set of *nuisance parameters* θ, defined as all quantities necessary to compute $\Pr(D|T)$ beyond the data and tree topology (Wheeler 2012). The three most consequential nuisance parameters are: (1) the model of character transformation; (2) branch parameters (branch lengths);

and (3) the distribution of rates of change among characters.

3.2.2 The General Reversible Markov Model

Note that this section was already explained in the previous chapter, where additional details are provided to help understand the equations. Here, some information is repeated to aid study of the material at the expense of page count.

The most general reversible description of character change for n states is specified by an instantaneous rate matrix R and a stationary frequency vector $\boldsymbol{\pi}$ (Tavbré 1986; Yang 1994). The symmetry conditions imposed on R are:

$$\forall i,\ R_{ii} = 0; \quad \forall i, j,\ R_{ij} = R_{ji}; \quad \sum_{i=1}^{n}\sum_{j=1}^{n} \pi_i \cdot \pi_j \cdot R_{ij} = 1$$

These matrices are combined in Tavaré's (1986) rate matrix Q:

$$Q_{ij} = \begin{cases} R_{i,j} \cdot \pi_j & \text{if } i \neq j \\ -\sum_{m=1}^{n} R_{i,m} \cdot \pi_m & \text{if } i = j \end{cases}$$

The probability of change between states i and j over time t is then:

$$P_{i,j}(t) = \sum_{m=1}^{n} e^{\lambda_m t} \cdot U_{m,i} \cdot U^{-1}_{j,m}$$

where λ_m are the eigenvalues of Q, U is the matrix of eigenvectors, and U^{-1} its inverse (Strang 2006). The time-reversibility constraint allows efficient computation of tree likelihoods (Wheeler 2012).

This formulation accommodates at most $n-1$ independent frequency parameters and $\binom{n}{2} - 1$ independent rate parameters.

3.2.3 Named Models as Special Cases

All substitution models in current use are restrictions of the general reversible process through additional symmetry requirements (Wheeler 2012). The hierarchy from least to most parameterized for nucleotide data is documented in Swofford et al. (1996) and includes, among others:

- **JC69** (Jukes and Cantor, 1969): a single type of substitution with equal base frequencies. This is the most restricted model and the only one with zero free parameters (beyond branch lengths).
- **K2P** (Kimura, 1980): distinguishes transitions from transversions with equal base frequencies.
- **HKY85** (Hasegawa et al., 1985): transitions versus transversions with unequal base frequencies.
- **GTR** (Lanave et al., 1984; Tavaré, 1986): the most general time-reversible model with six rate categories and four unequal base frequencies.

Models other than GTR have analytical solutions for $P_{i,j}(t)$, while GTR requires numerical eigenvalue decomposition (Wheeler 2012).

3.3 Rate Variation Among Sites

3.3.1 Invariable Sites

Molecular sequence data commonly show pronounced heterogeneity in substitution rates among positions. Two distributional models are regularly applied to accommodate this variation.

The parameter *proportion of invariable sites I* represents the frequently observed class of positions that never change (Hasegawa et al. 1985). A separate parameter is added to represent the fraction of sites available for substitution.

3.3.2 Rate Heterogeneity and the Discrete Gamma Distribution

To account for the fact that different positions in a sequence evolve at distinct rates, the *discrete Gamma distribution* is employed (Yang 1994). This model adds rate classes governed by a single shape parameter, α, which defines the degree of variation among sites.

The probability density function (PDF) of the Gamma distribution is defined as:

$$g(x;\alpha,\beta) = \frac{\beta^{\alpha} x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)} \tag{3.1}$$

Where:

- x: represents the substitution rate.
- α (alpha): the shape parameter. The smaller the value of α, the greater the heterogeneity (many slow sites and few fast ”hotspots”).
- β (beta): the scale parameter. In phylogenetic practice, one sets $\beta = \alpha$ to ensure that the mean of the distribution is 1 ($\alpha/\beta = 1$), maintaining the interpretation of branch length units as the expected number of substitutions per site.
- $\Gamma(\alpha)$ (Gamma of alpha): the Gamma function, which acts as a normalization constant so that the area under the curve equals 1.

Implementation via discretization:

Since integrating over a continuous distribution is computationally prohibitive, the curve is divided into k classes (usually 4 to 8) of equal areas. The mean rate of each partition is used as a multiplier for the tree's branch lengths.

It is fundamental to note that this is a **rate mixture model**: we do not know *a priori* which rate class belongs to which site. Therefore, the likelihood of site i is calculated as the weighted average of likelihoods across all k rate classes:

$$L_i = \sum_{j=1}^{k} \frac{1}{k} L_i(r_j) \tag{3.2}$$

Where r_j is the representative rate of the j-th class. This method allows capturing the phylogenetic signal of both conserved and variable sites simultaneously without drastically increasing the number of model parameters.

3.4 Expansion to Five States: Models with Indels

3.4.1 Five-State Jukes-Cantor Model

For dynamic homology analyses, the four-state nucleotide model can be expanded to include a fifth state representing insertion-deletion (indel) events (McGuire et al. 2001; Wheeler et al. 2006). Under the five-state Jukes-Cantor model (JC69+Gaps), the transition probability is:

$$P_{ij}(t) = \begin{cases} \frac{1}{5} + \frac{4}{5}e^{-\mu t} & \text{if } i = j \\ \frac{1}{5} - \frac{1}{5}e^{-\mu t} & \text{if } i \neq j \end{cases}$$

This approach treats indels as atomic events and emphasizes computational tractability over mechanistic realism.

3.4.2 Birth-Death Models

A more mechanistically grounded alternative is the birth-death model of Thorne et al. (1991) (TKF91, TKF92), which couples a Poisson birth-death process with standard four-nucleotide substitution models to compute the probability of transforming one sequence into another. TKF91 and TKF92 represent the most mechanistically explicit treatment of indels, where the probability of n insertions over time t is:

$$p_n(t) = e^{-\mu t}\left[1 - \lambda\beta(t)\right]\left[\lambda\beta(t)\right]^{n-1}$$

where $\beta(t) = (1 - e^{(\lambda-\mu)t})/(\mu - \lambda e^{(\lambda-\mu)t})$, with λ being the insertion rate and μ the deletion rate.

3.5 Forms of Likelihood and Their Relationships

Wheeler (2012) and Barry and Hartigan (1987) distinguish several forms of likelihood relevant to phylogenetic inference:

1. **Average Maximum Likelihood (AML)**: sums over all possible assignments of vertex states weighted by their probabilities. This is the standard "ML" method used in empirical analyses.

2. **Most Parsimonious Maximum Likelihood (MPML)**: assigns specific vertex states to maximize overall likelihood. Branch probabilities are constant across characters, so MPML generally does not select the same tree as parsimony (Wheeler 2012).

3. **Evolutionary Path Likelihood (EPL)**: proposed by Farris (1973), this form specifies the entire sequence of intermediate states between vertices. The tree maximizing EPL is precisely the most parsimonious tree (Wheeler 2012), a result also derived by Goldman (1990) and Tuffley and Steel (1997).

4. **Integrated Maximum Likelihood (IML)**: integrates over nuisance parameters θ given their distribution $\Phi(\theta|T)$. This corresponds to Bayesian MAP estimation when tree priors are flat (Steel and Penny 2000; Wheeler 2012).

The relationship among these forms is summarized in Wheeler (2012). The apparent paradox between Felsenstein's claim of ML consistency and Farris's result of MP-as-ML dissolves once we recognize that the two authors discussed different forms of likelihood.

3.6 Protein Modeling and Structural Constraints

3.6.1 Amino Acid Substitution Matrices

Protein evolution is more complex than nucleotide evolution. Empirical amino acid substitution matrices, such as PAM and WAG, were estimated from large databases of aligned protein sequences. These matrices capture average evolutionary trends but assume stationarity and homogeneity of the substitution process, which are biologically unrealistic.

More realistic are structurally constrained substitution (SCS) models, which account for the physical constraints imposed by protein folding on the rates of amino acid substitutions. SCS models frequently incorporate site-dependent evolution, meaning that the rate matrix at one site depends on the states of neighboring sites or structural contacts. This feature dramatically increases biological realism but renders the standard likelihood function intractable because ML-based phylogenetic methods compute likelihoods under the assumption of site independence.

3.6.2 Mixture Models and Site-Dependent Evolution

Mixture models such as CAT and related variants incorporate variable substitution processes among protein regions by combining simpler models (Lartillot and Philippe 2004). These approaches allow different sites to use different substitution matrices, partially accounting for the structural and functional heterogeneity of real proteins. PhyloBayes implements site-dependent models through a Dirichlet process mixture, in which the number of substitution classes is inferred from the data rather than specified *a priori*.

Regarding model selection, the approach depends critically on whether models assume site-independent or site-dependent evolution. Site-independent empirical models (e.g., WAG, LG, JTT) can be compared using standard likelihood-based criteria: likelihood ratio tests, AIC, and BIC. These criteria are implemented in software such as ModelTest-NG, ProtTest3, ModelFinder (within IQ-TREE2), and PartitionFinder. Site-dependent models, however, cannot be evaluated using these traditional criteria because evolutionary dependence among sites is incompatible with standard phylogenetic likelihood functions, which compute site likelihoods independently. Del Amparo and Arenas (2022) developed ProteinModelerABC, an approach based on approximate Bayesian computation (ABC) that performs model selection among SCS models using summary statistics related to protein structural and thermodynamic properties (Del Amparo and Arenas 2022).

3.6.3 Advanced Structurally Constrained Models

Advanced SCS models adapt statistical physics to molecular evolution. They evaluate amino acid mutations using the fixation probability from a Moran birth-death process and fitness via a Boltzmann distribution based on protein-folding free energy (Ferreiro et al. 2026). These models dramatically increase biological realism: mutations are accepted or rejected based on their effect on protein thermodynamic stability rather than on a fixed empirical exchangeability matrix. However, site-dependent SCS models cannot provide site-independent exchangeability matrices and are

therefore incompatible with standard likelihood-based phylogenetic inference. They are currently more useful for simulation and for understanding evolutionary constraints than for direct use in tree reconstruction.

A critical practical limitation is that most existing SCS model implementations are not available in standard, freely available phylogenetic frameworks such as IQ-TREE2, RAxML-NG, or BEAST2. Implementing these models in accessible software would facilitate empirical comparisons and promote their adoption in phylogenomic analyses.

3.7 Model Selection

3.7.1 Classical Model Selection Criteria

Model selection has been recognized as a prerequisite for accurate phylogenetic inference. Incorrectly specified models produce biased estimates of tree topology and branch lengths (Abadi et al. 2019; Buckley 2002; Sanderson 2002). Standard criteria for model selection include:

- **Likelihood Ratio Test (LRT).** When two models are nested, twice the difference in log-likelihood is approximately distributed as χ^2:

  $$2(l_{T,T'}) = 2\log(l_T/l_{T'}) = 2(\log l_T - \log l_{T'})$$

  The confidence threshold for one degree of freedom is 1.9207 log-likelihood units, corresponding to a likelihood ratio of 6.826.

- **Akaike Information Criterion (AIC).** The AIC (Akaike 1974) penalizes parameters with a fixed cost:

  $$\mathrm{AIC} = -2\log l_T + 2p$$

  An additional parameter is favored only if it improves the likelihood by at least one log unit.

- **Bayesian Information Criterion (BIC).** The BIC (Schwarz 1978) penalizes extra parameters more strongly, with the penalty depending on data size n:

  $$\mathrm{BIC} = -2\log l_T + p\log n$$

These criteria are implemented in software such as ModelTest and are reviewed comprehensively in Posada and Buckley (2004).

3.7.2 Impact of Model Selection

Abadi et al. (2019) showed that model selection affects branch length estimates more strongly than topology inference. Furthermore, no standard criterion was designed to select the model best suited specifically for accurate branch length inference.

3.8 Computational Implementation

3.8.1 Computing Tree Likelihood: Felsenstein's Pruning Algorithm

For a single character x, the likelihood of an internal vertex i with descendant vertices j and k is (Wheeler 2012):

$$L_i(x) = \left(\sum_{x_j} p_{x_i,x_j}(t_j) L_j(x_j) \right) \times \left(\sum_{x_k} p_{x_i,x_k}(t_k) L_k(x_k) \right)$$

Character likelihoods are multiplied across the entire dataset to obtain tree likelihood. Branch weights are estimated numerically by optimizing the marginal likelihood for each branch while keeping all other parameters constant, commonly using Brent's method (1973) or Newton-Raphson (Ypma 1995). Because character data are assumed to be independent and identically distributed (i.i.d.), the joint likelihood of multiple characters is the product of their individual values.

3.8.2 Markov Chain Monte Carlo for Bayesian Inference

Direct numerical integration of the posterior is computationally prohibitive for non-trivial datasets. A 100-taxon GTR analysis involves at least 205 parameters. Markov Chain Monte Carlo (MCMC) methods based on the Metropolis-Hastings algorithm (Hastings 1970; Metropolis et al. 1953) provide a tractable alternative by sampling trees proportionally to their posterior probabilities.

The acceptance ratio between a current tree T and a proposed tree T^* is:

$$\alpha = \min \left(1, \frac{\pi(T^*)}{\pi(T)} \right)$$

where $\pi(T) = p(\theta)p(T|\theta)p(t_T|\theta, T)p(D|\theta, T, t_T)$ is the numerator of the posterior. The ratio cancels the marginal probability of data (the denominator of the full posterior), which is difficult to compute directly.

Coupled Metropolis-MCMC (MC^3)

The most commonly used strategy is coupled Metropolis-MCMC (MC^3) (Geyer 1991). Multiple chains run in parallel, each heated to a different temperature. The cold chain (chain 0) converges to the posterior; hotter chains explore parameter space more broadly. Chains exchange states periodically according to a second acceptance ratio:

$$\alpha = \min \left(1, \frac{\pi_i(T_j)\pi_j(T_i)}{\pi_i(T_i)\pi_j(T_j)} \right)$$

The heating schedule is:

$$\pi_j(T) \propto \pi_0(T)^{\frac{1}{1+\lambda(j-2)}}, \quad \lambda > 0$$

3.9 Limits of Interpretation of Results

3.9.1 Statistical Inconsistency of Parsimony: The Felsenstein Zone

The original motivation for ML in systematics was the statistical inconsistency of parsimony under certain combinations of parameters (Felsenstein 1978). In a four-taxon tree where branches leading to taxa A and B are long (probability of change p) while the internal branch and branches to C and D are short (probability of change q), parsimony favors the incorrect topology $(AC|BD)$ when:

$$p^2 \geq q(1-q)$$

The region of parameter space satisfying this inequality is the "Felsenstein zone" of inconsistency (Wheeler 2012).

3.9.2 Clade Prior Bias and Clade Posterior Probabilities

A critical interpretive limitation of Bayesian phylogenetics is that a uniform prior over tree topologies does not induce a uniform distribution over clades of different sizes. The prior probability of a clade of size m in a tree with n leaves is (Wheeler 2012):

$$\Pr(c_m) = \frac{\left(\prod_{i=2}^{m}(2i-3)\right) \cdot \left(\prod_{i=m+1}^{n}(2i-2m-1)\right)}{\prod_{i=2}^{n}(2i-3)}$$

Pickett and Randle (2005) first demonstrated this empirically. Steel and Pickett (2006) subsequently proved that it is impossible to construct a prior over topologies that induces a uniform prior over clades. The influence of this bias can be extreme: for 50-taxon trees, the prior probability ratio between clades of sizes 2 and 25 can reach a factor of 10^{13} (Wheeler 2012).

Goloboff and Pol (2005) illustrated a particularly pathological consequence: a taxon with completely missing data can be placed specifically in a tree (with 55% posterior clade probability) solely because of this clade prior bias, even though all its placements have identical posterior probability under the data. Yang (2006) extended this example to show that clade support values approaching 1 can be generated for groups absent from the true tree.

These results have a direct practical implication: posterior clade probabilities from Bayesian analysis cannot be directly interpreted as measures of phylogenetic support.

3.10 Recent Advances

3.10.1 Hidden Markov Models for Sequence Analysis

A significant advance over simple Markov substitution models is the Hidden Markov Model (HMM), which adds a layer of unobserved latent states (Wheeler 2012). In an HMM, a sequence of observations is generated by a sequence of hidden states, each of which emits observations according to an emission probability and transitions to other states according to a transition probability. The probability of a sequence x given a state path π, transition probabilities t, and emission probabilities e is:

$$p(x, \pi | t, e) = p(\pi_0) e_{\pi_0, x_0} \cdot \prod_{i=1}^{n-1} t_{\pi_{i-1}, \pi_i} e_{\pi_i, x_i}$$

Three canonical algorithms address the three fundamental questions of HMM analysis (Wheeler 2012):

1. **Viterbi Algorithm** (Viterbi, 1967): identifies the most likely hidden state path $\pi^* = \arg\max_\pi p(x, \pi|t, e)$ using dynamic programming in time $O(k^2 n)$ for k states and n observations.

2. **Forward-Backward Algorithm**: computes the posterior probability of each state at each position using the product of forward and backward probabilities conditioned on the total sequence probability.

3. **Baum-Welch Algorithm** (Baum et al., 1970): estimates transition and emission parameters by iterating the Forward-Backward algorithm on training sequences until improvement in sequence probability falls below a threshold.

HMMs have been applied to pairwise and multiple sequence alignment through profile HMMs (Krogh et al. 1994), which model aligned sequence families with position-specific emission and transition probabilities. Profile HMMs are widely used for gene family identification and functional motif detection.

3.10.2 Likelihood-Based Sequence Alignment

The recognition that sequence alignment and phylogenetic inference should be performed jointly under a probabilistic model represents a major conceptual advance. The pairwise alignment recursion under the JC69+Gaps model uses the modified Needleman-Wunsch framework to sum probabilities instead of minimizing costs (Wheeler 2012):

$$\text{cost}[i][j] = \log\left(e^{-(\text{cost}[i-1][j-1]+\sigma_{i,j})} + e^{-(\text{cost}[i-1][j]+\sigma_{\text{indel}})} + e^{-(\text{cost}[i][j-1]+\sigma_{\text{indel}})}\right)$$

Wheeler (2012) emphasizes the distinction between *dominant likelihood* (the single best alignment, analogous to MPML) and *total likelihood* (the sum over all alignments, analogous to AML). The dominant alignment may represent as little as one-third of the total likelihood, underscoring the importance of this distinction for downstream inference.

Wheeler's Direct Optimization (DO) approach (1996, 2006) extends this pairwise framework to the Tree Alignment Problem (TAP), where median sequences and tree topologies are co-optimized under ML.

3.10.3 Birth-Death Indel Models

The TKF91 and TKF92 models of Thorne et al. (1991, 1992) represent the most mechanistically explicit treatment of indels. These models couple a birth (insertion rate λ) and death (deletion rate μ) process with standard four-nucleotide substitution models. Bayesian TAP implementations using TKF91/92 include BEAST (Lunter et al., 2005) and BaliPhy (Redelings and Suchard, 2005). The computational cost is prohibitive: Wheeler (2012) notes that BaliPhy requires hundreds of hours on a personal computer to analyze 12 protein sequences of length 400.

3.11 Machine Learning in Substitution Model Selection

3.11.1 Motivation for ML Approaches

The computational cost of likelihood-based model selection is substantial. Machine learning (ML) methods can "learn complex relationships from input data without computing likelihoods," and this "facilitates the application of machine learning to complex models, especially scenarios in which standard statistical and Bayesian inference may be intractable" (Y. K. Mo et al. 2024).

3.11.2 Random Forests and Deep Neural Networks

ModelTeller (Abadi et al., 2020) uses a random forest (RF) regressor trained on over 50 summary statistics. These statistics are broadly categorized into four groups: properties of the alignment itself, properties of a distance-based tree approximation, parameters estimated under a parameter-rich substitution model, and similarity metrics of sequences within taxon subsets. The RF regressor classifies 24 candidate nucleotide substitution models according to their predicted accuracy for branch length estimation.

ModelRevelator (Burgstaller-Muehlbacher et al., 2023) takes a different approach. It uses two independent deep neural networks to perform model selection without computing likelihoods or reconstructing trees. The first network, NNmodelfind, is a residual neural network that takes as input a set of summary statistics computed from pairwise alignments. It selects among six nucleotide substitution models varying in complexity from JC to GTR. The second network, NNalphafind, is a bidirectional LSTM that takes base composition profiles as input and predicts whether rate heterogeneity should be incorporated and, if so, estimates the α parameter of the Gamma distribution.

3.11.3 Direct Alignment Encoding

An alternative to summary statistics is to encode alignments directly as numerical matrices and feed them into neural networks with appropriate inductive biases. For four-taxon trees, convolutional neural networks (CNNs) have been applied to alignments encoded as integers or one-hot-encoded. Burgstaller-Muehlbacher et al. (2023) showed that CNNs analyzing images of nucleotide alignments can achieve performance comparable to ML-based model prediction.

Mo et al. (2024) note that Phyloformer (Nesterenko et al., 2022), which uses self-attention mechanisms, accommodates variable input sizes through careful network design. This approach offers a path for networks that can accept alignments of arbitrary length and taxon number while preserving the representational richness of direct alignment encoding.

3.11.4 Likelihood-Free Inference for Intractable Models

The most compelling application of ML to substitution models concerns models whose likelihoods are intractable. SCS models that incorporate site-dependent evolution cannot be incorporated into standard likelihood functions because current phylogenetic methods compute site likelihoods independently (Yang 2006). Simulation-based inference (SBI) offers a principled alternative: model parameters are inferred by matching observed summary statistics to statistics computed from simulations (Cranmer et al. 2020). Deep learning methods have substantially improved SBI through neural posterior estimation (NPE), which uses neural networks to learn the posterior distribution

directly from simulated data. These methods are directly applicable to substitution models for which a generative simulator exists but no tractable likelihood function does.

A related direction involves extending CTMC substitution models to capture richer substitution dynamics. Magee et al. (2024) introduced random-effects substitution models that extend standard CTMCs and developed scalable gradient approximations to enable Hamiltonian Monte Carlo sampling from the posterior under these richer models (Magee et al. 2024). Although primarily Bayesian rather than ML-based, this work illustrates the general direction toward models with greater capacity to capture heterogeneous substitution dynamics.

3.11.5 Training Data Design and Generalization

Supervised ML for substitution model selection requires labeled training data, which in phylogenetics means simulated alignments for which the true generative substitution model is known. The quality of training data determines the generalization capacity of the model. The training-versus-empirical mismatch is a central concern: training data generated under one substitution model may not generalize to empirical datasets that evolved under a different model (Y. K. Mo et al. 2024). Strategies to improve generalization include generating training data across a broad range of model parameters, standard regularization techniques (dropout, weight decay, ensemble methods), adding noise to alignments during training, and domain adaptation methods that promote the extraction of domain-invariant features (Y. Mo and Siepel 2024).

3.11.6 Limitations of ML-Based Model Selection

The most fundamental limitation of supervised ML in phylogenetics is the gap between simulated training data and empirical data. A traditional model selection criterion that selects a misspecified model will still compute a well-defined AIC or BIC score, making the error transparent in principle. An ML model trained on misspecified simulations may confidently output an incorrect recommendation without any signal of failure. Additionally, ML models for substitution model selection are generally black-box predictors that do not provide an explanation of why a particular model was selected. Leuchtenberger and von Haeseler (2024) showed that simplifying complex neural network classifiers into more interpretable forms can yield theoretical insights, suggesting interpretability is not inherently incompatible with ML approaches (Leuchtenberger and Haeseler 2024). Most ML models for model selection have been trained and evaluated on four-taxon trees or small alignments; the scalability to the hundreds or thousands of taxa routinely analyzed in phylogenomics has not been established.

3.12 Protein Neural Networks and Language Models

3.12.1 Protein Language Models

Large language models (LLMs) trained on protein sequences represent a fundamentally different approach to encoding evolutionary information. Protein language models (PLMs) such as ESM-2 and ProtTrans treat amino acid sequences as sentences in a molecular language. They learn statistical dependencies among amino acid positions from millions of sequences, implicitly capturing coevolutionary constraints, structural contacts, and evolutionary conservation (Valentini et al. 2023).

Unlike traditional substitution models, PLMs do not impose an explicit Markovian structure on the substitution process. Instead, they learn a high-dimensional representation of sequence space that reflects the actual distribution of natural protein sequences.

Lupo et al. (2022) showed that protein language models trained on multiple sequence alignments (MSAs) learn phylogenetic relationships. This finding implies that the attention mechanisms in these models encode information about evolutionary relationships. Mo et al. (2024) cite this result in the context of alternative strategies for representing alignments, noting that language models can produce "networks that can accept variable input sizes, and are capable of incorporating relationships among sites and lineages simultaneously."

3.12.2 Phyloformer and Attention Architectures

Phyloformer (Nesterenko et al., 2022) uses self-attention to infer pairwise evolutionary distances from aligned amino acid sequences. The model encodes sequence pairs through an attention mechanism that alternates between site-level attention (sharing information across sites within each pair) and pair-level attention (sharing information across sequence pairs within each site).

3.13 Conclusion

Substitution models occupy a central role in phylogenetic reconstruction from molecular data. Their theoretical foundation lies in the time-reversible general Markov process, of which all named models are derived as special cases. Implementation requires both efficient computation of tree likelihoods via pruning algorithms and robust model selection criteria that penalize overparameterization. Bayesian inference adds prior distributions over all model parameters, requiring careful justification of those priors and critical evaluation of the resulting posterior summaries.

The interpretive limitations are substantial. Posterior clade probabilities are subject to non-uniform clade prior bias that can produce strongly supported but erroneous topological conclusions. The Felsenstein zone of parsimony inconsistency motivates ML analysis, but ML itself is not immune to misspecification. The treatment of indels and the choice of substitution model both carry significant downstream consequences for inference.

Recent advances focus on more realistic and comprehensive probabilistic models: HMMs for sequence alignment, birth-death indel models, extensions to genomic rearrangement and phylogenetic networks, and applications of machine learning and language models for model selection and distance estimation. These developments expand the scope of model-based inference while simultaneously increasing computational demands. The fundamental tension between biological realism and computational tractability remains the defining challenge of the field.

Chapter Review Outline

1. Tavaré rate matrix (1986) Q: $Q_{ij} = R_{ij}\pi_j$ $(i \neq j)$; $P_{ij}(t)$ via eigenvalue decomposition of Q (Tavbré 1986; Yang 1994).
2. Model hierarchy: JC69 (0 free parameters) → K2P (transition/transversion) → HKY85 (unequal π) → GTR (6 rates, 4 freq.) (Swofford et al. 1996).
3. Rate variation: invariable sites (I) + discrete Gamma (α, mean = 1); GTR+I + Γ is standard empirical model.
4. JC69+Gaps (5 states): $P_{ij}(t) = \frac{1}{5} - \frac{1}{5}e^{-\mu t}$ $(i \neq j)$; indels as atomic events (Wheeler et al. 2006).
5. Birth-death models TKF91/92 (Thorne et al., 1991): couple insertion (λ) / deletion (μ) with substitution; mechanistically explicit but computationally expensive.
6. Felsenstein's pruning algorithm: $L_i(x) = \left(\sum_{x_j} p_{x_i,x_j}(t_j)L_j(x_j)\right) \times \left(\sum_{x_k} p_{x_i,x_k}(t_k)L_k(x_k)\right)$; assumes i.i.d. characters.
7. MCMC / MC^3 (Metropolis-coupled): cold chain converges to posterior; hot chains explore; exchange by $\alpha = \min(1, \pi_i(T_j)\pi_j(T_i)/\pi_i(T_i)\pi_j(T_j))$.
8. Model selection: LRT (χ^2), AIC ($-2\log l + 2p$), BIC ($-2\log l + p\log n$); Abadi et al. (2019): selection affects branch lengths more than topology.
9. Felsenstein zone: parsimony inconsistent when $p^2 \geq q(1-q)$ (long-branch attraction).
10. Clade prior bias: uniform prior over topologies ≠ uniform prior over clades; prior probability ratio between clades of sizes 2 and 25 can reach 10^{13} (Steel & Pickett, 2006).
11. Recent advances: HMMs (Viterbi, Forward-Backward, Baum-Welch), likelihood-based alignment (dominant vs. total), protein language models (ESM-2, Phyloformer).
12. Mixture models: CAT (PhyloBayes) models site-dependent substitution via Dirichlet process; SCS models use protein-folding free energy (Boltzmann distribution) and fixation probability (Moran process) but render standard ML intractable.
13. ML model selection: ModelTeller (random forest, 50+ summary statistics), ModelRevelator (deep neural networks, no likelihood computation); Phyloformer (self-attention for variable-size inputs).
14. Likelihood-free inference: simulation-based inference (SBI) and neural posterior estimation (NPE) enable evaluation of intractable SCS models; ABC for protein model selection (ProteinModelerABC).
15. Limitations of ML approaches: simulation-to-empirical gap, black-box predictions, scalability to large datasets; domain adaptation and interpretability methods as mitigation strategies.

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Chapter 4

Diversity of Data Types and Sequencing Strategies for Phylogenomic Reconstruction

In recent years, an explosion of sequencing methodologies and selective DNA enrichment has made it possible to generate phylogenomic data from practically any organism. However, the choice of the appropriate data type for a given project represents a crucial decision that affects the quality, cost, and applicability of the results. This chapter reviews a comprehensive range of sequencing strategies and data acquisition approaches, comparing their strengths and limitations in specific phylogenomic contexts.

4.1 Whole Genome Sequencing (Short Read)

Whole genome sequencing with short reads offers complete representation of genomic content, including coding, non-coding, regulatory, and repetitive regions. This approach is particularly valuable for deep and intermediate phylogenomic analyses.

4.1.1 General Characteristics

Category: Complete genome

Locus Type: All (coding, non-coding, regulatory, repetitive)

Typical Number of Loci: Entire genome; tens of thousands of extractable gene regions

Optimal Phylogenetic Scale: All scales; best for deep and intermediate

DNA Quality: High (high molecular weight DNA preferred for good assembly)

DNA Quantity: Moderate to high ($\geq$ 500 ng typical)

Tissue: Any tissue with sufficient DNA

Museum/Herbarium Specimen Compatibility: Poor (fragmented DNA resulting in poor assembly)

Relative Cost: High (U$50–200+ per sample at useful depth)

Bioinformatic Complexity: Very high (assembly, annotation, orthology pipelines)

Orthology Assessment Difficulty: Moderate to high (paralogy from whole genome duplications and copy number variation)

Matrix Completeness Tendency: High if genomes well assembled; variable otherwise

Cross-study Combinability: Moderate (depends on assembly quality and gene prediction consistency)

Broad Taxonomic Resource: Broad: prokaryotes, fungi, animals, plants; Earth BioGenome Project expanding rapidly

Suitability for Non-model Organisms: Excellent in principle; cost and assembly complexity limit adoption in high-N studies

4.1.2 Strengths and Limitations

The main advantages of this approach include complete genomic representation, possibility of synteny and gene family analyses, and archival value. Limitations include high cost at adequate depth for assembly, large genomes still being challenging, and substantial bioinformatic burden.

Delsuc et al. (2005), Jarvis et al. (2014), Lewin et al. (2018), and Lozano-Fernandez et al. (2022)

4.2 Whole Genome Sequencing (Long Read: PacBio HiFi / ONT)

Long read technologies enable assemblies with chromosome-level quality, resolving complex structural variants and repeat evolution.

4.2.1 General Characteristics

Category: Complete genome

Locus Type: All (including repetitive, structural variants)

Typical Number of Loci: Entire genome; chromosome-level assemblies achievable

Optimal Phylogenetic Scale: All scales; enables structural variation and repeat evolution analysis

DNA Quality: Very high (ultra-HMW DNA required, > 20 kb)

DNA Quantity: High (> 1 μg for PacBio HiFi; > 400 ng for ONT ligation)

Tissue: Fresh or liquid nitrogen frozen tissue strongly preferred

Museum/Herbarium Specimen Compatibility: Very poor (long reads require intact DNA)

Relative Cost: Very high (U$200–1000+ per sample)

Bioinformatic Complexity: Very high (assembly + polishing + annotation)

Orthology Assessment Difficulty: Moderate to high (similar to short WGS once assembled)

Matrix Completeness Tendency: Very high (chromosome-level assemblies reduce missing data)

Cross-study Combinability: Moderate (assembly quality improves combinability)

Broad Taxonomic Range: Growing rapidly: Vertebrate Genomes Project, Darwin Tree of Life, i5K

Suitability for Non-model Organisms: Excellent for quality but prohibitively expensive for most population-level studies

4.2.2 Strengths and Limitations

Chromosome-level assemblies resolve repeats, structural variants, and complex rearrangements with superior representation of genomic architecture. The main limitations are higher cost, more demanding tissue requirements, and impracticality of museum/herbarium material.

Formenti et al. (2022), Guiglielmoni et al. (2024), and Rhie et al. (2021)

4.3 Transcriptomics (RNA-seq)

RNA-seq captures exons of expressed protein-coding genes, offering efficient access to hundreds or thousands of nuclear single-copy loci. However, RNA quality and tissue requirements limit specimen sources.

4.3.1 General Characteristics

Category: Reduced representation (expression-dependent)

Locus Type: Coding exons (expressed genes only)

Typical Number of Loci: 1,000–20,000+ genes depending on tissue/condition

Optimal Phylogenetic Scale: Intermediate to deep (>50 Mya); limited at shallow scales

RNA Quality: High quality required (RIN $\geq$7 preferred)

RNA Quantity: Moderate ($\geq$ 100 ng total RNA)

Tissue: Fresh or RNAlater-preserved tissue; tissue type affects gene recovery

Museum/Herbarium Specimen Compatibility: Very poor (RNA degrades rapidly)

Relative Cost: Moderate (U$30–100 per sample)

Bioinformatic Complexity: High (de novo assembly, isoform resolution, orthology inference)

Orthology Assessment Difficulty: High (paralogy, alternative splicing, tissue-specific expression)

Matrix Completeness Tendency: Variable; depends on tissue overlap between samples

Cross-study Combinability: Low to moderate (tissue must match between studies)

Broad Taxonomic Range: Broad: 1KP (plants), i5K (insects), many vertebrate datasets

Suitability for Non-model Organisms: Excellent for gene discovery; limited by RNA preservation requirements

4.3.2 Strengths and Limitations

The approach recovers large number of coding loci, does not require probe design, and possesses integrated functional annotation. Limitations include strict RNA/tissue quality requirements, creation of missing data by tissue-specific expression, and orthology complicated by paralogy and alternative splicing.

Dunn et al. (2008), Figuet et al. (2014), Initiative (2019), and Misof et al. (2014)

4.4 Ultraconserved Elements (UCEs)

UCEs combine conserved cores (predominantly non-coding) with variable flanking regions, offering excellent cross-study combinability.

4.4.1 General Characteristics

Category: Target capture (probe-based enrichment)

Locus Type: Mixed: conserved core (mostly non-coding) + variable flanking regions

Typical Number of Loci: 500–5,000+ loci depending on probe set and divergence

Optimal Phylogenetic Scale: Broad: deep to moderately shallow ($\sim$ 5–500+ Mya)

DNA Quality: Moderate (tolerates moderate degradation)

DNA Quantity: Low to moderate (as low as 10–50 ng with optimized protocols)

Tissue: Any tissue; dry specimens, ethanol-preserved, silica-gel-dried work

Museum/Herbarium Specimen Compatibility: Good (one of the best approaches for museum/herbarium specimens with degraded DNA)

Relative Cost: Low to moderate (U$15–50 per sample once probes purchased)

Bioinformatic Complexity: Moderate (PHYLUCE pipeline well established)

Orthology Assessment Difficulty: Low (UCEs generally single-copy and highly conserved)

Matrix Completeness Tendency: High (standardized probe sets recover consistent loci across taxa)

Cross-study Combinability: Very high (same probe set $\rightarrow$ directly combinable)

Broad Taxonomic Range: Vertebrates, arthropods, mollusks, nematodes, cnidarians, annelids; expanding to many animal clades

Suitability for Non-model Organisms: Excellent: works with degraded DNA, low input, broad probe sets available

4.4.2 Strengths and Limitations

Cross-study combinability is excellent, works with museum/degraded DNA, cost per sample is low after probe purchase, and bioinformatic pipeline well tested. Limitations include short alignments per locus, predominantly non-coding (limits functional interpretation), and probe sets must be developed per major clade.

Branstetter et al. (2017), Faircloth et al. (2012), McCormack et al. (2013), and Zhang et al. (2019)

4.5 Anchored Hybrid Enrichment (AHE)

AHE combines conserved anchor regions (often exonic) with variable flankers (intronic/intergenic), balancing conserved and variable regions.

4.5.1 General Characteristics

Category: Target capture (probe-based enrichment)

Locus Type: Mixed: conserved anchor regions + variable flankers

Typical Number of Loci: 300–600+ loci (vertebrate kits); expandable with custom designs

Optimal Phylogenetic Scale: Broad: deep to shallow ($\sim$ 1–500+ Mya)

DNA Quality: Moderate (tolerates moderate degradation)

DNA Quantity: Low to moderate ($\sim$ 50–250 ng)

Tissue: Any tissue with extractable DNA

Museum/Herbarium Specimen Compatibility: Good (degraded DNA compatible with optimized protocols)

Relative Cost: Moderate (U$50–100 per sample through FSU center; includes library prep)

Bioinformatic Complexity: Moderate (pipeline established through Center for Anchored Phylogenomics)

Orthology Assessment Difficulty: Low to moderate (anchor loci selected for single-copy status)

Matrix Completeness Tendency: High (standardized enrichment $\rightarrow$ consistent recovery)

Cross-study Combinability: High (consistent probe sets within clades)

Broad Taxonomic Range: Vertebrates well developed; arthropods, plants, fungi; kits for >dozen clades

Suitability for Non-model Organisms: Excellent: broad probe design philosophy

4.5.2 Strengths and Limitations

Loci longer than UCEs on average result in more signal per locus. Good balance of conserved and variable regions. Commercial pipeline available. Limitations include centralization at FSU, proprietary probe designs, and cost per sample higher than UCE when self-processed.

Breinholt et al. (2018), Hime et al. (2021), Lemmon et al. (2012), and Prum et al. (2015)

4.6 Angiosperms353 / Universal Probe Sets

This universal set of 353 target genes was developed specifically for application across the diversity of angiosperms, with excellent herbarium specimen compatibility.

4.6.1 General Characteristics

Category: Target capture (universal probe set)

Locus Type: Protein-coding nuclear genes (low copy)

Typical Number of Loci: 353 target genes (+ recoverable flanking regions)

Optimal Phylogenetic Scale: Broad: ordinal to population level in angiosperms

DNA Quality: Moderate (herbarium specimens tested successfully)

DNA Quantity: Low to moderate (as low as $\sim$ 10 ng with degraded DNA protocols)

Tissue: Any tissue; silica-dried leaves, herbarium leaves work well

Museum/Herbarium Specimen Compatibility: Excellent (specifically tested and validated with herbarium material >100 years old)

Relative Cost: Low to moderate (U$15–40 per sample)

Bioinformatic Complexity: Moderate (HybPiper pipeline well documented and widely adopted)

Orthology Assessment Difficulty: Moderate (low-copy genes selected; WGD in many angiosperm lineages complicates paralogy detection)

Matrix Completeness Tendency: High (universal design ensures broad recovery; $\sim$ 200–350 genes typically recovered)

Cross-study Combinability: Very high (universal set designed for cross-study combinability)

Broad Taxonomic Range: Angiosperms (flowering plants) universally; also tested in gymnosperms and some bryophytes

Suitability for Non-model Organisms: Excellent for plants: no custom probe design necessary, ready-to-use kit

4.6.2 Strengths and Limitations

Universal across all flowering plants, no custom probe design necessary, excellent herbarium compatibility, and botanical community standard. Limitations include restriction to angiosperms, 353 loci may be insufficient for shallow-scale resolution in some groups, and paralogy in polyploid lineages.

Baker et al. (2021, 2022), Johnson et al. (2019), and Zuntini et al. (2024)

4.7 Custom Exon Capture / Lineage-Specific

Allows optimized probe design for specific clades, capturing chosen coding exons.

4.7.1 General Characteristics

Category: Target capture (probe-based enrichment)

Locus Type: Coding exons (selected by clade)

Typical Number of Loci: 200–10,000+ exons depending on design

Optimal Phylogenetic Scale: Flexible: designed for specific clade and scale

DNA Quality: Moderate (degraded DNA tolerated)

DNA Quantity: Low to moderate

Tissue: Any tissue with extractable DNA

Museum/Herbarium Specimen Compatibility: Good (similar to UCE/AHE for degraded DNA)

Relative Cost: Moderate to high (probe design + synthesis adds upfront cost)

Bioinformatic Complexity: Moderate to high (requires custom bioinformatic pipeline)

Orthology Assessment Difficulty: Moderate (depends on gene selection criteria)

Matrix Completeness Tendency: Moderate to high (depends on probe-target divergence)

Cross-study Combinability: Low (different studies target different genes)

Broad Taxonomic Range: Lineage-specific by definition; available for many vertebrate, insect, plant clades

Suitability for Non-model Organisms: Good (can be designed for any clade with some genomic resources)

4.7.2 Strengths and Limitations

Customization for clade of interest optimizes informativeness. Limitations include lack of standardization between studies (poor combinability), requirement for genomic/transcriptomic data for probe design, and upfront cost of probe synthesis.

Bi et al. (2012), Bragg et al. (2016), and Portik et al. (2016)

4.8 RADseq / ddRADseq / GBS

Restriction enzyme-based sequencing captures random genomic fragments, ideal for population-scale studies at shallow divergence.

4.8.1 General Characteristics

Category: Reduced representation (restriction enzyme-based)

Locus Type: Anonymous genomic fragments (random with respect to function)

Typical Number of Loci: Thousands–hundreds of thousands of loci containing SNPs

Optimal Phylogenetic Scale: Shallow: population to genus level (<10–20 Mya)

DNA Quality: Moderate to high (fresh DNA preferred for consistent digestion)

DNA Quantity: Moderate (100–500 ng typical)

Tissue: Fresh or well-preserved tissue preferred

Museum/Herbarium Specimen Compatibility: Poor (requires high-quality DNA for consistent digestion)

Relative Cost: Low (U$10–30 per sample; no probe purchase necessary)

Bioinformatic Complexity: Moderate (established pipelines: Stacks, ipyrad, dDocent)

Orthology Assessment Difficulty: Low at shallow scales (anonymous loci treated as independent)

Matrix Completeness Tendency: Low to moderate (locus dropout increases with divergence)

Cross-study Combinability: Very low (restriction sites not conserved between divergent taxa)

Broad Taxonomic Range: Universal in principle; widely used in population genomics of fish, birds, amphibians, insects, plants

Suitability for Non-model Organisms: Excellent for population-level studies with fresh tissue; poor for deep phylogenetics

4.8.2 Strengths and Limitations

Lower cost per sample, no probe design, massive SNP recovery, excellent for population genomics and phylogeography. Limitations include severe locus dropout with divergence, poor reproducibility between studies, and very poor with degraded/museum DNA.

Baird et al. (2008), Eaton and Overcast (2020), Huang and Knowles (2016), and Peterson et al. (2012)

4.9 Genome Skimming (Low-coverage WGS)

Low-coverage sequencing captures high-copy genomes: organellar (plastomes, mitogenomes) and nuclear ribosomal DNA. Deep genome skimming (DGS) can recover hundreds of nuclear genes.

4.9.1 General Characteristics

Category: Complete genome (low coverage)

Locus Type: High copy: organellar genomes (plastomes, mitogenomes) + nuclear ribosomal DNA

Typical Number of Loci: Complete organellar genomes + nrDNA operon; deep DGS can recover hundreds of nuclear genes

Optimal Phylogenetic Scale: Intermediate to deep for organellar genomes

DNA Quality: Low to moderate (works with degraded DNA for organellar recovery)

DNA Quantity: Low (even <1 ng can yield organellar data)

Tissue: Any tissue; herbarium, museum, even subfossil material

Museum/Herbarium Specimen Compatibility: Excellent (one of the best approaches for degraded specimens)

Relative Cost: Very low (U$5–20 per sample at 1–5× coverage)

Bioinformatic Complexity: Low to moderate (assembly of high-copy elements relatively straightforward)

Orthology Assessment Difficulty: Low for organellar genomes (uniparental inheritance, no paralogy)

Matrix Completeness Tendency: Very high for organellar genomes; variable for nuclear genes

Cross-study Combinability: High for organellar genomes; moderate for nuclear

Broad Taxonomic Range: Universal: any eukaryote; plants especially well served (plastomes)

Suitability for Non-model Organisms: Excellent: lower cost and tissue requirements; ideal for herbarium/museum specimens

4.9.2 Strengths and Limitations

Lower cost, lower DNA requirements, excellent for degraded specimens, organellar genomes easily assembled, DGS approach can recover nuclear genes. Limitations include organellar genomes representing single non-recombinant locus (limited independent signal), and most of nuclear genome not recovered at low coverage.

Dodsworth (2015), Straub et al. (2012), and Weitemier et al. (2014)

4.10 Organellar Genome Phylogenomics (Plastomics / Mitogenomics)

Analysis of complete organellar genomes from any sequencing strategy, including protein-coding genes, rRNA, tRNA, and intergenic spacers.

4.10.1 General Characteristics

Category: Organellar genome (from any sequencing strategy)

Locus Type: Coding genes + rRNA + tRNA + intergenic spacers from organellar genomes

Typical Number of Loci: $\sim$ 80 coding genes (plastome); $\sim$ 13 (animal mitogenome); $\sim$ 40+ (plant mitogenome)

Optimal Phylogenetic Scale: Deep to moderately shallow for plastomes; shallow for animal mitogenomes

DNA Quality: Low (organellar DNA high copy; recoverable from degraded samples)

DNA Quantity: Very low

Tissue: Any tissue

Museum/Herbarium Specimen Compatibility: Excellent

Relative Cost: Very low (often byproduct of other sequencing)

Bioinformatic Complexity: Low (assembly and annotation tools well established: GetOrganelle, NOVOPlasty, MITOS)

Orthology Assessment Difficulty: Very low (no paralogy concern in typical cases)

Matrix Completeness Tendency: Very high (complete organellar genomes standard)

Cross-study Combinability: Very high (organellar genomes universally comparable)

Broad Taxonomic Range: Universal across eukaryotes; thousands of plastomes and mitogenomes in GenBank

Suitability for Non-model Organisms: Excellent: works for any eukaryote; minimal requirements

4.10.2 Strengths and Limitations

Easier to assemble and annotate, massive existing datasets for comparison, excellent for barcoding and specimen identification. Limitations include single non-recombinant locus (single independent gene tree), maternal inheritance does not capture complete species history, cytoplasmic-nuclear discordance (hybridization, introgression).

Jansen et al. (2007), Li et al. (2021), Smith (2015), and Twyford and Ness (2017)

4.11 BUSCO Gene-Based Phylogenomics

Post hoc extraction of single-copy orthologous genes from assembled genomes or transcriptomes.

4.11.1 General Characteristics

Category: Post hoc extraction (from genomes or transcriptomes)

Locus Type: Coding genes (single-copy orthologs)

Typical Number of Loci: 300–5,000+ depending on lineage-specific BUSCO set

Optimal Phylogenetic Scale: Broad: effective from shallow to deep scales

DNA Quality: N/A (depends on source data: genome or transcriptome quality)

DNA Quantity: N/A (depends on source sequencing strategy)

Tissue: N/A (depends on source data)

Museum/Herbarium Specimen Compatibility: N/A (depends if genome/transcriptome can be generated)

Relative Cost: No additional cost (extracted from existing assemblies)

Bioinformatic Complexity: Low to moderate (automated BUSCO pipeline; downstream tree inference standard)

Orthology Assessment Difficulty: Moderate (BUSCO genes not always strictly single-copy, especially post-WGD)

Matrix Completeness Tendency: Moderate (depends on genome/transcriptome completeness)

Cross-study Combinability: High (BUSCO sets standardized per lineage)

Broad Taxonomic Range: Universal: BUSCO sets available for 193 lineages (eukaryotes, bacteria, archaea, viruses)

Suitability for Non-model Organisms: Excellent if genome/transcriptome data exist

4.11.2 Strengths and Limitations

No additional sequencing cost, standardized ortholog sets, automated pipeline, applicable to any existing genome/transcriptome. Limitations include dependence on input assembly quality, BUSCO genes not always single-copy, functional bias toward housekeeping genes.

Manni et al. (2021), Sahbou et al. (2022), Simão et al. (2015), and Waterhouse et al. (2018)

4.12 Assembly-Free / k-mer Methods (AAF, CASTER, Read2Tree)

Approaches that avoid traditional assembly, using direct k-mer comparisons or mapping relative to reference sequences.

4.12.1 General Characteristics

Category: Alignment-free or read-based

Locus Type: Complete genome (all sequence content used as k-mer spectra) or read mapping to reference

Typical Number of Loci: N/A (distance-based or relative mapping; not locus-based in traditional sense)

Optimal Phylogenetic Scale: Emerging: demonstrated at intermediate to deep scales

DNA Quality: Variable (works with raw reads; tolerates moderate quality)

DNA Quantity: Variable (benefits from moderate coverage; >5× useful)

Tissue: Any tissue yielding sequenceable DNA

Museum/Herbarium Specimen Compatibility: Potentially good (works from raw reads without assembly)

Relative Cost: No additional cost if sequencing data already exist

Bioinformatic Complexity: Low (avoids assembly, alignment, and orthology steps)

Orthology Assessment Difficulty: N/A (orthology implicit in k-mer overlap or mapping)

Matrix Completeness Tendency: N/A (not matrix-based in traditional sense; completeness = coverage)

Cross-study Combinability: Moderate (requires consistent sequencing strategy)

Broad Taxonomic Range: Universal in principle; CASTER tested across eukaryotes

Suitability for Non-model Organisms: Excellent in principle: minimal computational expertise; no reference genome necessary

4.12.2 Strengths and Limitations

Faster and simpler pipeline: raw reads → tree. Avoids assembly, alignment, and orthology inference. Limitations include limited model-based analysis (distances, not site patterns in most implementations), and some methods are recent and less well tested.

Balaban et al. (2025), Dylus et al. (2024), and Fan et al. (2015)

4.13 Comparative Considerations

The selection among these data types involves complex cost-benefit considerations. For population-scale phylogeographic studies, RADseq offers best value for cost. For deep phylogenomic studies in non-model organisms with large genomes, RNA-seq offers better balance of cost and informativeness. For studies requiring robust cross-study combination across multiple groups, UCEs or Angiosperms353 offer maximum standardization. For obtaining high-quality reference genomic assemblies, long-read sequencing with combined strategies is necessary.

Each data type captures different phylogenomic signals and resolution at different nodes in species trees. An increasingly adopted approach involves multi-strategy data approach, in which different data types provide complementary context for robust phylogenetic reconstruction.

Chapter Review Outline

1. Short-read WGS: complete genome, ≥500 ng HMW DNA, U$50–200+/sample; poor for museum; bioinformatics very high.
2. Long-read WGS (PacBio HiFi / ONT): chromosome-level assembly, ultra-HMW DNA >20 kb, U$200–1 000+; fresh tissue mandatory.
3. RNA-seq: expressed exons (1 000–20 000 genes), RIN≥7, U$30–100; tissue-specific variation creates missing data; paralogy from WGD and splicing.
4. UCEs: non-coding conserved cores + variable flankers; 500–5 000 loci; degraded DNA tolerated; PHYLUCE pipeline; very high combinability.
5. AHE (Anchored Hybrid Enrichment): exonic anchors + intronic flankers; 300–600+ loci; FSU center; loci longer than UCEs.
6. Angiosperms353: 353 universal genes for angiosperms; validated with herbarium >100 years; HybPiper pipeline; botanical standard.
7. RADseq/ddRADseq/GBS: restriction enzymes, thousands of SNPs; U$10–30/sample; ideal <10–20 Mya; severe dropout with divergence; very low combinability.
8. Genome skimming: low coverage (1–5×), U$5–20; recovers organellar + nrDNA; deep DGS → hundreds of nuclear genes; excellent for degraded.
9. Organellar phylogenomics: plastome (∼80 genes), animal mitogenome (∼13 genes); single non-recombinant locus; cytoplasmic-nuclear discordance.
10. BUSCO: post hoc extraction of single-copy orthologs; 300–5 000+ genes; 193 sets per lineage; no additional sequencing cost.
11. k-mer methods (AAF, CASTER, Read2Tree): raw reads → tree without assembly/alignment; emerging but less well tested.

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Chapter 5

RNA-seq in Phylogenomic Studies

5.1 Introduction and Motivations

RNA sequencing (RNA-seq) has emerged as a powerful and widely accessible approach for collecting phylogenomic data in diverse organisms, particularly those lacking reference genomes or established genomic resources. The application of RNA-seq in phylogenomic studies offers multiple important advantages: efficient access to hundreds or thousands of single-copy nuclear coding loci, simultaneous generation of gene expression and gene duplication data, and the ability to leverage large repositories of publicly available transcriptomes generated for other purposes. These properties make RNA-seq especially valuable for non-model organisms and for groups with large genomes that would be impractical to sequence in their entirety.

However, RNA-seq presents a distinct set of technical and analytical challenges that researchers must navigate. This chapter reviews the motivations for using RNA-seq in phylogenomics, characterizes the major technical and analytical challenges, and describes strategies that researchers have developed to overcome them.

5.2 Motivations for RNA-seq in Phylogenomics

5.2.1 Access to Multiple Coding Loci

RNA-seq provides comprehensive access to expressed genes, potentially capturing thousands of single-copy coding loci in a single reaction. For non-model organisms, this represents a considerable advantage over methods requiring prior probe design or knowledge of reference genomes. The amount of phylogenomic information available in a single transcriptome library can rival or exceed that obtained through targeted capture methods following investment in custom probe design.

5.2.2 Cost Economics in Non-Model Organisms

For taxa lacking well-annotated reference genomes, the cost of developing custom probe sets for targeted capture approaches can be prohibitive. In contrast, RNA-seq offers a lower cost entry point. A single transcriptome library with moderate coverage (for example, 50 million reads) can cost $30–100 for preparation and sequencing, comparable to the cost of a targeted capture sample without requiring the initial investment in probe design.

5.2.3 Integrated Expression Data

A distinctive feature of RNA-seq is that transcript abundance measurements (mapped read counts) are inherently captured during sequencing. This enables researchers to simultaneously extract (1) sequence data for phylogenetic inference and (2) gene expression data that characterize expression patterns across different tissues or developmental stages. This duality of data makes collection more efficient compared to sequencing methods yielding only sequence.

5.2.4 Access to Public Repositories

The NCBI Sequence Read Archive (SRA) and similar repositories host tens of millions of publicly available transcriptome libraries, frequently submitted for non-phylogenomic purposes (for example, gene expression studies, draft genome characterization). For phylogenomic researchers, this offers an opportunity to reuse existing data—enabling retrospective phylogenomic studies combining transcriptome data from multiple laboratories and projects. This data reuse accelerates construction of taxonomic phylogenies and reduces overall project costs.

5.3 Technical Challenges: Tissue Preservation and RNA Quality

5.3.1 Rapid RNA Degradation

RNA is chemically unstable compared to DNA, subject to degradation through multiple mechanisms. Ubiquitous RNases (RNA-cleaving enzymes) are especially prevalent in natural environments, contaminating tools, water, and air. Once tissue is collected, RNA degradation begins immediately through spontaneous hydrolysis and endogenous nuclease activity, unless immediate preservation measures are initiated.

RNA quality is commonly measured through the RNA Integrity Number (RIN), which ranges from 10 (completely intact RNA) to 1 (degraded RNA). For phylogenomic applications, RIN values ≥ 7 are generally considered acceptable, although higher RIN (8–10) is preferred. Significant RNA degradation results in loss of complete full-length transcripts and introduction of alignment artifacts that complicate subsequent orthology.

5.3.2 Tissue Preservation Methods

Several preservation strategies have been developed to maintain RNA integrity:

- **Silica-gel**: Fresh tissue is immediately placed in blue silica-gel desiccant, which absorbs moisture and dehydrates tissue rapidly. This approach is low-cost, portable, and effective for field use. RNA extracted from samples preserved in silica-gel often exhibits reasonable RIN (7–9), although some tissues are more prone to degradation.

- **RNAlater**: A chemical preservation solution that stabilizes RNA through a proprietary mixture of components that inhibit nucleases. Tissue is immersed in RNAlater, which penetrates tissue within 24 hours and protects RNA for extended periods (months to years) at room temperature. RNAlater typically preserves higher quality RNA than silica-gel, but is more expensive and less portable in field settings.

- **Liquid Nitrogen**: Flash-freezing fresh tissue in liquid nitrogen offers superior preservation if storage at -80°C is maintained. This approach is best for research environments with freezing infrastructure, but less practical for field collection.
- **Desiccated Drying**: Drying under low humidity in buffer solution preserves RNA reasonably well and can be implemented in field settings with minimal equipment.

5.3.3 Tissue-Specific Variation

Transcriptomic composition varies dramatically between tissues. A muscle transcriptome is dominated by transcripts related to muscle contraction, while a brain transcriptome is enriched in genes related to neuroactivity. For phylogenomics, this creates a potential source of missing data: a gene may be absent from a transcriptome of a specific tissue simply because it is not being highly expressed in that context.

Researchers frequently mitigate this variation through combining multiple tissues (for example, combining data from muscle, brain, and digestive glands) to improve gene coverage and matrix completeness. Alternatively, standardizing on a single tissue (for example, always using muscle tissue) sacrifices completeness in favor of comparability across studies.

5.4 Analytical Challenges: Orthology Determination and Paralog Contamination

5.4.1 Increased Orthology Complexity

Orthology determination—the identification of sequences truly homologous derived by speciation and not by gene duplication—is complicated in RNA-seq data for multiple reasons:

- **Whole-Genome Duplications (WGD)**: Many organisms have undergone whole-genome duplications in their evolutionary histories. For example, vertebrate lineages experienced two rounds of WGD near the base of vertebrate radiation. These duplications amplify every gene in the genome, potentially creating hundreds of pairs of paralogous genes. Distinguishing true orthologs from paralogs resulting from WGD requires careful filtering and synteny analysis.
- **Intraspecific Structural Variation**: Some genes may be present in single copy in some individuals of a population but duplicated in others, due to polymorphic structural variations. This introduces heterogeneity in orthology determination.
- **Alternative Splicing**: Many genes produce multiple isoforms through alternative splicing. Different isoforms can have very different exonic structure, complicating alignment and orthology. It is often unclear which isoform represents the most evolutionarily informative.

5.4.2 Orthology Pipelines

To overcome these challenges, researchers have developed sophisticated automated pipelines:

OrthoMCL groups homologous sequences through reciprocal BLAST search, best-reciprocal-hit filtering, and Markov clustering. For RNA-seq data, OrthoMCL can be combined with additional filters removing truncated sequences, low-expression isoforms, or apparent copies based on sequence similarity within predefined thresholds.

InParanoid identifies ortholog pairs between two genomes while placing secondary paralogs in context. InParanoid is particularly useful for multi-species RNA-seq data where multiple copies of a gene may be present in one or more species.

HybPiper was originally developed for hybridized targeted capture data but adapts well to RNA-seq data, assembling novel transcripts from reads, identifying exons and introns, and performing alignment to known reference sequences. This pipeline reduces isoform artifacts through careful selection of reference sequences and similarity thresholds.

5.4.3 Paralog Contamination and Contamination Capture

A particular challenge is "paralog contamination capture"—the situation in which an orthology pipeline inadvertently includes a paralogous gene in a phylogenomic data matrix. This can occur when:

- A species has a duplicated copy and the duplicated copy is accidentally selected as "ortholog" for use in matrix
- A contaminant sequence from another species is accidentally assigned as ortholog
- Alternative isoforms of the same gene are treated as separate genes

Captured paralogs can introduce misleading systematic signal in phylogenomic matrices, resulting in incorrect tree topologies. Rigorous verification through reverse BLAST, individual gene tree analysis, and orthology tests can mitigate this risk.

5.5 Analytical Challenges: Gene Tree Discordance and Alignment

5.5.1 Incomplete Lineage Sorting (ILS)

ILS—the retention of ancestral polymorphism—occurs when multiple alleles of a gene segregate in an ancestral population for a period sufficient that they can become fixed in different descendants. This results in gene histories that differ from species history. In RNA-seq data with multiple genes, most genes will likely exhibit some degree of discordance due to ILS, especially in rapid radiations and lineages with large effective population size.

5.5.2 Introgression

Historical hybridization followed by backcrossing introduces DNA segments from one species into the genome of a related species. If a gene was introgressed, its gene tree will reflect the introgression history rather than the speciation history. This can result in highly discordant gene topology.

5.5.3 Gene Duplication and Loss

Some genes undergo duplication in one lineage followed by loss in the sister lineage. If only one of the co-orthologs is analyzed, the resulting gene tree may reflect the duplication-loss history rather than the speciation history.

5.5.4 Alignment Quality and Masking

Sequence alignments are the foundation of all phylogenetic inference. Low-quality alignments introduce systematic noise that can confound phylogenomic signals. For RNA-seq data specifically, several sources of low-quality alignment are common:

- **Low-Complexity Regions**: Low-complexity sequences (for example, homopolymers, dinucleotide repeats) often align ambiguously and should be masked before phylogenetic analysis.
- **Rapidly Evolving Sequences**: Rapidly evolving sequences (for example, immunological protein surface regions) can have alignment positions with low confidence, particularly at deep phylogenomic distances.
- **Alternative Splicing Artifacts**: Different isoforms can introduce gaps and misalignments if alternative splice sites are not adequately resolved.

Masking strategies can remove problematic sites while retaining informative sites. Methods include ALISCORE (identifies ambiguously aligned sites), ZORRO (assigns confidence weights to alignments), and manual filtering based on sequence complexity.

5.6 Strategies for Overcoming Challenges

5.6.1 Automated Processing Pipelines

Several automated processing pipelines have been developed to streamline the trajectory from raw RNA-seq data to phylogenomic matrices:

TranscriptomeTools: A suite of scripts that coordinates de novo transcript assembly, reference alignment, ortholog extraction, and multiple alignment.

PHYLONOME: A web pipeline that accepts raw FASTQ data, performs de novo assembly and orthology, returning multiple alignment of orthologs.

SequenceMatrix: A program with graphical interface for sequence data management, enabling rapid combination of multiple gene regions with automatic partitioning strategies for model testing and phylogenetic reconstruction.

5.6.2 Multi-Method Analysis of Discordance

Given the multiple sources of gene tree discordance, an increasingly adopted approach involves multi-method analysis that treats discordance explicitly:

- **Concatenated Analysis**: Methods that concatenate multiple genes into a supermatrix and perform direct phylogenetic reconstruction (for example, maximum likelihood, Bayesian inference) on the concatenated matrix. These approaches often employ phylogenomic models that allow rate heterogeneity among gene partitions.
- **Individual Gene Tree Analysis**: Methods that reconstruct individual gene tree for each gene then synthesize gene trees into a final species tree. Examples include ASTRAL (from "Accurate Species TRee ALgorithm"), which resolves discordance through tree search that maximizes the number of consistent trios across gene trees.

- **Explicit Models**: Evolutionary models that explicitly model processes such as ILS, introgression, and duplication-loss can be fit to gene tree data to estimate demographic and historical parameters.

5.6.3 Enrichment of Targeted Capture Derived from RNA-seq

A complementary strategy involves using initial RNA-seq data to inform design of custom probes for subsequent targeted capture. For example, researchers could:

1. Sequence transcriptomes of representatives of the taxa in question
2. Identify conserved genes found consistently across taxa that would be good capture targets
3. Design probes based on these transcriptome sequences
4. Use probes to enrich DNA libraries (including degraded DNA from museum specimens) of additional taxa

This approach combines advantages of RNA-seq (low initial cost, access to multiple genes without reference genome knowledge) with advantages of targeted capture (compatibility with degraded DNA, low per-sample cost after probe design).

5.7 Empirical Applications and Case Studies

5.7.1 Invertebrates

RNA-seq was particularly impactful in invertebrate phylogenomics, where reference genomes were historically scarce. Misof et al. (2014) utilized transcriptome data from >100 insect species to produce the first robustly resolved insect phylogenome, establishing relationships among all major groups of modern insects.

5.7.2 Plants

For plants, the One Thousand Plant Transcriptomes (1KP) initiative generated transcriptomes from 1000 plant species comprising the full diversity of major plant groups. Initiative (2019) demonstrated that 1KP data could be combined for X-ray analysis of plant evolution across deep timescales.

5.7.3 Quality Considerations in Retrospective Data

Analysis of datasets of public transcriptomes revealed substantial variation in quality across studies. Researchers reusing transcriptome data should carefully evaluate RIN quality, sequencing depth, and processing methods from previous studies before inclusion in phylogenomic analyses.

5.8 Future Directions and Perspectives

5.8.1 Long-Read RNA Sequencing

Long-read RNA sequencing (PacBio Iso-Seq, Oxford Nanopore direct RNA sequencing) offers the capability to sequence full-length transcripts without need for de novo assembly. This can resolve some alternative splicing ambiguities and improve orthology confidence in the future.

5.8.2 Single-Cell Transcriptomics

Single-cell RNA sequencing (scRNA-seq) opens entirely new avenues for phylogenomics, potentially allowing phylogenomic characters to be assigned to specific cell types and biological functions. However, substantial analytical challenges related to gene dropout and capture variability must be overcome.

5.8.3 Integration with Other Data Types

An increasing trend is integration of multiple data types (sequence, expression, phenotype) into unified phylogenetic modeling. RNA-seq provides a natural entry point for such integration due to the availability of integrated expression data.

Chapter Review Outline

1. RNA-seq accesses hundreds–thousands of nuclear coding loci without probes; cost \$30–100/sample; integrated expression data; reusable public repositories (SRA).
2. RNA quality: RIN≥7 required; degradation by RNases is rapid. Preservation: silica-gel (field, cheap), RNAlater (penetrates in 24h, months at room temp.), liquid nitrogen (-80°C, superior), desiccated drying.
3. Tissue-specific variation: transcriptomes differ drastically between tissues; combining multiple tissues improves coverage but complicates comparability.
4. Orthology complicated by: whole-genome duplications (WGD), intraspecific structural variation, alternative splicing (multiple isoforms).
5. Orthology pipelines: OrthoMCL (reciprocal BLAST + Markov clustering), InParanoid (ortholog pairs between genomes), HybPiper (de novo assembly + reference alignment).
6. Paralog contamination: inadvertent inclusion of paralog in matrix → misleading signal; mitigate with reverse BLAST, individual gene trees, orthology tests.
7. Gene tree discordance: ILS (ancestral polymorphism retention), introgression (historical hybridization), duplication–loss.
8. Alignment quality: mask low-complexity regions, rapidly evolving sequences and alternative splicing artifacts; tools: ALISCORE, ZORRO.
9. Multi-method approach: concatenation (supermatrix) vs. coalescent species tree (ASTRAL) vs. explicit models (ILS, introgression).
10. RNA-seq → custom probe design: transcriptomes inform subsequent targeted capture for degraded museum DNA.
11. Future directions: Iso-Seq / direct RNA (full-length transcripts), scRNA-seq (single cell), multi-data integration.

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Chapter 6

Repetitive Elements, Transposable Elements, and GC Content Bias

6.1 Introduction

Repetitive sequences constitute a substantial and frequently underestimated fraction of eukaryotic genomes. In the human genome, approximately 50% of the sequence is composed of repetitive elements (Koning et al. 2011; Lander et al. 2001), with considerably higher proportions in many plant and insect organisms. These elements—including transposable elements, satellite DNA, and segmental duplications—present unique challenges and opportunities for phylogenomic studies. This chapter synthesizes definitions, evolutionary origins, obstacles posed for genomic analyses, and sequencing and bioinformatics strategies for dealing with them.

The recognition that eukaryotic genomes are dominated by repetitive sequences dates to the reassociation kinetics experiments of Britten and Kohne (1968), who demonstrated that a large fraction of animal DNA consists of reiterated sequences. Since then, whole-genome sequencing has confirmed that repetitive elements account for substantial proportions of genome size across the tree of life.

6.2 Definitions of Key Genomic Concepts

6.2.1 Repetitive Elements

Repetitive elements (REs) are nucleotide sequences that occur in multiple copies throughout a genome. In eukaryotic organisms, estimates suggest that approximately 50% of the human genome is composed of repetitive sequences (Haubold and Wiehe 2006). At the broadest level, REs can be grouped by their dispersal patterns into two classes: tandem repeats, whose units are organized head-to-tail adjacent to one another, and interspersed repeats, whose units are distributed at different genomic locations.

Tandem repeats include satellite DNA, minisatellites, and microsatellites (also called short tandem repeats, or STRs). Satellite DNA is found in pericentromeric, subtelomeric, and interstitial regions, forming constitutive blocks of heterochromatin essential for structures such as centromeres and telomeres. Interspersed repeats are predominantly transposable elements, which alone represent approximately 45% of the human genome (Ruiz-Ruano et al. 2023).

Functionally, some repetitive elements are essential for genome maintenance. Telomeric repeats

TTAGGG protect chromosome ends, while α-satellite repeats form the structural basis of centromeres. Beyond structural roles, REs contribute regulatory sequences including promoters, enhancers, splicing sites, and insulator elements that actively reshape cellular transcriptomes (Chuong et al. 2017; Lu et al. 2020; Rebollo et al. 2012).

6.2.2 Transposable Elements

Transposable elements (TEs) are a subset of repetitive elements defined by their capacity to change genomic position, or "transpose." McClintock (1950) first identified these elements in maize, and now TEs are recognized as nearly ubiquitous components of eukaryotic genomes and among the most abundant encoding sequences in nature (Hayward and Gilbert 2022; Wells and Feschotte 2020).

The most fundamental classification distinguishes two major classes based on transposition intermediates (Wicker et al. 2007). Class I elements, or retrotransposons, replicate through an RNA intermediate that is reverse-transcribed back to DNA before integration—a "copy-and-paste" mechanism. Class II elements, or DNA transposons, move directly as DNA through a "cut-and-paste" mechanism mediated by the enzyme transposase, though some subclasses (e.g., Helitrons, Polintons/Mavericks) employ alternative replicative strategies (Feschotte and Pritham 2007; Kapitonov and Jurka 2006).

Class I retrotransposons are further subdivided into long terminal repeat (LTR) retrotransposons, which are structurally related to retroviruses, and non-LTR elements, which include long interspersed nuclear elements (LINEs) and short interspersed nuclear elements (SINEs). In the human genome, the L1 family of LINEs constitutes approximately 17% of the genomic sequence (Finnegan 1989), while Alu elements—the most abundant SINEs—represent approximately 10–13% (Batzer and Deininger 2002; Dewannieux et al. 2003).

LTR retrotransposons are structurally related to retroviruses; indeed, endogenous retroviruses (ERVs) are considered a subclass of LTR retrotransposons that likely originated from ancestral germline retroviral infections (Jern and Coffin 2008). Non-LTR retrotransposons integrate via target-primed reverse transcription (TPRT), a mechanism in which the element's endonuclease nicks the target DNA and uses the exposed 3′ hydroxyl as a primer for reverse transcription *in situ* (Luan et al. 1993). Additionally, YR retrotransposons represent a third category that shares reverse transcriptase homology with LTR retrotransposons but possesses a tyrosine recombinase domain more closely related to those found in certain DNA transposons (Goodwin and Poulter 2003), illustrating the modular and chimeric nature of TE evolution.

TEs can be classified as autonomous, encoding their own transposition machinery, or non-autonomous, dependent on enzymatic products of other TEs for mobilization. SINEs, for example, do not encode their own reverse transcriptase and are mobilized in trans by L1 machinery.

6.2.3 Non-Coding Regions and "Junk DNA"

Non-coding DNA refers to genomic sequences that do not encode proteins. In the human genome, protein-coding exons constitute only approximately 1.5–2% of the total sequence, with the remaining 98–99% being non-coding. The term "junk DNA" was formalized by Ohno (1972), who argued that large genomes inevitably accumulate sequences—passively propagated over evolutionary time—that do not encode proteins.

It is important to distinguish between "non-coding" and "non-functional," a confusion that has been a persistent source of debate, particularly following the ENCODE project (Consortium 2012; Doolittle 2013; Graur et al. 2013). Non-coding DNA encompasses a broad array of functional elements, including genes for non-coding RNAs (rRNAs, tRNAs, snRNAs, snoRNAs, miRNAs, lncRNAs, piRNAs), regulatory sequences (promoters, enhancers, silencers, insulators), origins of replication, and structural elements such as centromeres and telomeres.

True "junk DNA" in the rigorous evolutionary sense refers to sequences without demonstrated biological function, including the majority of pseudogenes and remnants of degraded transposons that evolve at neutral rates. The fraction of the genome under purifying selection—and therefore probably functional—is considerably smaller, perhaps 5–15%.

6.2.4 GC Content Bias

The term "GC content bias" encompasses two related but distinct phenomena: a biological dimension concerning the non-uniform distribution of guanine (G) and cytosine (C) nucleotides across genomes, and a technical dimension concerning systematic errors introduced during high-throughput sequencing.

Biological Variation in GC Content

GC content—the proportion of guanine and cytosine bases in a DNA sequence—varies substantially both between and within genomes (Benjamini and Speed 2012). In the human genome, GC content is distributed across successive stretches (isochores) that vary from approximately 35% in GC-poor regions to approximately 60% in GC-rich regions. Across species, genomic GC content varies from below 25% to above 75% and is shaped by a complex interplay of mutation, selection, recombination, and genetic drift.

A major driver of GC content heterogeneity in eukaryotes is GC-biased gene conversion (gBGC), a DNA repair mechanism associated with recombination that preferentially fixes G/C alleles over A/T alleles at heterozygotic sites during meiosis (Duret and Galtier 2009; Eyre-Walker 1993; Galtier et al. 2001; Lassalle et al. 2015). Because gBGC generates substitution patterns indistinguishable from those produced by positive selection favoring higher GC content, it can confound molecular evolutionary and phylogenetic analyses (Romiguier et al. 2010).

The distribution of transposable elements correlates with GC content: SINEs such as Alu tend to accumulate in genomic regions rich in GC, while L1 LINEs are enriched in AT-poor regions (Richard et al. 2008).

Technical GC Bias in High-Throughput Sequencing

In the context of high-throughput sequencing, GC content bias refers to the systematic dependence between read coverage and GC content of sequenced fragments (Browne et al. 2020; Chen et al. 2013; Dohm et al. 2008). This effect is well documented on Illumina sequencing platforms, where the number of reads mapped to a genomic region depends considerably on its nucleotide composition, with both high-GC and high-AT fragments typically being underrepresented.

Empirical evidence strongly implicates PCR amplification during library preparation as the primary source of this technical bias. The severity of GC bias varies across library preparation protocols and sequencing platforms.

Mitigation strategies include PCR-free library preparation, use of additives such as betaine (for GC-rich regions) or trimethylammonium chloride (for GC-poor regions), selection of less-biased PCR polymerase mixtures, and computational correction methods applied post-sequencing.

6.3 Evolutionary Origins of Repetitive DNA

6.3.1 Origins of Transposable Elements

The deep evolutionary origins of TEs remain debated. The conservation of an RNA recognition motif across various enzymes crucial for TE integration suggests pre-eukaryotic emergence of the central transposition machinery. However, whether TEs originated once in the last universal common ancestor, arose independently multiple times, or spread across lineages through horizontal transfer remains unresolved.

Two factors complicate phylogenetic reconstruction of TE evolutionary history. First, many TE lineages have undergone horizontal transfer between even distantly related species (Schaack et al. 2010), decoupling TE phylogeny from host phylogenies. Second, TEs can exhibit modular or chimeric evolution, in which functional domains are exchanged between distinct superfamilies.

6.3.2 Non-LTR Retrotransposons: LINEs and SINEs

Long interspersed nuclear elements (LINEs) are autonomous non-LTR retrotransposons that encode the enzymatic machinery required for their own retrotransposition. LINE-1 (L1) elements are the dominant autonomous retrotransposon in mammalian genomes, comprising approximately 17% of the human genome. L1 elements are ancient (Malik and Eickbush 2001), with L1-like clades recognized across vertebrates.

Short interspersed nuclear elements (SINEs) are non-autonomous retrotransposons that depend on enzymes encoded by LINEs for their mobilization (Dewannieux et al. 2003). SINEs are typically derived from small structural RNA genes. Most mammalian SINEs are derived from tRNAs, with the notable exception of Alu elements, which originated from the 7SL RNA gene that encodes the RNA component of the signal recognition particle. Alu elements are restricted to primates and arose approximately 65 million years ago through fusion of two monomeric precursors (Batzer and Deininger 2002).

6.3.3 DNA Transposons

DNA transposons constitute approximately 3% of the human genome, with hAT and Mariner/Tc1 superfamilies predominating (Cordaux and Batzer 2009). Although their transposition is non-replicative, DNA transposons accumulated large numbers of copies through transposition during DNA replication and through homologous recombination-mediated repair of double-strand breaks at excision sites. In the human genome, all known DNA transposon families are considered molecular fossils, with no elements currently active.

6.3.4 Tandem Repeats: Satellites, Minisatellites, and Microsatellites

Satellite DNA are arrangements of sequences repeated in tandem that can span megabases without interruption, typically located in centromeric, pericentromeric, and subtelomeric chromosomal regions (Garrido-Ramos 2017). The term “satellite” derives from the observation that these sequences, due

to their biased base composition, form distinct secondary bands in cesium chloride density-gradient centrifugation.

Evolutionary origins of satellite DNA families are diverse and often lineage-specific (Plohl et al. 2012). Centromeric satellite arrays evolve rapidly, showing little sequence conservation between distantly related species despite conserved centromere function (Henikoff et al. 2001; Malik and Henikoff 2009).

An intriguing hypothesis for the origin of centromeric repeats proposes that centromeres were derived from telomeres during early eukaryotic chromosome evolution. This model posits that linearization of an ancestral circular chromosome was accompanied by capture of retrotransposons at broken chromosome ends, with subsequent divergence of telomere-like internal repeats giving rise to proto-centromeric sequences.

Satellite DNA undergoes concerted evolution (Dover 1982), by which repetition units within a species are homogenized through recombinational mechanisms (unequal crossing over, gene conversion, rolling-circle amplification), producing low intraspecific but high interspecific sequence divergence.

Microsatellites (also known as simple sequence repeats or short tandem repeats) consist of 1–6 bp motifs repeated in tandem, typically spanning a few hundred base pairs (Ellegren 2004). They are abundant in eukaryotic genomes, comprising approximately 3% of the human genome. The primary mechanism generating microsatellite length variation is replication slippage (also called slipped-strand misalignment), in which DNA polymerase transiently dissociates during replication of a repetitive template and reanneal out of frame, leading to insertion or deletion of one or more repeat units.

Minisatellites (variable number tandem repeats, VNTRs) have larger repetition units of 10–100 bp and are frequently enriched near telomeres. Their expansion is driven primarily by unequal crossing over and gene conversion rather than replication slippage, reflecting the larger unit length that favors recombination-based mechanisms.

6.3.5 Interplay Between Repeat Classes

The traditional classification of repeats into interspersed and tandem classes belies extensive reciprocal interactions between these compartments. Transposable elements can serve as substrates for the *de novo* emergence of satellite DNA families: when TE copies are clustered, unequal crossing over and recombination-based amplification can expand them into tandem arrays, effectively converting interspersed repeats into satellite DNA (Meštrović et al. 2015). Conversely, TEs accumulate preferentially within satellite-rich heterochromatic regions, where they may be silenced but contribute to heterochromatin dynamics.

The life cycle of a TE lineage can be described in four phases: invasion (horizontal transfer or activation of a previously dormant element), amplification (proliferation throughout the host genome), maturation and silencing (host defense mechanisms such as DNA methylation and piRNA pathways suppress transposition), and death or degeneration (accumulation of mutations renders the element non-functional). This four-phase model applies across diverse TE superfamilies and host lineages, though the duration and intensity of each phase vary considerably (Kidwell and Lisch 2002).

6.4 Obstacles Posed by Genomic Repetitions for Genomic Studies

6.4.1 Ambiguity and Multi-Mapping

The core of de novo assembly is identification of overlaps between reads to form longer contiguous sequences (contigs). When a read originates from a repetitive region, it may match perfectly to multiple locations across the genome. Most assembly algorithms (such as de Bruijn Graphs) depend on the uniqueness of k-mers (sequences of length k). If a repetitive element is longer than the k-mer size, the assembler cannot determine which "copy" of the repetition a specific read belongs to.

This results in "collapsed" assemblies, where multiple distinct genomic loci are incorrectly merged into a single representative sequence, or the assembler simply terminates the contig, leading to high fragmentation.

6.4.2 Graph Complexity and Tangles

Modern assemblers represent genomic data as a graph where nodes are sequences and edges are overlaps. In a repetition-free genome, the graph would ideally be a linear path. However, repetitive DNA introduces "tangles" or cycles into the graph.

In a de Bruijn graph, each repetition creates a node with multiple incoming and outgoing edges. Resolution of these paths requires information that spans the entire repetition to "fill" flanking unique sequences. If read length (L) is shorter than repetition length (R), the path through the graph is mathematically non-deterministic without auxiliary information.

6.4.3 Structural Variations and Misassembly

Repetitive regions are hotspots for structural evolution. Highly similar repetitions (for example, paralogous genes) may be confused with allelic variations in diploid organisms. An assembler may erroneously connect two distant unique regions because they share a common repetitive boundary, leading to large-scale inversions or translocations in the final structure.

6.5 Specific Challenges of Assembly and Annotation

Assembly of genomic sequences from short read data represents one of the most significant computational challenges in bioinformatics, primarily due to the presence of repetitive DNA elements (Chaisson et al. 2015; Phillippy et al. 2008; Treangen and Salzberg 2011). Repetitive sequences constitute a large fraction of eukaryotic genomes and are the primary source of assembly fragmentation and errors.

The difficulty arises from three fundamental issues: ambiguous mapping, graph-based complexity, and read length limitations.

The essential challenge is a "puzzle" where many pieces are identical. Without reads long enough to capture both the repetitive element and "anchor" unique sequences on both sides, the physical orientation and copy number of these elements remains computationally unresolvable. This is why the transition from Short-Read Sequencing (Illumina) to Long-Read Sequencing (PacBio, Oxford Nanopore) was revolutionary for achieving "telomere-to-telomere" assemblies (Logsdon et al. 2020).

6.6 Sequencing and Bioinformatics Solutions

6.6.1 Long-Read Sequencing Technologies

Third-generation sequencing, or long-read, technologies have transformed the landscape of repetition analysis by producing reads that routinely exceed 10,000 bp and can extend beyond 100,000 bp in ultra-long configurations. Two commercial platforms dominate this space: Pacific Biosciences (PacBio) and Oxford Nanopore Technologies (ONT).

PacBio's SMRT (Single Molecule Real-Time) sequencing, particularly the high-fidelity (HiFi) mode, generates reads with median lengths of 15–20 kb and base-level accuracy exceeding 99.9% through circular consensus sequencing. HiFi reads have demonstrated very high accuracy for mapping repetition polymorphisms and resolving complex tandem repeat structures. ONT sequencing, in contrast, excels at producing ultra-long reads that can exceed 100 kb, making it particularly valuable for spanning even the largest repetitive arrays.

Each platform offers complementary strengths. HiFi reads provide the accuracy needed for reliable variant calling within repetitions, while ultra-long ONT reads provide the contiguity to span larger and more complex repetitive structures.

6.6.2 Bioinformatics Tools for Repetition Identification and Masking

Before genome annotation can proceed, repetitive elements must be identified and masked to prevent spurious gene predictions and false-positive BLAST alignments.

RepeatMasker is the most widely used tool for this purpose (Smit et al. 2013). It screens input DNA sequences against curated libraries of known repetitions, including the Dfam database of Hidden Markov Models of transposable element (Storer et al. 2021).

RepeatModeler complements RepeatMasker by enabling de novo identification of repetition families from unannotated genome assemblies (Flynn et al. 2020). More recently, the EDTA (Extensive De Novo TE Annotator) pipeline has emerged as a comprehensive alternative that integrates RepeatModeler, LTR_retriever, and other specialized tools to achieve more complete and accurate transposable element annotation (Ou et al. 2019).

For tandem repetition genotyping from long-read data, several specialized tools have been developed. TRGT (Tandem Repeat Genotyping Tool) was specifically designed for PacBio HiFi data and provides robust genotyping and visualization of tandem repeats at genome-wide scale (Dolzhenko et al. 2024). LongTR uses a clustering strategy combined with partial-order alignment and hidden Markov models to infer consensus haplotypes.

MotifScope represents a newer algorithmic approach for characterization and visualization of tandem repeat motifs, identifying more motifs and more accurately representing underlying repeat sequences than previous methods (L. Zhang et al. 2025).

6.6.3 Assembly Strategies for Repetition-Rich Genomes

Assembly of repetition-rich genomes has been revolutionized by the combination of long-read sequencing with modern assembly algorithms. The fundamental insight is that repetitive sequence can be resolved when reads are longer than the repetition unit. Current state-of-the-art assemblers, including hifiasm (Cheng et al. 2021), Verkko (Rautiainen et al. 2023), and LJA, all incorporate strategies for handling complex repetition structures within assembly graphs.

For near telomere-to-telomere assembly of homozygous genomes, current best practice involves using PacBio HiFi reads and ultra-long ONT reads. HiFi reads are used to construct an initial assembly graph consisting of unitigs that contain no long exact repetitions, with possible connections between them determined by repetition structure. Highly repetitive regions become represented as complex subgraphs, or "tangles," which are then resolved using ultra-long ONT reads that span these regions.

The landmark achievement of the Telomere-to-Telomere Consortium in completing the first gap-free human genome assembly (T2T-CHM13) demonstrated that this combined approach could resolve even the most recalcitrant repetition regions (Nurk et al. 2022; Rhie et al. 2021), including centromeric satellite arrays, rDNA clusters, and segmental duplications.

6.6.4 Pangenome Approaches to Repetition Variation

A single linear reference genome cannot capture the full spectrum of repetition variation present in a population. The Human Pangenome Reference Consortium (Wang et al. 2022) addresses this limitation by constructing graph-based pangenome references that encode population-level diversity, including variation in repetitive regions (W.-W. Liao et al. 2023).

Pangenome graphs are particularly powerful for resolution of variable number tandem repeats (VNTRs), which have hypervariable repeat number and composition. The Repetition-Pangenome-Graph (RPGG) data structure was developed to encode both population diversity and internal repetition structure of VNTR loci from multi-haplotype-resolved assemblies, enabling accurate VNTR composition estimation even from short-read data.

Initial analyses have demonstrated substantial improvements in variant discovery, with a 34% reduction in small variant errors and a 104% increase in the detection of structural variants per haplotype compared to GRCh38-based workflows (W.-W. Liao et al. 2023). The most recent data release incorporates phased haplotypes from 232 individuals, selected to represent diverse ancestries, with the goal of expanding to over 700 haploid genomes (Wang et al. 2022). This population-level approach has yielded an average of over 29,000 structural variants per individual, compared to fewer than 16,000 when using a linear reference alone (Groza et al. 2024).

6.6.5 Combination of Strategies

Given the complexity of dealing with repetitions, an increasingly adopted strategy involves combining multiple sequencing technologies and bioinformatics tools. For example:

1. PacBio HiFi sequencing for accuracy and reliable repetition variant calling
2. Ultra-long ONT sequencing for repetition structure contiguity
3. Strand-seq mapping for haplotype phasing resolution
4. Masking tools such as RepeatMasker for annotation
5. Genotyping tools such as TRGT for tandem repeat characterization
6. Pangenome construction to capture population-level variation

6.7 Emerging Challenges and Future Directions

Despite the transformative advances in long-read sequencing and pangenome construction, several challenges remain. The cost and complexity of generating multiple complementary data types (PacBio HiFi, ONT ultra-long, Strand-seq) for a single genome remain prohibitive for many non-model organisms. Computational resources required for pangenome graph construction and variant calling at population scale are substantial. Multiallelic VNTRs present particular difficulties for pangenome graphs, as their combinatorial complexity can make graph representations intractable.

At the single-cell level, the convergence of long-read sequencing with single-cell omics is opening new avenues for understanding repeat element dynamics, including somatic retrotransposition and allele-specific expression of TE-derived regulatory elements. The development of tools like RAmbler, which resolves complex repeats in specific human chromosomes (Chakravarty et al. 2025), and improvements in long-read RNA sequencing methods further extend the analytical toolkit for repeat biology.

The adoption of the T2T-CHM13v2.0 assembly as the standard reference, together with the maturation of the HPRC pangenome, will provide a more complete foundation for understanding repeat biology and its implications for genome evolution, gene regulation, and human disease.

6.8 Implications for Phylogenomics

Repetitive elements present unique challenges and opportunities for phylogenomic studies. While aligned matrices traditionally exclude repetitive regions due to alignment ambiguity, recent work has demonstrated that repetitions—particularly transposable elements—can provide informative phylogenomic signal when treated appropriately.

The structure, copy number, and sequence of repetitions evolve under complex selective and demographic pressures, reflecting the evolutionary history of lineages. Understanding the dynamics of repetitive element and developing methods for incorporating repetition signal into phylogenetic analysis remains a productive area of investigation.

Chapter Review Outline

1. Human genome: ∼50% repetitive; ∼45% TEs; protein-coding exons only ∼1.5–2% (Koning et al. 2011; Lander et al. 2001).
2. TE classification: Class I (retrotransposons, "copy-and-paste" via RNA): LTR (retrovirally related), LINEs (L1 ≈ 17% human genome), SINEs (Alu ≈ 10–13%, derived from 7SL RNA, ∼65 Mya, restricted to primates). Class II (DNA transposons, "cut-and-paste"): hAT, Mariner/Tc1; ∼3% of human genome; all fossils in humans.
3. Autonomous vs. non-autonomous: SINEs depend on L1 machinery in trans.
4. Satellite DNA: tandem repeats megabases long (centromeres, telomeres); concerted evolution (Dover, 1982) via unequal crossing over and gene conversion.
5. Microsatellites (STRs, 1–6 bp): ∼3% human genome; variation by replication slippage. Minisatellites (VNTRs, 10–100 bp): enriched near telomeres; expansion by recombination.
6. Biological GC bias: gBGC (GC-biased gene conversion) generates patterns indistinguishable from positive selection (Galtier et al. 2001); SINEs accumulate in GC-rich regions, LINEs in AT-poor regions.
7. Technical GC bias: coverage depends on GC composition on Illumina; primary source = PCR amplification; mitigation: PCR-free library, betaine, computational correction.
8. Assembly obstacles: ambiguous k-mers → repetition collapse; tangles in de Bruijn graph; if $L_{\text{read}} < R_{\text{repetition}}$, path non-deterministic.
9. Long-read solutions: PacBio HiFi (15–20 kb, >99.9% accuracy, variant calling) + ONT ultra-long (>100 kb, contiguity); T2T-CHM13 assembly (Nurk et al. 2022).
10. Tools: RepeatMasker (Dfam HMMs), RepeatModeler (de novo), EDTA (integrated LTR_retriever), TRGT (tandem genotyping, PacBio HiFi).
11. Pangenome: graph-based reference (HPRC) captures population variation in repetitions; RPGG for VNTRs.
12. Interplay between repeat classes: TEs serve as substrates for *de novo* satellite DNA emergence via unequal crossing over; four-phase TE life cycle (invasion, amplification, silencing, degeneration).
13. Emerging challenges: cost of multi-technology sequencing, pangenome computational requirements, multiallelic VNTR complexity; single-cell long-read approaches for somatic TE dynamics; T2T-CHM13v2.0 as standard reference.

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Chapter 7

Information Technology Applied to Phylogenetic Problems

Abstract

Phylogenetic inference — the reconstruction of evolutionary relationships among organisms — has been profoundly shaped by advances in information technology since its formalization as a computational discipline in the 1960s. This review examines the historical interaction between computing hardware, software design, programming language choice, and phylogenetic methodology. Beginning with the first implementations of parsimony and likelihood algorithms on mainframe computers, the narrative traces the development of seminal software packages (PHYLIP, Hennig86, PAUP*, POY, TNT, MrBayes, RAxML, BEAST, IQ-TREE, and PhyG), the Bayesian MCMC revolution of the early 2000s, and the ongoing challenge of scaling analyses to genome-scale datasets. Central debates — including parsimony versus likelihood, trees versus networks, and the search for a unified optimality criterion — are examined alongside the computational obstacles that constrain them. Three recent contributions from Wheeler and collaborators are used as case studies: the PhyG software for phylogenetic graph analysis (Wheeler et al. (2024)), Thompson sampling for adaptive search optimization (Wheeler (2024)), and the Phylogenetic Minimum Description Length (PMDL) criterion (Wheeler and Varón (2025)). The review concludes by identifying persistent unresolved obstacles, including NP-hardness of tree search, incomputability of algorithmic complexity, the difficulty in bridging cladistic and model-based analytical traditions, and the emerging role of machine learning in phylogenetics.

Keywords: phylogenetics, computational complexity, heuristic search, phylogenetic networks, minimum description length, MCMC, maximum likelihood, software

7.1 Introduction

Computational phylogenetics emerged from a synthesis between evolutionary biology and computer science. Before the 1960s, phylogenetic reconstruction was predominantly manual, based on morphological characters and comparative pattern analysis. The problems were small — typically tens of taxa — and could be solved by visual inspection or hand calculations. Early contributions from Hennig (1950), Remane (1952), and Mayr and Linsley (1954) established conceptual principles, but lacked the computational speed to explore them systematically.

Initial mathematical formalization came from Edwards and Cavalli-Sforza (1964), who applied maximum likelihood methods to genetic data, and from Fitch and Margoliash (1967), who developed the minimum evolution method. However, widespread adoption of computational methods in phylogenetic analyses awaited the availability of accessible software. The catalyst was Felsenstein (1973), whose program PHYLIP (Phylogenetic Inference by Maximum Likelihood) and its sequence of developments provided computationally viable implementations of parsimony and likelihood on mainframes and, later, on personal computers.

This chapter examines the history of phylogenetic software design, the methodological debates that shaped implementation choices, and the computational obstacles that remain unresolved. The focus is on how information technology — hardware, programming languages, and search techniques — has either facilitated or constrained methodological advances in phylogenetics.

7.2 Historical Overview of Phylogenetic Software

7.2.1 The Early Era: Mainframes and the Birth of PHYLIP (1973–1985)

Felsenstein's PHYLIP (*PHYLogeny Inference Package*), released in 1973 with significant updates in 1978 and 1985, was the first truly portable and extensible phylogenetic software. Written in FORTRAN and later converted to C, PHYLIP ran on mainframes (UNIX, VMS) and, crucially, on personal computers once they became available. The modular architecture — with separate programs for parsimony (PARS), likelihood (DNAML, PROML), distance (NEIGHBOR), and tree drawing (DRAWTREE) — established a design standard that persists to this day.

PHYLIP's genius was twofold: (1) it implemented mathematically rigorous but computationally demanding algorithms in efficient language (C), allowing application to data sizes that were previously intractable; and (2) it provided a consistent and well-documented user interface that democratized phylogenetic analysis. Researchers without programming experience could now execute complex analyses.

PHYLIP's limitation was its tree search. The parsimony algorithm employed ('fitch' originally, later refined branch-and-bound search) was enumerative and consequently exponential in the number of taxa. For data with more than 20–30 taxa, exhaustive search became computationally prohibitive. Heuristic methods (random sequential addition, nearest-neighbor search) were adopted for speed, but always with the cost of possible suboptimality.

7.2.2 Proliferation of Specialized Software (1985–2000)

The 1980s and 1990s saw an explosion of new programs, each optimized for a different set of criteria and data:

- **PAUP* (Phylogenetic Analysis Using Parsimony)** (Swofford 1993): The intellectual commercial successor to PHYLIP, developed by David Swofford. PAUP* implemented advanced search algorithms (branch-and-bound search with efficient pruning, nearest-neighbor search with multiple starting points) and was the first to integrate parsimony and maximum likelihood in a single platform. Its graphical interface and powerful command language made it the gold standard for molecular phylogenetics until the emergence of MrBayes. The downside: PAUP* was commercial (paid license) and, for a time, the phylogenetic community was divided between PHYLIP users (free, open) and PAUP* users (paid, faster).

- **HENNIG86 and TNT** (Goloboff et al. 2008): Originally developed by James Farris in 1986 and based on Farris's earlier HENNIG program, HENNIG86 was optimized specifically for parsimony and emphasized algorithms for implicit tree enumeration. Its successor, TNT (Tree analysis using New Technology) by Goloboff et al., revolutionized parsimony search with the implementation of efficient branch-and-bound search with critical cost pruning and intelligent use of memory caching to iteratively explore the neighborhood of current solutions. TNT remains among the fastest parsimony searchers and is the standard for studies with many taxa (>500).

- **POY (Phylogenetic Analysis Using Parsimony)** (Wheeler 2001): A distinct program aimed at data that are not aligned a priori, particularly DNA and protein sequences where alignment is uncertain. POY employed simultaneous searches in both alignment and tree space, treating alignment as a variable dynamic search state, not fixed. This approach was computationally much more expensive than fixed alignment plus tree search, but offered the theoretical advantage of avoiding artifacts from initial alignment.

- **RAxML (Randomized Axelerated Maximum Likelihood)** (Stamatakis 2006): Released in 2006 by Alexandros Stamatakis, RAxML was a game changer. It implemented maximum likelihood search with rapid convergence heuristics (branch-and-bound search, nearest-neighbor search) and, crucially, efficient parallelization for distributed memory systems (MPI). RAxML could run on HPC clusters and scale to hundreds of cores. Its rapid search algorithm (with frequent likelihood function evaluation) and branch rearrangement parallelization made it the de facto standard for large-scale likelihood analyses in the post-2006 era. RAxML is written in C with MPI extensions, demonstrating how careful design can harness hardware parallelism.

- **BEAST (Bayesian Evolutionary Analysis by Sampling Trees)** (Drummond and Rambaut 2007): Representing a fundamentally new paradigm, BEAST was specialized for Bayesian analysis of hierarchical data, particularly those with dating uncertainty and multiple lineages. The software implemented MCMC (Markov Chain Monte Carlo) to sample the joint posterior distribution of trees and model parameters. Its architectural innovation was enabling flexible prior relaxation, complex substitution models, and hierarchical data structures (calibration dates, multiple loci, multiple individuals per species). BEAST introduced a new computational paradigm: instead of searching for a global optimal tree (as in parsimony or likelihood), MCMC sampled the posterior distribution, and uncertainty was captured naturally.

7.2.3 The Bayesian Revolution and MCMC (2000–2015)

MrBayes (Ronquist and Huelsenbeck 2003), developed by Fredrik Ronquist and John Huelsenbeck, was the catalyst for widespread adoption of Bayesian methods in molecular phylogenetics. MrBayes implemented MCMC with intelligent proposal solutions to the problem of exploration in tree space (topology plus branch lengths plus model parameters). The Metropolis-Hastings algorithm was combined with well-calibrated tree moves (subtree pruning and regrafting) that improved chain mixing.

The Bayesian revolution had profound computational implications. Unlike heuristic search (which attempts to find one peak), MCMC attempts to sample the neighborhood of many peaks,

exploring parameter space more exhaustively. This typically requires more iterations for convergence — not infrequently chains with 1 to 10 million generations. The tradeoff was that phylogenetic uncertainty was captured naturally: the posterior distribution of trees represented uncertainty from both stochastic analysis and data.

The computational disadvantage of MCMC was obvious: a Bayesian analysis typically took 10–100× more CPU time than a comparable maximum likelihood analysis, because the algorithm had to explore more of the space. For datasets with hundreds or thousands of taxa, Bayesian analyses became impractical on single-processor machines. Parallelization of MrBayes on MPI systems helped, but did not fully resolve the problem.

7.2.4 Modern Software: Efficient Paradigms (2015–Present)

The computational limitations of complex Bayesian methods, combined with the data revolution (next-generation sequencing producing millions of loci), drove a new wave of optimizations.

IQ-TREE (Inferred Quotient Tree) (Minh et al. 2020): Developed by Bui Quang Minh and collaborators, IQ-TREE implemented a suite of innovations:

1. UFBoot fast model selection algorithm that tests substitution models without respecifying the tree between tests, saving optimization iterations.
2. Model space search with information criteria (BIC, AICc) enabling automatic model selection.
3. Fast support calculation algorithm (aBayes, aLRT, SH-aLRT) that avoided resampling.
4. Embarrassingly parallel analysis of multiple loci (file-by-file parallelism).
5. SIMD compilation (SIMD instructions such as AVX2, AVX512) that vectorized critical likelihood operations.

IQ-TREE quickly became the gold standard for phylogenetic likelihood analyses on genomic datasets.

RAxML-NG (Kozlov et al. 2019): A modern rewrite of classic RAxML, with code refactored for cache efficiency, native support for advanced SIMD (AVX3), and modular API enabling integration with workflow frameworks (Nextflow, Snakemake). RAxML-NG maintains RAxML's status as one of the fastest maximum likelihood searchers.

BEAST 2 (Bouckaert et al. 2019): A complete rewrite of BEAST with new MCMC engine, new proposed operators for better mixing, and, crucially, integration with the BEAGLE library (Ayres et al., 2019) for GPU-accelerated likelihood calculation.

PhyG (Wheeler et al. 2024): Innovative software for phylogenetic graph analysis, enabling explicit representation of reticulation events (hybridization, lateral transfer) that do not fit the standard tree model. PhyG implements search in graph space (highly combinatorially complex) using sophisticated heuristics including Thompson sampling.

7.3 Central Debates Shaped by Computation

7.3.1 Parsimony Versus Likelihood

One of the oldest debates in phylogenetics is whether parsimony (minimizing evolutionary change) or likelihood (maximizing probability of data under an evolutionary model) is methodologically

preferable. This debate has deep theoretical roots, but also computational limitations.

Parsimony is computationally simpler: Fitch's dynamic programming algorithm (Fitch 1971) computes parsimony in time $O(n \cdot m)$ for n taxa and m characters. Finding the minimum parsimony tree is NP-hard, but rapid search heuristics (branch-and-bound search in TNT, neighborhood search in PAUP*) can find good local minima in reasonable time even for large data.

Likelihood is computationally more expensive. Felsenstein's dynamic programming algorithm for computing likelihood is also $O(n \cdot m)$ per evaluation, but each tree search move requires reevaluation — and for complex networks of evolutionary models (multiple partitions, variable-rate models, molecular clocks), the constant factor is much larger. Consequently, any likelihood analysis that is not a small dataset requires significant parallelization or hardware acceleration (GPU).

The historical trend reflects computational cost. In the 1980s–1990s, parsimony dominated because data were small, machines were slow, and parsimony was fast. With the advent of faster machines, multiprocessors, and later GPUs, likelihood and Bayesian methods became practical. Today, most analysts prefer likelihood or Bayesian methods, because they capture model and data uncertainty more transparently, but parsimony persists in contexts where search efficiency is critical (very large data, high number of taxa).

7.3.2 Networks Versus Trees

Most classical phylogenetics assumes a tree topology: each modern species descends from a common ancestor, and relationships form a bifurcating tree structure. However, reticulation (events where lineages do not descend linearly from an ancestor but instead recombination, hybridization, or lateral gene transfer occur) violates the tree model.

Phylogenetic network methods such as quasi-median networks (Holland and Moulton 2004), split networks (Bryant and Moulton 2004), and more recently PhyG (Wheeler et al. 2024) explicitly represent reticulation. The computational complexity of network inference is substantially higher than tree inference. Search space in reticulated graphs is exponentially larger, and network likelihood is computationally more expensive to evaluate.

The computational constraint kept tree methods dominant for decades. Only recently, with faster hardware and better heuristics (Thompson sampling in PhyG), have network methods become practical for moderately-sized datasets. Nevertheless, most practical phylogenetics remains in trees because networks are computationally expensive and require much larger datasets for statistical resolution of reticulation.

7.3.3 Unified Optimality Criterion

Since the birth of computational phylogenetics, the field has debated: which optimality criterion is correct? Parsimony minimizes evolutionary steps, maximum likelihood maximizes probability under a model, Bayesian methods sample posterior distributions. Each has theoretical justification, but they are theoretically inconsistent and frequently yield different trees.

A recent approach is minimum description length (MDL) (Wheeler and Varón 2025), which formalizes the idea that the best phylogenetic hypothesis is one that provides the most efficient compression of the data — intuitively, one that captures pattern with the least ad hoc information. For phylogenetics, PMDL (Phylogenetic Minimum Description Length) (Wheeler and Varón 2025) defines a description length that includes:

$$\text{PMDL} = (\text{desc. length tree}) + (\text{desc. length data} \mid \text{tree}) \tag{7.1}$$

In principle, PMDL unifies parsimony (which minimizes steps) and likelihood (which minimizes residual data length). The computational barrier: computing PMDL requires evaluating the algorithmic complexity (Kolmogorov) of the model, which is incomputable in general. Heuristic approximations are under development, but PMDL remains more of a conceptual idea than a practical routine method.

7.4 Unresolved Computational Obstacles

7.4.1 NP-Hardness of Tree Search

The problem of finding the minimum parsimony tree is NP-hard (Graur et al. 1991). This means that no known deterministic algorithm can solve the problem in polynomial time for arbitrarily large instances. Consequently, any search algorithm is heuristic: it can find a good solution, but cannot guarantee global optimality.

This has profound practical implications. When an analyst runs a parsimony analysis and finds k equally parsimonious trees, there is no guarantee that no even more parsimoniously optimal trees exist outside the explored search space. Different starting points, different search strategies, and different parameterizations may discover different sets of optimal trees.

For maximum likelihood, the situation is even worse: the problem of finding the maximum likelihood tree is also NP-hard (Chor and Tuller), and the likelihood function is multimodal (many local maxima). Search heuristics such as branch-and-bound can get stuck in local maxima. Bayesian methods (MCMC) have the advantage of stochastic exploration, but convergence to true posterior distributions is guaranteed only asymptotically.

7.4.2 Incomputability of Algorithmic Complexity

As mentioned above, formalizing unified optimality criteria requires quantifying algorithmic complexity of data (Kolmogorov description length). However, this quantity is incomputable in general (Turing's halting theorem). This places a fundamental mathematical limit on any optimality scheme that attempts to be universal.

The implication: no general-purpose phylogenetic program can compute a true universal measure of "improvement" of one tree over another. Any optimality criterion (parsimony, likelihood, approximate MDL) is of necessity a heuristic or partial approximation.

7.4.3 Conceptual Framework: Cladistics Versus Model-Based Methods

A profound and still-unresolved debate is between the cladistic tradition (focus on synapomorphies and analysis of discrete characters) and model-based methods (focus on probabilistic evolutionary models of sequence change). These traditions have distinct histories and philosophers, and their fundamental principles are sometimes in tension.

From a computational perspective, classical cladistics (parsimony analysis of morphological data) is computationally simpler: heuristic search in discrete space. Model-based methods (likelihood analysis of DNA sequences under Markov models) are computationally more complex but theoretically more rigorous about model uncertainty.

A potential bridge is that offered by recent software such as IQ-TREE: automatic model selection combined with flexible support analysis that does not assume a specific model. But fundamentally, the two conceptual frameworks have not yet been unified in a single theoretical framework.

7.4.4 Emerging Role of Machine Learning

Machine learning (deep neural networks, ensemble methods) has begun to be applied to phylogenetic problems:

- **Alignment prediction:** Neural networks have been trained to predict optimal alignments directly from raw sequences, bypassing classical dynamic programming algorithms.
- **Topology prediction:** Graph neural network architectures (graph neural networks, GNNs) can be trained on known trees to predict topology from new data.
- **Model selection:** Machine learning models can predict which substitution model is most appropriate for a given dataset, potentially faster than likelihood ratio tests.

The advantage: speed. A trained neural network model can make predictions in milliseconds, while classical recalculations take minutes. The disadvantage: interpretability and statistical rigor. Machine learning models are often "black boxes" — it is unclear what was learned or whether the model extrapolates well to data outside the training domain. For phylogenetics, where data are often unique and under unknown evolutionary pressure, extrapolation is risky.

The future role of machine learning is probably as a stack acceleration tool (rapid screening of many models, prediction of good initial estimates for optimization) before refinement via classical methods, rather than replacement of classical methods.

7.5 Case Study: Recent Contributions from Wheeler and Collaborators

Three recent contributions illustrate how computational design continues to shape modern phylogenetics.

7.5.1 PhyG: Phylogenetic Graph Analysis

Wheeler et al. (2024) presented PhyG, software for phylogenetic graph inference that explicitly represents reticulation. PhyG's principal computational innovation is the implementation of Thompson sampling for graph space exploration.

Thompson sampling is a multi-armed bandit exploration algorithm: in the absence of prior knowledge, it distributes search effort proportional to the promise of different search space regions based on past success. In phylogenetic search context, this allows PhyG to explore alternative graphs with an adaptive risk-reward strategy: regions of space that have produced good graphs receive more simulation, while less promising regions receive less.

7.5.2 Thompson Sampling for Adaptive Optimization

Wheeler (2024) explored Thompson sampling as a general strategy for heuristic search in phylogenetics. The Thompson approach is well known in machine learning (contextual bandits, exploration-exploitation tradeoff), but its application to phylogenetic tree search is relatively novel.

The computational advantage: Thompson sampling is parallelizable. Different threads or processes can maintain copies of the belief distribution about promising regions of search space and explore in parallel, periodically converging to share knowledge. This enables exploration of much larger search space than serial methods, and potentially finds better global minima.

7.5.3 Phylogenetic Minimum Description Length (PMDL)

Wheeler and Varón (2025) formalized the minimum description length criterion for phylogenetics (PMDL), seeking a unified optimality criterion that reconciles parsimony and likelihood. PMDL captures the idea that the best phylogenetic hypothesis is one that explains the data with minimal total description (lower model complexity plus lower data residual).

Computational implementation of PMDL is challenging because it requires approximations of algorithmic complexity of models (Kolmogorov). Heuristic versions are under development, but full impact remains to be seen.

7.6 Discussion: Future of Phylogenetic Software

The history of phylogenetic software illustrates general principles of scientific software design:

1. **Hardware optimization matters:** The choice of language, parallelization, and code vectorization have multiplicative effects on performance. RAxML, IQ-TREE, and PhyG all aggressively optimize for modern hardware (multicore, SIMD, GPU).

2. **Computational constraints shape methodology:** Theoretical debates (parsimony vs. likelihood, trees vs. networks) do not occur in a vacuum; computational practicality influences which method is accepted as standard.

3. **Modular architecture enables innovation:** PHYLIP established the standard of specialized modules for different tasks (parsing, analysis, visualization). This design facilitates adoption of new algorithms and allows users to compose customized analyses.

4. **Openness and reproducibility:** Open-source software (PHYLIP, RAxML, IQ-TREE, MrBayes) with complete documentation and source code access enabled confidence, independent validation, and stimulation of community development. In contrast, commercial or closed software often did not receive the same adoption or innovation drive.

The next generation of phylogenetic software will likely continue to leverage hardware acceleration (GPUs, TPUs), machine learning integration for stack acceleration heuristics, and modular designs enabling reproducible and traceable analyses via workflow systems (Snakemake, Nextflow). Unification of optimality criteria and resolution of cladistic versus model-based debates will probably come from theoretical research before computational innovation.

7.7 Conclusion

Computational phylogenetics emerged from synthesis between evolutionary biology and computer science. The history of its software development is a history of four decades of computational optimization, algorithmic design, and resolution of methodological disputes through implementation. The seminal softwares — PHYLIP, PAUP*, TNT, RAxML, MrBayes, IQ-TREE, BEAST, and PhyG — each shaped how modern phylogenetics is practiced, and each reflected compromises between theoretical rigor, computational efficiency, and usability.

Obstacles remain unresolved, particularly NP-hardness of tree search and incomputability of universal optimal criteria. These are not deficiencies of particular software or current hardware, but fundamental mathematical limitations. Future progress will probably come from synthesis of approaches — combining brute-force computational power (better hardware) with refined algorithms (better heuristics) and, potentially, machine learning as an acceleration tool, while always maintaining statistical rigor and interpretability.

Chapter Review Outline

1. Software chronology: PHYLIP (Felsenstein, 1973, FORTRAN→C) → HENNIG86/TNT (Farris/Goloboff, parsimony) → PAUP* (Swofford, parsimony+ML, commercial) → POY (Wheeler, dynamic alignment) → MrBayes (Ronquist & Huelsenbeck, Bayesian MCMC) → RAxML (Stamatakis, 2006, MPI) → BEAST (Drummond, 2007, hierarchical MCMC) → IQ-TREE (Minh, UFBoot, SIMD) → RAxML-NG → BEAST 2 (BEAGLE/GPU) → PhyG (Wheeler, graphs).
2. Parsimony: Fitch algorithm $O(n \cdot m)$; search NP-hard; TNT = fastest searcher (>500 taxa).
3. ML: Felsenstein algorithm $O(n \cdot m)$ per evaluation; constant factor much larger than parsimony; requires parallelization/GPU.
4. Bayesian Revolution (2000–2015): MrBayes implemented Metropolis-Hastings + subtree pruning and regrafting moves; 10–100× more CPU than ML; MC^3 with hot chains.
5. IQ-TREE: UFBoot, automatic model selection (BIC/AICc), fast support (aBayes, SH-aLRT), SIMD (AVX2/AVX-512), file-by-file parallelism.
6. BEAGLE 3: library for GPU likelihood calculation; speedup up to 100×; constraint = GPU memory.
7. Phylogenetic networks: quasi-median networks (Holland and Moulton 2004), Neighbor-Net (Bryant and Moulton 2004), PhyG (Wheeler et al. 2024); search space exponentially larger than trees.
8. Thompson sampling (Wheeler, 2024): adaptive exploration of search space (multi-armed bandits); parallelizable; used in PhyG.
9. PMDL (Wheeler & Varón, 2025): Phylogenetic Minimum Description Length criterion; unifies parsimony and likelihood; barrier = incomputability of Kolmogorov complexity.
10. Unresolved obstacles: NP-hardness (no guarantee of global optimum), incomputability of algorithmic complexity, tension between cladistic vs. model-based traditions.
11. Emerging ML: neural networks for alignment/topology/model prediction; fast but "black box"; future role = heuristic stack acceleration.

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Chapter 8

Design of High-Performance Computing Systems for Phylogenomics

Abstract

Modern phylogenomic analyses involve hundreds to thousands of loci extracted from genome-scale data. Each locus must be individually aligned, trimmed, and subjected to gene tree estimation before downstream species tree inference. This scaling expansion, driven by targeted capture sequencing, transcriptomics, and whole-genome approaches, has rendered desktop-scale computing inadequate for most research groups. This chapter examines design and implementation of high-performance computing (HPC) systems specifically optimized for phylogenomics, covering from workload profiling audit through financial optimization, resource allocation strategy, and multi-user management. We present a cluster proposal tailored for a university bioinformatics department and discuss general principles applicable to laboratories ranging in scale from simple workstations to regional clusters.

8.1 Introduction

A single IQ-TREE analysis on a partitioned alignment of several hundred loci can consume tens of thousands of CPU hours. Bayesian analyses with BEAST or MrBayes add further orders of magnitude when MCMC convergence requires extended chain lengths. Furthermore, questions about whether standard clade support metrics (bootstrap, SH-aLRT, aBayes) are calibrated with respect to alignment sensitivity — a question actively being investigated — demand large-scale replication across alternative alignments and parameter configurations. Such studies are computationally infeasible without dedicated HPC resources.

General university research clusters, while valuable, often have scheduling policies optimized for fairness at campus scale and not for the specific characteristics of phylogenomic workloads. A dedicated departmental cluster enables optimization of hardware heterogeneity, storage topology, and scheduling policies specifically for phylogenomics.

8.2 Computational Profile of Phylogenomic Workflows

The design of an HPC system for phylogenomics must be informed by careful analysis of computational bottlenecks at each pipeline stage. Phylogenomic workflows are heterogeneous: some stages

are memory-limited, others are CPU or I/O constrained, and some are increasingly well-suited for GPU acceleration.

8.2.1 Phase 1: Quality Control and Read Pre-processing

Tools such as *fastp* (Chen et al. 2018) and Trimmomatic (Bolger et al. 2014) perform adapter removal, quality trimming, and read filtering. These tasks are predominantly I/O-bound, with modest CPU requirements. *fastp* is notable for its multi-threaded I/O handling and typically saturates at 4–8 threads. The key bottleneck is storage throughput: a single Illumina HiSeq/NovaSeq lane can produce 100–400 GB of compressed FASTQ data, and QC operations involve reading and rewriting these files in their entirety.

Resource Profile: I/O-limited; 4–8 CPU cores per job; 4–16 GB of RAM; high-throughput scratch storage critical.

8.2.2 Phase 2: Assembly, Mapping, and Locus Extraction

Depending on study design, reads may be assembled *de novo* (*e.g.*, SPAdes, Trinity) or mapped to a reference (*e.g.*, BWA-MEM2, HybPiper for targeted capture data). *De novo* assembly, particularly with Trinity for transcriptomic data, is both CPU-intensive and memory-hungry, with peak RAM usage frequently exceeding 100 GB for moderate-sized datasets. Reference-based approaches are more modest but still benefit from fast storage for BAM file I/O.

Resource Profile: CPU- and memory-intensive; 16–32 cores; 64–256 GB of RAM; fast scratch storage.

8.2.3 Phase 3: Orthology Inference

Tools such as OrthoFinder (Emms and Kelly 2019) perform all-vs-all DIAMOND/BLAST searches followed by MCL clustering. The DIAMOND search phase is embarrassingly parallel and benefits from high core counts, while the MCL clustering phase is memory-intensive for large proteome sets (>50 species).

Resource Profile: CPU-intensive (parallelizable); 16–64 cores; 32–128 GB of RAM.

8.2.4 Phase 4: Multiple Sequence Alignment

MAFFT (Katoh and Standley 2013), MUSCLE, and other alignment tools are CPU-bound, with most implementations threading efficiently to 4–16 cores. However, because a typical phylogenomic dataset involves hundreds to thousands of individual loci, the dominant parallelism strategy is to run many independent alignment jobs concurrently rather than allocate many cores to a single alignment. This makes alignment an ideal workload for SLURM job arrays.

Resource Profile: CPU-bound; 4–16 cores per individual alignment; 2–8 GB of RAM per alignment; high job throughput via arrays.

8.2.5 Phase 5: Alignment Trimming and Filtering

Tools such as trimAl (Capella-Gutiérrez et al. 2009) and BMGE (Criscuolo and Gribaldo 2010) are lightweight, typically completing in seconds per locus. The primary consideration is I/O latency rather than computational power.

Resource Profile: Minimal CPU/RAM; I/O-latency-sensitive.

8.2.6 Phase 6: Gene Tree Estimation (Maximum Likelihood)

Maximum likelihood tree inference with IQ-TREE (Minh et al. 2020) or RAxML-NG (Kozlov et al. 2019) is the most CPU-intensive phase of the pipeline for ML-based approaches. Both tools implement efficient multi-threaded likelihood evaluation using vectorized SIMD kernels (AVX2/AVX-512). IQ-TREE 2 demonstrates high parallelism efficiency, achieving near-linear scaling to 4 cores and useful speedups to 8–16 cores for individual analyses. For phylogenomic studies, the optimal strategy is often to infer gene trees in parallel (one per locus), with each tree search allocated 4–8 cores.

Resource Profile: CPU-bound; benefits from high clock speed and AVX-512; 4–16 cores per gene tree; 2–16 GB of RAM; high job throughput via SLURM arrays.

8.2.7 Phase 7: Gene Tree Estimation (Bayesian)

Bayesian inference with MrBayes (Ronquist et al. 2012) or BEAST/BEAST2 (Bouckaert et al. 2019) is substantially more expensive than ML inference due to MCMC sampling requirements. The BEAGLE library (Ayres et al. 2019) provides GPU-accelerated likelihood computation, achieving speedups of up to 100× for large datasets on NVIDIA GPUs. BEAGLE 3 includes OpenCL and CPU-threaded implementations, but the CUDA-based implementation on modern NVIDIA hardware remains the performance leader. GPU memory is the primary constraint: double-precision nucleotide analyses with four rate categories require approximately 2 GB of GPU memory per 1,000 OTUs and 10,000 unique site patterns.

Resource Profile: GPU-accelerated via BEAGLE/CUDA; 24–48 GB of GPU memory recommended; moderate CPU for non-BEAGLE code paths.

8.2.8 Phase 8: Species Tree Inference and Concordance Analysis

Summary coalescent methods such as ASTRAL-III (Zhang et al. 2018) are fast and memory-efficient, typically completing in minutes even for datasets with hundreds of loci and species. Concatenation-based (supermatrix) approaches using IQ-TREE or RAxML-NG on the complete partitioned alignment are more demanding and represent a second major computational bottleneck. Concordance factor analysis (gCF, sCF) in IQ-TREE requires re-evaluation of all gene trees against the species tree and is moderately CPU-intensive.

Resource Profile: Variable; ASTRAL is lightweight; concatenated ML is CPU-intensive (similar to Phase 6 but on larger data); concordance analysis is moderate.

8.3 Proposed Hardware Architecture

8.3.1 Design Philosophy

The cluster follows a heterogeneous design philosophy: rather than purchasing uniform nodes, we propose three distinct node types optimized for different computational profiles. This approach maximizes cost-effectiveness by avoiding over-provisioning of expensive resources (GPUs, large RAM) on nodes where they are not needed.

Table 8.1: Recommended Hardware Components for Phylogenomic Cluster

Component	Qty	CPU	RAM	Storage	GPU
Compute Node (Type A)	4	2× EPYC 9554 (64C)	512 GB DDR5	2× 2TB NVMe	—
High-Memory Node (Type B)	1	2× EPYC 9554 (64C)	1.5 TB DDR5	4× 4TB NVMe	—
GPU Node (Type C)	1	1× EPYC 9454 (48C)	256 GB DDR5	2× 2TB NVMe	2× RTX A6000 (48GB)
Management/Login Node	1	1× EPYC 9124 (16C)	64 GB DDR5	2× 1TB NVMe (RAID 1)	—
NAS Server	1	1× Xeon entry	64 GB ECC	12× 18TB HDD (RAID6) + 2× 4TB NVMe	—
Networking (25GbE Switch)	1	—	—	—	—
UPS + Rack + Cabling	1	—	—	—	—
Estimated Total Cost: $163,000–$185,000					

8.3.2 Recommended Components

8.3.3 CPU Selection Rationale

The AMD EPYC 9554 (64 cores, 3.75 GHz boost, 256 MB L3 cache) provides excellent balance of core count, clock speed, and cache size for phylogenomic workloads. Both IQ-TREE and RAxML-NG use vectorized AVX2/AVX-512 likelihood kernels, and the Zen 4 architecture in the EPYC 9004 series provides native AVX-512 support. The large L3 cache is particularly beneficial for tree search operations, which exhibit irregular memory access patterns when traversing likelihood arrays across tree internal nodes. The dual-socket configuration yields 128 cores per Type A node (512 total across four nodes), enabling simultaneous execution of 32–64 gene tree jobs per node at 4–8 cores each.

8.3.4 GPU Selection Rationale

The NVIDIA RTX A6000 provides 48 GB of GDDR6 memory per card, which is critical for BEAGLE-accelerated Bayesian analyses. GPU memory is the binding constraint for BEAGLE: a nucleotide analysis with 500 OTUs and 50,000 unique site patterns in double precision requires approximately 10 GB of GPU memory. Two RTX A6000 cards enable simultaneous analysis of multiple partitions or independent MCMC chains, and BEAST can partition datasets across multiple BEAGLE instances, one per GPU.

8.3.5 Storage Architecture

The proposed storage architecture follows a three-tier design:

1. **Tier 1 — NVMe SSD (fast scratch):** Use for active analysis directories. Raw read processing, assembly, and alignment are I/O-intensive. NVMe units (ideally in RAID-0 or ZFS stripe for throughput) prevent storage from becoming a bottleneck. Aim for at least 4–8 TB of fast scratch per node.

2. **Tier 2 — SATA SSD or HDD RAID (warm storage):** Intermediate results, completed alignments, and tree files that need to be accessible but are not under active computation. 20–50 TB is a reasonable starting point.

3. **Tier 3 — Cold/archival (tape, cloud cold-tier):** Raw sequenced reads (FASTQ) after QC and assembly. These are large but rarely re-accessed. Compressed storage here is cost-effective.

8.3.6 Network Fabric

For a multi-node cluster, if your workflow parallelizes *across* nodes (*e.g.*, MPI-based analyses like ExaBayes; Aberer et al. (2014)), you need low-latency interconnect. InfiniBand (HDR, 200 Gb/s) is

the gold standard. If cross-node MPI jobs are rare, 25 GbE or even 10 GbE is sufficient and much more affordable.

For most phylogenomic labs, InfiniBand can be entirely avoided. The dominant parallelism pattern is "many independent jobs across loci," which requires a job scheduler — not MPI.

8.3.7 Financial Optimization: Total Cost of Ownership

Hardware purchase price is only 40–60% of the total cost of 5 years of ownership. The remainder is electricity, cooling, maintenance, and staff time.

Electricity and Cooling

A dual-socket server with high-end CPUs consumes 800–1500 W under load. At an average U.S. institutional rate of $0.10/kWh, a single node running 1 kW continuously costs $876/year in electricity alone — before cooling overhead. Cooling typically adds 30–60% to electricity cost (expressed as Power Usage Effectiveness, PUE). A well-designed server room achieves PUE 1.3–1.5; a repurposed office closet can reach 2.0+.

Buy vs. Cloud

For sustained and predictable workloads (daily pipeline runs, always-active shared resource), owned hardware almost always beats cloud computing in cost per compute-hour after year 1–2. For bursty workloads (occasional large assembly, annual grant deadline crunch), cloud instances (AWS, Google Cloud, Azure) avoid idle hardware costs. AWS Spot / GCP Preemptible instances are 60–90% cheaper than on-demand and tolerable for embarrassingly parallel phylogenomic jobs that checkpoint well.

A hybrid approach is increasingly the recommended strategy for academic genomics labs: own a modest base system for daily work, burst to cloud during peaks.

5-Year Total Cost of Ownership Projection

Table 8.2: 5-Year Total Cost of Ownership Projection

Category	**Year 1**	**Years 2–5 (annual)**
Hardware acquisition	$163,000–$185,000	—
Electricity (8 kW sustained)	$7,000	$7,000
Cooling (supplemental, est.)	$2,000	$2,000
System administration (0.25 FTE)	$20,000	$20,000
Warranty/replacement fund	—	$5,000
Cloud burst (contingency)	$3,000	$3,000
TOTAL	**$195,000–$217,000**	**$37,000/yr**
5-Year Total Cost: $343,000–$365,000		
Cost per user per year (15–25 users): $2,700–$4,900		

8.4 Software Stack and Environment Management

8.4.1 Operating System and Job Scheduling

All nodes will run Rocky Linux 9 (RHEL-compatible), which provides long-term stability and broad compatibility with bioinformatics software. The SLURM workload manager (Team 2023) will handle job scheduling, resource allocation, and accounting.

SLURM is the dominant scheduler in academic HPC and provides native support for GPU-aware scheduling (GRES), job arrays (essential for phylogenomic batch processing), fair-share scheduling across user groups, and detailed accounting for resource usage reporting.

8.4.2 Containerization and Reproducibility

Software environments will be managed via Apptainer containers (formerly Singularity) and the Lmod environment module system. Apptainer containers allow users to package complete software stacks (including specific versions of IQ-TREE, RAxML-NG, BEAST, MAFFT, etc.) in portable and reproducible images that can be shared and archived alongside publications.

Unlike Docker, Apptainer runs unprivileged and integrates natively with SLURM, making it suitable for shared HPC environments. Pre-built container images for common phylogenomic workflows will be maintained by the system administration team.

8.4.3 Pipeline Orchestration

Users will be encouraged to implement reproducible workflows using Snakemake (Mölder et al. 2021) or Nextflow, both integrating with SLURM to automatically decompose phylogenomic pipelines into parallelizable job steps. A departmental Snakemake template for standard phylogenomic workflows (QC → assembly → orthology → alignment → trimming → gene trees → species tree) will be provided as a starting point.

8.5 Resource Allocation Strategy

8.5.1 SLURM Partition Design

The cluster will be divided into SLURM partitions that map to hardware topology and computational profiles identified above:

Table 8.3: Recommended SLURM Partitions

Partition	Nodes	Max Walltime	Use Case
standard	Type A (4 nodes)	7 days	ML tree search, alignment
highmem	Type B (1 node)	14 days	*De novo* assembly, large concatenated analyses
gpu	Type C (1 node)	14 days	BEAST/MrBayes with BEAGLE
quick	All Type A	4 hours	QC, trimming, ASTRAL, testing

8.5.2 Fair-Share Scheduling

SLURM's fair-share scheduling algorithm will be configured with hierarchical accounts corresponding to research groups. Each PI will receive a baseline allocation (*e.g.*, 40% of cluster capacity for primary phylogenomics group, with remainder divided among collaborating groups). Fair-share

decay ensures that groups that have recently consumed disproportionate resources are deprioritized in the queue, preventing any single project from monopolizing the cluster.

8.5.3 Resource Limits and Accountability

Per-user CPU and memory limits will be enforced via SLURM fair-share scheduling. Without these, a single user running 10,000 IQ-TREE jobs will starve all others. Maximum wall-time limits per partition prevent zombie jobs from hoarding resources. SLURM accounting (`sacct`) will be used to monitor usage, justify hardware purchases in grant reports, and identify bottlenecks.

8.6 Recommended Resource Requirements by Analysis Type

Table 8.4: Recommended Resource Requests by Analysis Type

Task	Cores	RAM	Partition	Notes
fastp (per sample)	4–8	8 GB	quick	Job array; 5 min/sample
MAFFT L-INS-i (per locus)	4	4 GB	quick	Job array; sec–min/locus
IQ-TREE gene tree	4–8	4 GB	standard	Job array; 10 min–hrs/locus
IQ-TREE concat. (partitioned)	32–64	64–128 GB	standard	Single large job; hrs–days
RAxML-NG (partitioned)	32–64	64–128 GB	standard	Consider MPI for >64 cores
Trinity assembly	32	256–1024 GB	highmem	Requires Type B node
BEAST + BEAGLE (per chain)	4–8 + 1 GPU	32 GB + 48GB VRAM	gpu	Use `--gres=gpu:1`
ASTRAL-III	1–4	16 GB	quick	Usually $<$ 30 minutes

8.7 Multi-User Management

8.7.1 Account Structure

User accounts will be managed through LDAP, integrated with the university's existing identity management system. SLURM accounts will follow a hierarchical structure: a top-level "phylogenomics" account with sub-accounts for each PI group. Graduate students and postdocs will be added to their PI's sub-account, inheriting fair-share allocations and partition access.

8.7.2 Storage Quotas

Each user will receive a 50 GB home directory (backed up nightly) and 2 TB scratch allocation on the NAS. Group-level shared directories will be provisioned at 10–20 TB per PI, depending on active project needs. Scratch directories unused for more than 60 days will be marked for archival. An automated monthly report will notify users of storage consumption and approaching quota limits.

8.7.3 Security and Access Control

SSH key-based authentication will be required for all access. Two-factor authentication will be enforced for administrative operations. User data will be segregated by Unix group permissions, and sensitive datasets can be placed in restricted directories accessible only to authorized users. All nodes will be behind the departmental firewall, with access only via VPN or campus network.

8.7.4 Training and Documentation

A departmental wiki will provide documentation covering SLURM job submission, container usage, example SLURM scripts for common phylogenomic tasks, and resource allocation best practices. Bi-annual workshops will be offered for new graduate students. Power users will be designated as "cluster champions" to provide peer-level support and contribute to documentation.

8.8 Monitoring and Alerts

You cannot optimize what you do not measure.

- **Resource Monitoring:** Deploy a lightweight stack such as Prometheus + Grafana or simply use SLURM's built-in accounting plus summaries of `squeue`/`sacct`. Track CPU utilization, memory high-water marks, disk I/O throughput, and GPU utilization over time.
- **Thermal and Hardware Health:** Use IPMI/iDRAC/iLO (available on all server-grade hardware) to monitor temperatures, fan speeds, and disk health. Configure email alerts for SMART disk warnings and thermal throttling events.
- **Job Efficiency Audits:** Periodically review completed jobs. Users who request 500 GB of RAM but use 30 GB, or request 64 cores but run single-threaded, waste scheduler resources. Provide feedback — most users simply copy-paste job scripts without adjusting parameters.

8.9 Scalability Roadmap: Think in Phases

Most labs cannot (and should not) build a complete cluster on day one. A sensible progression:

1. **Phase 1 — Single high-spec workstation** (dual-socket, 256–512 GB RAM, 64–128 cores, NVMe scratch, 1 GPU). Install SLURM even here. This handles the majority of phylogenomic workflows for a 3–5 user lab.
2. **Phase 2 — Add second node** when CPU contention becomes chronic. Connect via 10/25 GbE, share data via NFS.
3. **Phase 3 — Dedicated storage server** when data volumes exceed local disks. Centralize warm and cold storage.
4. **Phase 4 — Heterogeneous cluster** with specialized nodes (high-memory, GPU, high-clock) as grants and workloads justify.
5. **Ongoing — Cloud bursting** for transient peak demand instead of over-provisioning permanent hardware.

8.10 Conclusion

Well-designed HPC systems for phylogenomics amplify the analytical capacity of research groups by orders of magnitude. The investment — initial capital, electricity, staff — is significant, but returns quickly in the form of more analyses per researcher, more trained students, and higher-quality publications. For laboratories operating in institutions with limited central HPC support, a dedicated departmental cluster is often the most pragmatic path to sustained analytical productivity.

Chapter Review Outline

1. 8 workflow phases: (1) QC/pre-proc. (I/O, fastp 4–8 cores), (2) assembly/mapping (CPU+RAM, Trinity >100 GB), (3) orthology (OrthoFinder, parallel DIAMOND), (4) alignment (MAFFT, SLURM arrays), (5) trimming (trimAl/BMGE, lightweight), (6) ML gene tree (IQ-TREE/RAxML-NG, AVX-512, 4–8 cores/locus), (7) Bayesian gene tree (BEAST+BEAGLE/GPU, 24–48 GB VRAM), (8) species tree (lightweight ASTRAL; heavy concatenation).
2. Heterogeneous design: 3 node types. Type A (4 nodes): 2×EPYC 9554 (128 cores), 512 GB DDR5. Type B (1 node): 1.5 TB RAM (assembly). Type C (1 node): 2×RTX A6000 (48 GB VRAM each).
3. CPU: EPYC 9554 — 64 cores, 3.75 GHz boost, 256 MB L3, native AVX-512 (Zen 4).
4. GPU: RTX A6000 for BEAGLE/CUDA; 10 GB VRAM for 500 OTUs × 50,000 patterns in double precision.
5. 3-tier storage: Tier 1 NVMe (fast scratch), Tier 2 SATA/HDD RAID (warm), Tier 3 cold/archival (tape/cloud).
6. Network: 25 GbE sufficient if jobs are independent per-locus; InfiniBand (HDR 200 Gb/s) only for cross-node MPI (ExaBayes).
7. SLURM: partitions `standard` (Type A, 7d), `highmem` (Type B, 14d), `gpu` (Type C, 14d), `quick` (4h); hierarchical fair-share per PI.
8. 5-year TCO: hardware $163–185k + electricity $7k/yr + cooling $2k/yr + sysadmin 0.25 FTE $20k/yr + contingency = **$343–365k total**; $2,700–4,900/user/yr.
9. Software: Rocky Linux 9, Apptainer (Singularity) for containers, Snakemake/Nextflow for pipelines, Lmod for modules.
10. Scalability roadmap: Phase 1 (single workstation + SLURM) → Phase 2 (2nd node, NFS) → Phase 3 (dedicated NAS) → Phase 4 (heterogeneous cluster) → ongoing cloud bursting.

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Chapter 9

Omics Data Management

Abstract

Computational analyses are now central to virtually all omics research, yet the computational component is frequently the weakest link in scientific reproducibility. This chapter examines the theoretical foundations of reproducibility, the FAIR principles (Findable, Accessible, Interoperable, Reusable) for research data and software management, and practical implementation of these principles in phylogenomics and multi-omics contexts. Special attention is given to technical computing limitations—hardware-dependent behavior, memory and storage constraints, the cloud computing paradox—that affect reproducibility even when best practices are followed. We provide a practical checklist for making phylogenomic projects FAIR and discuss connections between HPC system design and FAIR compliance.

9.1 Introduction: The Reproducibility Problem in Computational Biology

Computational analyses are now central to virtually all omics research, yet the computational component is frequently the weakest link in scientific reproducibility. The problem has multiple interacting dimensions.

9.1.1 Software Opacity and Fragility

Many bioinformatics tools are poorly documented, version-sensitive, and distributed without containerized environments, meaning that installing the same tool six months later on a different operating system may produce different results—or fail entirely. Martín del Pico et al. (2024) evaluated FAIRness across research software in life sciences and found that reproducibility and verifiability of computational components remain systemically deficient. Zafeiropoulos et al. (2021) documented this from the facility side: their HPC system accumulated over 200 software packages with frequent and non-periodic versioning and broken dependency chains, motivating their transition to containerized pipelines.

9.1.2 Parameter and Configuration Sensitivity

Powers et al. (2022) demonstrated that something as apparently routine as thread allocation per task or the choice of compression level in GATK can materially change runtime and, in the case

of scatter-gather configurations, potentially introduce artifacts. These configuration decisions are rarely reported in methods sections.

9.1.3 Neglect in Data Management

Raw sequenced data, intermediate files, and final results are frequently stored in ad hoc directory structures without metadata, without versioning, and without data lifecycle policy. When a student graduates, their analysis history may be effectively lost.

9.1.4 Scale of the Problem

Kim et al. (2018) conducted a case study in reproducibility in bioinformatics and found that even well-documented analyses could not be reliably replicated due to environment differences and dependency issues. This is not a marginal concern—it affects the fundamental reliability of published results.

9.2 The FAIR Principles: A Framework for Data Stewardship

9.2.1 Origin and Scope

The FAIR Guiding Principles were formalized by Wilkinson et al. (2016) as a set of guidelines to improve Findability, Accessibility, Interoperability, and Reusability of digital assets—including data, metadata, and the infrastructure that supports them. A key emphasis of the original paper is *machine actionability*: the ability of computational systems to find, access, interoperate with, and reuse data with minimal or no human intervention, motivated by the accelerating volume and complexity of research data.

9.2.2 The Fifteen Sub-Principles

Findable (F)

Data must be discoverable by both humans and machines.

- F1: (Meta)data are assigned globally unique and persistent identifiers (*e.g.*, DOIs, accession numbers).
- F2: Data are described with rich metadata.
- F3: Metadata explicitly include the identifier of the data they describe.
- F4: (Meta)data are registered or indexed in a searchable resource (*e.g.*, NCBI, ENA, Zenodo).

Accessible (A)

Once found, data must be retrievable under clear conditions.

- A1: (Meta)data are retrievable by their identifier using a standardized communication protocol (*e.g.*, HTTP, FTP).
- A1.1: The protocol is open, free, and universally implementable.

- A1.2: The protocol allows for authentication and authorization where necessary.
- A2: Metadata remain accessible even when the data themselves are no longer available. This is critical for long-term scientific preservation.

Interoperable (I)

Data must be integrable with other data and compatible with applications and workflows.

- I1: (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation (*e.g.*, RDF, OWL, standardized ontologies).
- I2: (Meta)data use vocabularies that themselves follow the FAIR principles.
- I3: (Meta)data include qualified references to other (meta)data, enabling construction of knowledge graphs.

Reusable (R)

Data must be well-described so they can be replicated and combined in different contexts.

- R1: (Meta)data are richly described with a plurality of accurate and relevant attributes.
- R1.1: (Meta)data are released with a clear and accessible data use license (*e.g.*, CC-BY, CC0).
- R1.2: (Meta)data are associated with detailed provenance (who generated the data, how, when, with what parameters).
- R1.3: (Meta)data conform to relevant community domain standards (*e.g.*, MIGS/MIMS for genomics, ISA-Tab for multi-omics).

9.2.3 FAIR Applies to Three Entities

The principles address three distinct types of entities: **data** (or any digital object), **metadata** (information about a digital object), and **infrastructure** (systems that host, index, and serve these objects). For example, principle F4 requires that both metadata and data be registered in a searchable resource—the infrastructure component. This three-way scope is important because many researchers treat FAIR as merely a data labeling exercise, when in fact it demands corresponding investments in infrastructure and metadata services.

9.3 FAIR Beyond Data: Principles for Research Software

The original FAIR principles target data and metadata. Recognizing that research software is equally critical and equally fragile, Barker et al. (2022) proposed FAIR4RS—an adaptation of FAIR principles for research software. The key adaptations include:

- Software should have persistent identifiers (such as DOIs via Zenodo or Software Heritage).
- Software should be described with rich metadata including dependencies, runtime environment, and version history.

- Software should use formats and standards accepted by the community, interoperate with other tools, and be released under clear licenses.

Martín del Pico et al. (2024) operationalized this with FAIRsoft, providing a practical evaluation framework that assesses research software across these dimensions. Periodic evaluations of bioinformatics tools revealed persistent gaps in documentation, licensing, and source code availability.

For omics research, the implication is direct: a phylogenomic pipeline is only as reproducible as its least FAIR component. If IQ-TREE is well-versioned and deposited in Bioconda but your custom filtering script lives unversioned on a lab desktop, the overall analysis is not reproducible.

9.4 Implementing FAIR in Omics Data Management

9.4.1 Data Repositories and Identifiers

In genomics and phylogenomics, primary repositories already impose portions of the FAIR principles:

- **NCBI SRA / ENA / DDBJ** for raw sequenced reads—accession numbers serve as persistent identifiers (F1); metadata fields capture organism, sequencing platform, and study design (F2, R1); and the International Nucleotide Sequence Database Collaboration ensures global indexing (F4).
- **TreeBASE / Dryad / Zenodo** for phylogenetic trees, alignments, and supplementary data—DOIs provide persistence (F1), and Zenodo integration with GitHub enables archival of versioned software.
- **NCBI GenBank / UniProt / Ensembl** for assembled sequences and annotations.

The gap is typically not at the repository level but at the lab level: researchers frequently fail to deposit intermediate analysis products (filtered alignments, partition schemes, model selection outputs, bootstrap trees) that would be necessary to truly reproduce a phylogenomic study.

9.4.2 Metadata: The Perennial Weak Link

Rich metadata is the critical link in FAIR, yet it remains the most neglected aspect in practice. For phylogenomics, essential metadata that is frequently missing includes:

- Exact software versions and command-line parameters for each pipeline step.
- Alignment masking or filtering criteria (*e.g.*, Gblocks settings, trimAl thresholds).
- Model selection results (which substitution model was chosen per locus, and by which criterion).
- Description of the computational environment (operating system, library versions, container image hash).
- Provenance linkage between raw data, intermediate products, and final results.

Niehues et al. (2024) demonstrated a practical approach to this problem by wrapping a multi-omics analysis workflow as a FAIR Digital Object, embedding metadata, data, and computational workflow together in a single reusable package. They used RO-Crate as the packaging format, Snakemake as the workflow engine, and Git for version control—providing a concrete model for how phylogenomic labs could structure their own analyses.

9.4.3 Workflow Managers as FAIR Enablers

Workflow managers serve a dual purpose: they enforce reproducibility by codifying analysis steps, and they generate machine-readable provenance metadata as a side effect of execution. The two dominant tools in bioinformatics are:

- **Snakemake** (Mölder et al. 2021): Python-based, integrates with Conda and Singularity/Apptainer, generates execution DAGs and logs.
- **Nextflow** (Di Tommaso et al. 2017): Groovy-based, designed for portability across HPC, cloud, and local environments; the nf-core community (Ewels et al. 2020) provides curated, peer-reviewed bioinformatics pipelines.

Both produce records of what was executed, with what parameters, in what order, and on what data—directly addressing FAIR principles R1.2 (provenance) and R1.3 (community standards).

9.4.4 Containers as Reproducibility Infrastructure

Containers freeze a software environment into a portable image. In the context of FAIR:

- **Singularity/Apptainer** is preferred for HPC because it executes without root privilege and integrates with SLURM. Zafeiropoulos et al. (2021) adopted Singularity as their standard container format for its compatibility with Docker images and suitability for multi-user HPC environments.
- **Docker** is more common in cloud and development contexts; nf-core pipelines ship as Docker images.
- Container image hashes (SHA256 digests) serve as persistent identifiers for the exact computational environment, satisfying F1 for software environments.

9.5 Technical Computing Limitations and Their Impact on Reproducibility

9.5.1 Hardware-Dependent Behavior

Computational results in bioinformatics are not always bit-for-bit identical across different hardware, even with identical software. Sources of non-determinism include:

- **Floating-point arithmetic:** Different CPU architectures (Intel vs. AMD, different SIMD instruction sets such as AVX2 vs. AVX-512) may produce slightly different floating-point results, which can propagate through likelihood calculations in phylogenetic inference. Powers et al. (2022) demonstrated that the Genomics Kernel Library (GKL) in GATK uses AVX-512 instructions for PairHMM and Smith-Waterman kernels, reducing HaplotypeCaller runtime from 35 hours to under 1 hour—but this acceleration is architecture-dependent.
- **Thread-order non-determinism:** Multi-threaded programs that reduce results across threads (*e.g.*, summing log-likelihoods) may accumulate floating-point values in different orders depending on thread scheduling, producing slightly different sums. IQ-TREE, RAxML-NG, and BEAST all exhibit this behavior to varying degrees.

- **Random seed sensitivity:** Heuristic tree search algorithms depend on initial trees generated from random seeds. Reporting and fixing the random seed is essential for reproducibility.

9.5.2 Memory and Storage Constraints

As documented by Zafeiropoulos et al. (2021), storage is the most universal and persistent constraint across all types of bioinformatics analysis. Pipeline tools generate unpredictably large intermediate files, causing job failures when quotas are reached. This has direct reproducibility implications: if a job fails mid-way and is restarted with different parameters or on a different data subset, the final result may differ from what would have been produced by an uninterrupted execution.

Ghedira et al. (2024) further noted that in resource-limited environments, lack of adequate storage may force researchers to prematurely delete intermediate files, losing the provenance chain necessary for reproducibility.

9.5.3 The Cloud Computing Paradox

Cloud computing (AWS, GCP, Azure) offers elasticity and avoids hardware maintenance, but introduces its own reproducibility challenges:

- Instance types change over time; a VM configuration available today may not exist in two years.
- Spot/preemptible instances can be terminated mid-computation.
- Data egress costs may discourage sharing of large datasets.
- Software environments in cloud VMs are ephemeral unless containerized.

Langmead and Nellore (2018) reviewed cloud computing for genomics and argued that a hybrid approach—owning base infrastructure and bursting to cloud during peaks—provides the best balance of cost, control, and reproducibility. However, this requires disciplined use of containers and workflow managers to ensure that results are consistent across local and cloud environments.

9.6 Practical Checklist: Making a Phylogenomic Project FAIR

The following checklist translates FAIR principles into concrete actions for a typical phylogenomic study:

9.6.1 Before Analysis

Register your project in a repository (*e.g.*, OSF, Zenodo) and obtain a DOI.

Deposit raw sequenced reads in NCBI SRA or ENA before beginning analysis.

Define your computational environment in a container (Singularity/Docker) or Conda `environment.yml` file.

Write your pipeline in a workflow manager (Snakemake or Nextflow) rather than an ad hoc series of shell scripts.

9.6.2 During Analysis

Record all software versions, parameters, and random seeds in version-controlled configuration files with Git.

Use SLURM accounting database (`sacct`) to log computational resource usage.

Store intermediate files in a structured directory hierarchy with descriptive naming conventions.

Use checksums (MD5 or SHA256) on key input and output files to verify data integrity.

9.6.3 After Analysis

Deposit alignments, trees, model selection results, and partition schemes in a public repository (Dryad, Zenodo, TreeBASE).

Include a `README.md` or metadata file describing the provenance of each deposited file.

Release your analysis pipeline (Snakemake/Nextflow + container definition) under an open-source license on GitHub/GitLab.

Archive the repository in Zenodo to obtain a DOI for the software.

Apply a clear data license (CC-BY or CC0) to all deposited data.

9.7 Key Concepts for Further Study

9.7.1 Ontologies and Controlled Vocabularies in Omics

FAIR interoperability (I1, I2) requires shared vocabularies. Key resources include:

- **Gene Ontology (GO):** Standardized terms for gene function across species.
- **Sequence Ontology (SO):** Terms for sequence features and annotations.
- **Environment Ontology (ENVO):** Terms for environmental contexts (relevant for metagenomics/metabarcoding).
- **EDAM Ontology:** A comprehensive ontology of bioinformatics operations, data types, formats, and topics—directly relevant for describing computational workflows.

9.7.2 Data Management Plans (DMPs)

Funding agencies (NIH, NSF, ERC, UKRI) increasingly require DMPs that specify how data will be handled, shared, and preserved. The NIH 2023 Data Management and Sharing Policy explicitly references FAIR principles. A DMP should address: what data will be generated, what metadata standards will be used, where data will be deposited, who will have access, and how long data will be preserved.

9.7.3 The Distinction Between Open and FAIR

FAIR does not require data to be open or free. Principle A1.2 explicitly allows for authentication and authorization. Sensitive data (*e.g.*, human genomic data under IRB/ethics restrictions) can be FAIR—discoverable and described with rich metadata—while remaining access-controlled. Metadata should always be open (A2), even if the data themselves are restricted.

9.7.4 Computational Provenance Standards

- **W3C PROV:** A general-purpose provenance model.
- **RO-Crate:** A lightweight approach to packaging research data with metadata, used by Niehues et al. (2024) for multi-omics workflows.
- **CWLProv / WES:** Standards for recording provenance from workflow executions in the Common Workflow Language ecosystem.

9.8 Connections to HPC System Design

The FAIR principles and HPC system design considerations from our companion chapter (*Design of High-Performance Computing Systems for Phylogenomics*) intersect at several points:

- **Storage tiering** (§4 of companion) directly supports FAIR data lifecycle management: scratch for active analysis, warm storage for intermediate results with metadata, cold storage/repositories for FAIR-compliant and archived deposits.
- **SLURM accounting** (§8 of companion) generates computational provenance automatically—job IDs, resource usage, wall times, and exit codes are logged and can be attached as metadata to analysis results.
- **Containerization** (§9 of companion) is simultaneously an HPC software management strategy and a FAIR reproducibility mechanism.
- **Training** (§10 of companion) is essential for both HPC use and FAIR adoption; researchers who do not understand SLURM will not write reproducible job scripts, and researchers who do not understand FAIR will not deposit metadata.
- **Workflow managers** (§9 of companion) enforce efficient resource utilization on HPC (allowing SLURM to schedule tasks optimally) and FAIR compliance (generating machine-readable provenance records).

9.9 Conclusion

Reproducibility in omics research is not a purely theoretical exercise. It has direct practical implications for scientific trust, the ability of other researchers to build upon your work, and the long-term validity of discoveries. The FAIR principles provide a framework for transforming historical neglect in data management into systematic practice. Implementation requires investment in infrastructure (repositories, indexing, file systems with lifecycle management), documentation

(rich metadata, community standards), and researcher behavior (use of workflow managers, code versioning, data deposition).

For phylogenomics specifically, adoption of FAIR practices—from deposition of raw reads through deposition of alignments and final trees—transforms an individual study into a reusable contribution to collective phylogenetic science. Each carefully documented and deposited analysis becomes a starting point for future research, comparison, or validation. This is the future of phylogenomics—not studies in vacuo, but synthesis of accumulated knowledge, enabled by rigorous data management.

Chapter Review Outline

1. Reproducibility problem: software opacity, parameter/configuration sensitivity (Powers et al., 2022), data management neglect, loss of history when students graduate.
2. FAIR (Wilkinson et al., 2016): Findable, Accessible, Interoperable, Reusable; emphasis on **machine actionability**; applies to data, metadata *and* infrastructure.
3. 15 sub-principles: F1 (persistent IDs: DOIs), F2 (rich metadata), F3 (ID in metadata), F4 (indexed/searchable); A1 (standardized protocol), A1.1 (open), A1.2 (authentication), A2 (metadata persist without data); I1 (formal language: RDF/OWL), I2 (FAIR vocabularies), I3 (qualified references); R1 (rich attributes), R1.1 (clear license: CC-BY/CC0), R1.2 (provenance), R1.3 (community standards: MIGS/MIMS).
4. FAIR $\neq$ open: A1.2 allows access control; metadata always open (A2).
5. FAIR4RS (Barker et al., 2022): adaptation for research software; DOIs via Zenodo/Software Heritage; metadata for dependencies and environment.
6. Repositories: NCBI SRA/ENA/DDBJ (raw reads), TreeBASE/Dryad/Zenodo (trees/alignments), GenBank/UniProt/Ensembl (assembled sequences).
7. Workflow managers: Snakemake (Python, Conda/Singularity, DAGs) and Nextflow (Groovy, nf-core); generate provenance automatically (R1.2, R1.3).
8. Containers: Apptainer (rootless, SLURM integration) for HPC; Docker for cloud; SHA256 hash = persistent ID for environment.
9. Technical reproducibility limitations: floating-point arithmetic (AVX2 vs. AVX-512), thread-order non-determinism, random seed sensitivity.
10. Cloud paradox: elasticity but instance types change, spot may be terminated, egress costs, ephemeral environments; solution = hybrid approach (Langmead and Nellore 2018).
11. Key ontologies: GO (gene function), SO (sequence features), ENVO (environments), EDAM (bioinformatics operations). Provenance: W3C PROV, RO-Crate, CWLProv.

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Chapter 10

Artificial Intelligence in Contemporary Biomedical Science

10.1 Introduction

Artificial intelligence (AI) has rapidly transformed into a central methodological framework within modern biomedical science. In the last decade, AI applications in biomedicine have expanded to encompass stem cell differentiation modeling, target identification for gene therapy, microbial image synthesis, vaccine epitope design, protein structure prediction, and clinical decision support systems, among many others (Jumper et al. 2021; Topol 2019). This expansion has been driven by three converging developments: exponential growth of multimodal biomedical datasets, maturation of deep learning architectures, and widespread availability of high-performance computational resources.

Recent work in cellular therapy demonstrates how AI can model dynamic gene regulatory networks to guide stem cell differentiation, integrating synthetic biology with predictive modeling (Choudhury et al. 2025). Similarly, computational biology and machine learning approaches have enhanced workflows in gene therapy, including target identification, vector design, and outcome prediction (Danaeifar and Najafi 2025). In structural biology, the development of AlphaFold2 (Jumper et al. 2021) marked a crucial moment, demonstrating that deep learning could solve one of the field's oldest problems — the protein folding problem — with near-experimental accuracy. More recently, generative approaches have been applied to design novel proteins and nucleic acid constructs with defined functional properties (Ferruz et al. 2022).

Despite these advances, the rapid expansion of AI methodologies has raised important theoretical and practical questions that the scientific community has not yet fully resolved. Central among these are the following: Are current AI systems genuinely explanatory, or are they sophisticated pattern recognizers without mechanistic insights? Can deep learning models be trusted in high-risk clinical contexts where errors carry life-altering consequences? How should uncertainty be quantified, propagated, and communicated to end users? And to what extent do biases encoded in training data compromise the generalizability and fairness of AI-guided recommendations?

These questions collectively define the contemporary methodological debate in biomedical AI. The present chapter examines this debate through the major paradigms — supervised and unsupervised learning, deep learning and representation learning, generative models and synthetic data, Bayesian and probabilistic inference, large language models and hybrid systems and AutoML — with particular attention to both demonstrated scientific potential and unresolved challenges.

Our treatment emphasizes epistemic responsibility: the responsibility of researchers not only to understand what their models predict, but why, under what conditions, and with what degree of confidence.

10.2 Supervised and Unsupervised Machine Learning in Biomedicine

10.2.1 Supervised Learning: Applications and Scope

Methods of supervised learning — including support vector machines (SVMs), random forests, gradient boosted trees, and feedforward neural networks — have become central to analyses in genomics, transcriptomics, proteomics, and clinical informatics (Libbrecht and Noble 2015). In genomics, supervised classifiers trained on labeled sequence data have been applied to identification of pathogenic single nucleotide polymorphisms (SNPs), prediction of splice sites, and classification of regulatory elements. In transcriptomics, machine learning models trained on RNA-seq data enable differential expression analysis, cell type deconvolution, and prediction of phenotype at the transcriptome level.

In gene therapy research, supervised learning has been applied to all phases of the therapeutic pipeline. At the target identification stage, models trained on curated disease-gene association databases — such as DisGeNET or the OMIM catalog — predict novel therapeutic targets from multi-omic profiles (Danaeifar and Najafi 2025). At the vector design stage, classifiers predict the efficiency and specificity of adenoassociated virus (AAV) serotypes for particular tissue tropism based on capsid sequence features. At the outcome prediction stage, models trained on patient clinical and genomic data estimate therapeutic response, risk of adverse event, and long-term durability of gene correction.

Proteomics has similarly benefited from supervised learning approaches. Mass spectrometry data processed through machine learning pipelines now enable high-throughput protein identification, quantification, and mapping of post-translational modifications, enabling large-scale profiling of disease-associated proteomes (Orre et al. 2019). In clinical imaging, supervised convolutional neural networks (CNNs) achieved performance comparable to radiologists on tasks such as tumor detection, classification, and segmentation across multiple cancer types (Litjens et al. 2017).

10.2.2 Unsupervised Learning and Discovery of Latent Structure

Unsupervised learning techniques — including clustering algorithms, dimensionality reduction methods, and hierarchical modeling — enable discovery of latent biological structure in high-dimensional omic datasets without the need for pre-labeled training data. Principal component analysis (PCA), t-distributed stochastic neighbor embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP) have become standard tools for visualization of single-cell RNA-seq (scRNA-seq) data, enabling researchers to identify transcriptionally distinct cell populations with unprecedented resolution (Becht et al. 2018; McInnes et al. 2018).

Weighted gene co-expression network analysis (WGCNA) exemplifies a biologically motivated unsupervised approach that reconstructs networks of co-regulated genes from expression data, associating network modules with clinical phenotypes or experimental conditions (Langfelder and Horvath 2008). These module-trait associations have been used to nominate gene regulatory hubs as therapeutic targets across a range of diseases, including neurodegenerative disorders, cancer, and

metabolic disease.

In the context of cellular therapy, unsupervised methods have been applied to characterize the heterogeneity of stem cell populations, identify rare subpopulations with distinct differentiation potentials, and map trajectories of cell state transitions during reprogramming (Saelens et al. 2019). Integration of multimodal single-cell data — combining transcriptomics, chromatin accessibility (scATAC-seq), and proteomics — presents ongoing challenges for unsupervised analysis, motivating the development of new methods for dimensionality reduction and data integration such as MOFA+ and Seurat v4 (Argelaguet et al. 2020; Y. Hao et al. 2021).

10.2.3 Debates: Generalizability, Bias, and Uncertainty

The debate surrounding supervised and unsupervised approaches centers on three interrelated concerns: generalizability across populations and contexts, encoding of systematic bias, and inadequate quantification of predictive uncertainty. Many biomedical machine learning models are trained on datasets derived from non-representative population samples — often predominantly European ancestry — which substantially limits their external validity and clinical utility in underrepresented groups (Obermeyer et al. 2019; Popejoy and Fullerton 2016). This concern is particularly acute for polygenic risk scores, which have been demonstrated to exhibit substantial performance disparities across ancestry groups when derived from eurocentric genome-wide association study (GWAS) cohorts (Martin et al. 2019).

Performance metrics in published studies of biomedical machine learning frequently emphasize aggregate accuracy, area under the receiver operating characteristic curve (AUROC), or related summary statistics, without sufficient attention to calibration — the degree to which predicted probabilities correspond to empirical event frequencies. Well-calibrated uncertainty estimates are essential for clinical deployment, where a model expressing 90% confidence in an incorrect diagnosis carries very different implications from one expressing 55% confidence (Guo et al. 2017). Prospective validation in diverse and independent cohorts under realistic operational conditions remains the exception rather than the rule in published literature (Wynants et al. 2020).

10.3 Deep Learning and Representation Learning

10.3.1 Architectures and Biomedical Applications

Deep learning has enabled automated and hierarchical extraction of features from complex, high-dimensional biological data, fundamentally altering research workflows in imaging, genomics, and structural biology. Convolutional neural networks (CNNs) have been applied to automated histopathological image analysis, demonstrating that deep architectures can identify morphological features predictive of cancer subtype, prognosis, and treatment response at a scale that would be impractical for human annotators (Esteva et al. 2017; Kather et al. 2019). In hematology, CNNs trained on microscopy images of blood cells enable automated classification of hematopoietic cell types and identification of abnormal populations relevant to leukemia diagnosis (Matek et al. 2019).

In the domain of cellular therapy, deep learning supports quality control of stem cell cultures by detecting morphological deviations from expected differentiation trajectories in time-lapse image data. Mesenchymal stem cell (MSC) preparations, critical for a range of regenerative medicine applications, have been characterized using deep CNNs trained to predict immunomodulatory

potency from phase-contrast microscopy images, reducing dependence on costly and time-consuming functional assays (Kusumoto et al. 2021).

Recurrent architectures, including long short-term memory (LSTM) networks and transformer-based models, have been applied to genomic sequence modeling, prediction of gene expression from chromatin state, and analysis of clinical time series. The transformer architecture, originally developed for natural language processing, has been adapted for genomics in models such as Enformer (Avsec et al. 2021), which predicts gene expression and chromatin accessibility from DNA sequence with substantially improved long-range interaction modeling compared to previous convolutional architectures.

A notable application of representation learning in microbiology involves generation of synthetic microbial colony datasets using deep learning style transfer (Pawłowski et al. 2022). By transferring visual style features from real colony images to synthetically generated backgrounds, the authors were able to substantially expand training corpora for automated colony detection models without requiring additional manual annotation, while maintaining competitive detection performance. This work illustrates how representation learning techniques can directly address the widespread bottleneck of limited labeled data in biomedical image analysis.

10.3.2 Interpretability and the Black Box Problem

Despite these impressive empirical achievements, deep neural networks remain largely opaque to mechanistic interpretation. The representations learned and encoded in hidden layers of a deep network are not, in general, directly mappable to scientifically meaningful constructs, making it difficult to understand why a given prediction was made or to identify the biological features driving model behavior. This interpretability deficit is a central concern in clinical translation, where regulatory agencies, clinicians, and patients may reasonably demand explanations for algorithmic recommendations (Rudin 2019).

A range of post-hoc interpretability methods have been developed to partially address this limitation, including gradient-based saliency maps, layer-wise relevance propagation (LRP), integrated gradients, and SHAP values (SHapley Additive exPlanations) (Lundberg and S. I. Lee 2017; Sundararajan et al. 2017). These methods provide approximate attributions of model predictions to input features, but are not without limitations. Saliency-based methods have been shown to be sensitive to model architecture and initialization, producing inconsistent attributions across ostensibly equivalent models (Adebayo et al. 2018). Furthermore, attribution of features at the input level does not necessarily illuminate the internal computational mechanisms by which predictions are generated.

There is growing recognition within the field that intrinsically interpretable models — which are designed to be transparent by construction, rather than explained post hoc — may be preferable to black box alternatives in high-risk domains, even at some cost to predictive performance (Rudin 2019). The debate continues regarding the degree to which interpretability and predictive accuracy are fundamentally in tension, and whether advances in architecture design can produce models that are both powerful and mechanistically transparent.

10.4 Generative Models and Synthetic Data

10.4.1 Generative Adversarial Networks and Variational Autoencoders

Generative models introduced a new paradigm in biomedical AI, enabling synthesis of realistic data samples, design of novel molecular entities, and expansion of training datasets. Generative adversarial networks (GANs), introduced by Goodfellow et al. (2014), have been applied to synthesis of histopathological images, expansion of MRI datasets, generation of realistic electronic health record (EHR) data, and de novo design of molecular structures with desired physicochemical properties (Baowaly et al. 2019; Kadurin et al. 2017).

Variational autoencoders (VAEs) provide an alternative generative framework that imposes a structured probabilistic prior on the latent space, enabling controlled generation and interpretable interpolation between data points. VAEs have been applied in drug discovery to navigate chemical space toward regions predicted to exhibit desired properties of potency, selectivity, and pharmacokinetics (Gomez-Bombarelli et al. 2018). Hybrid architectures combining the representational power of GANs with the latent structure of VAEs have further expanded the generative toolkit available to biomedical researchers.

10.4.2 Protein and Nucleic Acid Design

Perhaps the most scientifically significant application of generative models in biomedicine is in structural and functional protein design. Building on the representational foundations established by AlphaFold2 (Jumper et al. 2021), generative protein language models such as ProtGPT2 (Ferruz et al. 2022) and ESMFold (Lin et al. 2023) have enabled design of novel protein sequences with predicted structural properties. These tools reduce the combinatorial search space of protein engineering from an astronomically large sequence space to a tractable design problem, with implications for enzyme engineering, therapeutic antibody design, and synthetic biology.

In vaccine development, generative and predictive modeling are increasingly integrated across the vaccine design pipeline. AI-guided epitope prediction tools identify immunogenic peptide candidates from pathogen proteomes, while mRNA structure optimization algorithms design sequences that maximize translational efficiency and stability for mRNA vaccine platforms (Imani et al. 2025). Diffusion model-based approaches — analogous to those used in image generation — have been applied to design protein linkers with specified target interfaces, representing a cutting-edge application of generative AI in therapeutic protein engineering (Watson et al. 2023).

10.4.3 Synthetic Data: Promises and Risks

Synthetic data generation addresses a critical bottleneck in biomedical AI: the scarcity of large, annotated, and privacy-compliant datasets. Privacy regulations — including the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union — impose substantial restrictions on sharing and secondary use of patient data, limiting the scale and diversity of training corpora available to researchers. Synthetic data generated by GAN or diffusion models can, in principle, be shared without direct privacy risk while preserving the statistical properties of the underlying clinical population.

Nevertheless, synthetic datasets are not a panacea. Generative models trained on non-representative

source data will faithfully reproduce, and may even amplify, the biases present in those data. Evaluation of synthetic data quality requires careful comparison to real data distributions in clinically meaningful dimensions, and there is evidence that naively evaluated realism may mask systematic distortions in rates of rare events, minority subgroup characteristics, and outcome distributions (Walonoski et al. 2018). Furthermore, the use of synthetic data for regulatory submission or clinical validation remains an evolving area, with guidance from agencies such as the FDA still being formalized (U.S. Food and Drug Administration 2023).

10.5 Bayesian Networks and AI-Guided Probabilistic Inference

10.5.1 Principled Modeling of Uncertainty

Bayesian approaches offer a mathematically principled framework for modeling uncertainty in biomedical AI, integrating prior domain knowledge with observed data through Bayes' theorem to produce posterior probability distributions over parameters and predictions. Unlike point-estimate discriminative models, Bayesian inference explicitly represents and propagates uncertainty, making it well-suited to high-risk biomedical applications where the cost of overconfident predictions is substantial (Ghahramani 2015).

In gene therapy, Bayesian networks have been applied to model the probabilistic dependencies among genomic, transcriptomic, and clinical variables, enabling principled target prioritization under uncertainty (Danaeifar and Najafi 2025). In epidemiology and pharmacovigilance, hierarchical Bayesian models are used to estimate adverse event rates from spontaneous reporting systems, sharing statistical strength across related drug-event combinations while appropriately representing the uncertainty associated with rare outcomes. In vaccine development, Bayesian adaptive trial designs use interim data to update prior beliefs about vaccine efficacy and safety, enabling more efficient use of clinical trial resources (Berry et al. 2010).

10.5.2 Structure Learning and Causal Inference

Beyond its utility for uncertainty quantification, Bayesian networks offer a framework for representation and learning of causal dependencies among variables — a property that aligns more closely with the explanatory goals of scientific investigation than purely associative models. Causal models based on directed acyclic graphs (DAGs) encode hypothesized causal relationships among biological variables and can, under appropriate assumptions about data-generating mechanisms, be used to predict the consequences of experimental interventions (Pearl 2009).

Causal discovery algorithms — such as the PC algorithm, FCI, and more recent variants augmented with deep learning — attempt to infer causal graph structure from observational data, a notoriously challenging problem requiring strong assumptions (Spirtes et al. 2000). In single-cell genomics, causal discovery has been applied to inference of gene regulatory networks from scRNA-seq data, with the goal of identifying target-transcription factor relationships that can be experimentally validated and therapeutically exploited (Skok Gibbs et al. 2022).

10.5.3 Hybrid Probabilistic-Deep Learning Models

The contemporary debate in this area centers on whether hybrid models that combine deep learning representations with probabilistic graphical model inference can achieve superior interpretability

and uncertainty quantification without sacrificing the representational power that has made deep learning so empirically successful. Approaches such as deep kernel learning, neural process models, and Bayesian neural networks represent attempts to integrate these paradigms (Wilson and Izmailov 2020). Gaussian process regression, a Bayesian nonparametric approach, has been applied to model gene expression dynamics as a function of pseudotime in single-cell trajectory analysis, providing principled uncertainty bounds on inferred differentiation trajectories (Lönnberg et al. 2017). The practical challenge of scaling Bayesian inference to the parameter regimes of modern deep learning architectures — often involving billions of parameters — remains an active area of research.

10.6 Large Language Models in Scientific Workflows

10.6.1 Applications in Biomedical Research

Large language models (LLMs) — including GPT-4 (OpenAI 2023), Gemini (Google DeepMind 2023), and domain-adapted models such as BioMedLM, BioGPT (Luo et al. 2022), and Med-PaLM 2 (Singhal et al. 2023) — are increasingly integrated into scientific workflows. These models, trained on vast corpora of biomedical literature, clinical notes, and genomic sequences, exhibit emergent capabilities including literature synthesis, hypothesis generation, answering clinical questions, and automation of bioinformatics pipelines. In biomedical named entity recognition (NER), relation extraction, and clinical note processing, LLMs have achieved state-of-the-art performance on established benchmarks, often without task-specific fine-tuning (J. Wei et al. 2022).

Protein language models (PLMs) represent a related family of models trained not on natural language but on large databases of amino acid sequences. Models such as ESM-2 (Lin et al. 2023) and ProtTrans (Elnaggar et al. 2022) learn contextualized representations of amino acid positions that encode evolutionary, structural, and functional information, and have been used to predict the effects of mutations on protein stability and function, classify protein families, and guide the generation of novel sequences. The success of PLMs illustrates how the transformer architecture originally developed for language can be productively adapted to biological sequence data.

10.6.2 Epistemic Limitations and Risks

Despite their capabilities, LLMs exhibit significant limitations that are particularly consequential in biomedical contexts. Hallucination — the generation of plausible but factually incorrect or fabricated information — is a well-documented failure mode of current autoregressive language models (Ji et al. 2023). In the context of scientific literature synthesis, LLMs have been observed generating plausible-appearing but nonexistent citations, attributing fabricated claims to legitimate-appearing authors and journals. This poses a non-trivial risk to research integrity when LLM outputs are incorporated into manuscripts or systematic reviews without independent verification.

LLMs also lack explicit epistemic grounding: they do not maintain formal representations of knowledge or inference mechanisms that distinguish known facts from plausible interpolations of training data. They cannot reliably distinguish between well-established scientific consensus and contested or preliminary findings, and may reproduce outdated information or misconceptions widely discussed in the training corpus. Clinical deployment of LLMs in decision support roles therefore requires robust safeguards, including retrieval-augmented generation (RAG) architectures that ground model outputs in verified and up-to-date knowledge bases, uncertainty-aware output

calibration, and mandatory human-in-the-loop review (Lewis et al. 2020).

Regulatory and ethical frameworks for LLM deployment in clinical settings are still nascent. The FDA has issued design guidance on AI-enabled devices but has not yet comprehensively addressed LLM-specific risks (U.S. Food and Drug Administration 2023). Questions of accountability when an LLM contributes to a clinical error, informed consent for AI-assisted decision-making, and equitable access to LLM-powered tools across healthcare settings remain largely unresolved.

10.7 Hybrid Systems and AutoML

10.7.1 Hybrid AI: Combining Paradigms

Hybrid AI systems integrate complementary paradigms — symbolic reasoning, probabilistic graphical models, and deep learning — with the goal of retaining the strengths of each while mitigating their individual weaknesses. Neurosymbolic AI approaches, for example, seek to combine the pattern recognition capabilities of deep networks with the rule-based compositional reasoning of symbolic AI, thereby enabling models that can learn from data while also respecting logical constraints and providing interpretable reasoning traces (Garcez and Lamb 2023; Marcus and Davis 2019).

In biomedical applications, hybrid systems have been developed for clinical decision support, integrating deep learning models for medical image analysis with knowledge graph-based reasoning over clinical ontologies such as SNOMED CT and the Gene Ontology. These architectures can ground predictions in structured biological knowledge, reducing the risk of predictions that are statistically plausible but biologically incoherent. Hybrid approaches are also prominent in drug discovery, where graph neural networks (GNNs) operating on molecular graphs are combined with physics-based force fields and quantum chemical calculations to predict binding affinities and ADMET properties with improved reliability (Wieder et al. 2020).

10.7.2 AutoML: Democratization and Its Pitfalls

Automated machine learning (AutoML) platforms — including AutoKeras, TPOT, H2O AutoML, and cloud-hosted offerings from major technology vendors — lower barriers to model development by automating the processes of feature engineering, algorithm selection, hyperparameter optimization, and architecture search. In biomedical research contexts where deep expertise in machine learning may not be available, AutoML enables researchers to deploy competitive predictive models without manual tuning, potentially accelerating discovery and broadening access to AI-assisted analysis (He et al. 2021).

However, the democratization provided by AutoML raises serious concerns regarding methodological transparency and scientific reproducibility. Researchers who deploy AutoML pipelines without fully understanding the underlying algorithmic choices — the selection of a particular learner, the application of specific preprocessing transformations, or the configuration of cross-validation procedures — may inadvertently violate key modeling assumptions or introduce subtle forms of data leakage. Published studies based on AutoML analyses may be difficult to reproduce if the AutoML system introduces non-determinism or if the specific pipeline configuration is not completely reported. There is risk that AutoML adds even greater speed to the existing tendency to report predictive performance metrics without adequate mechanistic interpretation or external validation (Kapoor and Narayanan 2023).

10.7.3 Federated Learning and Privacy-Preserving AI

An important variant of hybrid and distributed AI is federated learning, which enables model training across multiple institutions without centralizing sensitive patient data (Rieke et al. 2020). In federated learning, each participating institution trains a local model on its own data, and only model parameters — rather than raw data — are communicated to a central aggregation server. This architecture can substantially expand the effective training set available to biomedical AI models without violating patient privacy regulations and is particularly valuable for rare diseases where no single institution has sufficient cases for robust model training. However, federated learning introduces its own technical challenges, including heterogeneity of data distributions across sites (the so-called "non-IID" data), communication overhead, and susceptibility to adversarial model poisoning attacks (Kairouz et al. 2021).

10.8 Persistent Challenges

The preceding sections have identified a constellation of challenges that cut across AI paradigms and biomedical application domains. Taken collectively, they define the frontier of contemporary debate.

Interpretability and mechanistic understanding. The opacity of deep learning models remains the most frequently cited barrier to clinical translation. While post-hoc interpretability methods provide partial mitigation, they do not fully resolve the underlying problem: that many high-performing models are not designed to be mechanistically informative. Advances in intrinsically interpretable architectures, as well as theoretical understanding of what neural networks actually compute, are needed (Rudin 2019).

Reproducibility and methodological rigor. A substantial proportion of published biomedical AI studies have been shown to suffer from inadequate reporting of methodological details, inappropriate data-splitting practices, overfitting to benchmark datasets, and failure to validate on independent prospective cohorts (Kapoor and Narayanan 2023; Wynants et al. 2020). The development of field-specific reporting standards — analogous to CONSORT for clinical trials and STROBE for observational studies — has been proposed as a partial remedy (Mongan et al. 2020).

Bias, fairness, and equity. Systematic underrepresentation of diverse populations in biomedical training datasets has direct consequences for the fairness and generalizability of AI-derived clinical tools. Addressing this requires not only technical interventions — such as reweighting, adversarial debiasing, and equity-constrained optimization — but also structural changes to data collection practices and research infrastructure (Obermeyer et al. 2019).

Regulatory oversight and clinical validation. The regulatory landscape for AI-enabled devices is rapidly evolving, but significant gaps remain. Continuous learning systems — which update their parameters in response to new data after initial deployment — present particular regulatory challenges, as post-market model drift may render an approved device progressively less safe or effective without triggering re-evaluation (U.S. Food and Drug Administration 2023). International harmonization of regulatory standards is also needed as medical AI devices are increasingly deployed across jurisdictions.

Epistemological tension. Perhaps more fundamentally, there is an unresolved epistemological tension between the goal of predictive performance and the goal of scientific understanding. A model that accurately predicts patient outcomes without any mechanistic interpretability may be clinically useful but scientifically unsatisfying — and may fail silently in situations that differ from

its training distribution in ways that a mechanistic model could detect. Biomedical sciences have historically prioritized causal explanation over predictive empiricism, and there is legitimate debate regarding whether the broad adoption of black box AI represents a departure from that epistemic tradition that should be embraced or resisted (Lipton 2018).

10.9 Conclusion

Artificial intelligence has catalyzed transformative advances across gene therapy, cellular therapy, vaccine design, structural biology, and computational biomedicine more broadly. The empirical achievements documented across supervised learning, deep learning, generative models, Bayesian inference, and language models collectively demonstrate that AI is not merely a computational convenience but a genuine engine of scientific discovery, capable of identifying patterns, generating hypotheses, and accelerating experimental workflows at a scale and speed that transcends unaided human cognitive capacity.

However, predictive performance alone is insufficient for sustainable scientific and clinical integration. The biomedical community faces a constellation of unresolved challenges — interpretability, reproducibility, bias, regulatory adequacy, and epistemological coherence — that collectively demand a more deliberate and critically rigorous approach to AI methodology adoption. Deploying powerful but opaque models without adequate validation infrastructure, reporting standards, or governance frameworks risks a form of methodological excess that could ultimately undermine confidence in AI-assisted science.

The path forward requires coordinated effort across multiple dimensions. Technical research must advance interpretable architectures, principled uncertainty quantification, and robust methods for detecting distributional shift. Clinical validation must evolve to prioritize prospective, multi-site evaluation in diverse patient populations over retrospective benchmarking in convenience cohorts. Reporting standards must mandate transparent disclosure of methodological details sufficient for independent replication. Regulatory frameworks must adapt to the distinct challenges posed by continuous learning systems and LLM-based decision support. And the scientific community must engage in sustained reflection on the epistemological role of AI in biomedical investigation — not as a substitute for mechanistic understanding, but as a powerful complement to it.

The contemporary debate is therefore not whether AI should be used in biomedical science, but how it should be integrated into scientific methodology in a manner that preserves — and ultimately deepens — our epistemic standards for what it means to understand biological systems and provide safe, effective, and equitable clinical care.

Chapter Review Outline

1. Supervised learning: SVMs, random forests, gradient boosting, feedforward networks; applications to pathogenic SNPs, splice site prediction, regulatory element classification, AAV vector design, proteomics (mass spectrometry).
2. Unsupervised learning: PCA, t-SNE, UMAP for scRNA-seq; WGCNA for co-expression networks; MOFA+, Seurat v4 for multimodal integration.
3. Debates: generalizability (European ancestry biases in GWAS/polygenic scores), calibration vs. AUROC, rare prospective validation (Obermeyer et al. 2019; Popejoy and Fullerton 2016).
4. Deep learning: CNNs (histopathology, hematopoietic cell classification), LSTMs, Transformers; Enformer (gene expression from DNA sequence) (Avsec et al. 2021).
5. Interpretability: "black box"; post-hoc methods (saliency maps, LRP, SHAP, integrated gradients) are unstable (Adebayo et al. 2018); intrinsically interpretable models preferable in high-risk contexts (Rudin 2019).
6. Generative models: GANs (Goodfellow, 2014) for histopathological images and synthetic EHR; VAEs for chemical space navigation in drug discovery; diffusion models for protein linker design.
7. Protein design: AlphaFold2 (Jumper et al. 2021), ProtGPT2 (Ferruz et al. 2022), ESMFold; vaccine epitope prediction; mRNA optimization.
8. Synthetic data: address scarcity + privacy (HIPAA/GDPR), but amplify source biases; FDA regulation evolving.
9. Bayesian networks: principled uncertainty; DAGs for causal inference (PC algorithm, FCI); hybrid probabilistic–deep models (deep kernel, neural processes, Bayesian neural networks).
10. LLMs: GPT-4, BioGPT, Med-PaLM 2; hallucination = central risk; lack epistemic grounding; mitigation via RAG + human-in-the-loop.
11. AutoML (AutoKeras, TPOT, H2O): democratizes ML but double black box; risk of data leakage and irreproducibility. Federated learning: trains without centralizing data; challenges = non-IID data, adversarial poisoning.

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Chapter 11

Legal and Ethical Aspects of Genomic Data Use

11.1 Introduction

The contemporary genomic revolution, characterized by exponential reduction in sequencing costs and unprecedented increase in availability of DNA and proteome data at population scale, presents profound legal, ethical, and regulatory challenges. The contemporary researcher in phylogenomics and bioinformatics operates in a complex context marked by multiple regulatory frameworks governing the collection, storage, sharing, and use of genomic data. These frameworks vary substantially between national jurisdictions, often in tension with scientific norms of data openness and reproducibility.

This chapter examines the legal and ethical aspects central to contemporary genomic research, with particular emphasis on the Brazilian context. We cover the international treaties establishing the sovereign right of States over genetic resources within their borders; the specific Brazilian legislation regulating access to biodiversity, protection of personal data, and ethical use of animals in research; the international regulatory frameworks shaping the sharing of genomic data; and the emerging ethical considerations in genomic research with indigenous and traditional communities.

11.2 International Frameworks: Convention on Biological Diversity and Nagoya Protocol

11.2.1 The Convention on Biological Diversity (1992)

The Convention on Biological Diversity (CBD), adopted at the United Nations Conference on Environment and Development in Rio de Janeiro in 1992 and in force since December 1993, establishes an international legal framework for the conservation of biological diversity, the sustainable use of its components, and the fair and equitable sharing of benefits derived from the use of genetic resources (United Nations 1992). The CBD recognizes the sovereignty of States over genetic resources in their territories and establishes that access to genetic resources is a matter of internal competence of each contracting party.

The key principles of the CBD include: (1) recognition of state sovereignty over genetic resources within national jurisdictions; (2) right of States to legislate on access to genetic resources within their territories; (3) obligation to facilitate access to genetic resources on mutually agreed terms;

and (4) obligation to share benefits derived from the use of genetic resources in a fair and equitable manner. These principles form the foundation for all subsequent legislation on access to biodiversity at the global level.

For researchers, the CBD implies that collection of biological specimens or genomic data from organisms in a country requires prior informed consent (PIC) from the competent authorities of that country and results in obligations for benefit-sharing (United Nations 1992).

11.2.2 Nagoya Protocol on Access and Benefit Sharing (2014)

The Nagoya Protocol on Access and Benefit Sharing, adopted in 2010 and in force since October 2014, operationalizes and provides specific mechanisms for implementation of the principles established by the CBD (United Nations Environment Programme 2014). The protocol requires that signatory countries establish legislative, administrative, and policy measures to: (1) ensure access to genetic resources conditioned on prior informed consent; (2) require disclosure of the origin of genetic resources in applications for intellectual property rights; (3) establish requirements for mutually agreed terms documenting the conditions of access; and (4) monitor the use of genetic resources along supply chains.

For biologists and researchers in genomics, the Nagoya Protocol means that genomic data from organisms collected in signatory countries cannot be freely shared or published without attention to the benefit-sharing obligations established through mutually agreed terms. This requirement may impose substantial restrictions on data publication and the open science practice that has historically been the norm in genomics.

11.3 Brazilian Legislation: Biodiversity and Data Protection

11.3.1 Biodiversity Law (Lei nº 13.123/2015) and SisGen

Brazil, holder of approximately 13% of global biodiversity and signatory to both the CBD and the Nagoya Protocol, implemented the Biodiversity Law (Lei nº 13.123, of May 20, 2015) and subsequently Decree nº 8.772/2016, which regulate access to components of Brazilian biodiversity and the sharing of benefits derived from such access (Brasil 2015, 2016).

The Biodiversity Law establishes that any access to a component of Brazilian biodiversity for purposes of scientific research, bioprospecting, conservation, or sustainable use requires prior registration in the Genetic Heritage Management System (SisGen). SisGen is an electronic information system that consolidates data on access to components of Brazilian genetic heritage, derived products, and reproductive material obtained from access, with the purpose of: (1) identifying the origin of genetic materials; (2) tracking the access history; (3) facilitating benefit-sharing; and (4) complying with Brazil's obligations under the Nagoya Protocol.

Researchers in phylogenomics must register with SisGen before accessing, collecting, or using components of Brazilian biodiversity for research purposes. This includes collection of specimens for sequencing, obtaining tissues from organisms of Brazilian origin, and analysis of genomic sequences derived from Brazilian biodiversity. Failure to register with SisGen constitutes an environmental violation and can result in fines of up to R$ 200,000, in addition to possible revocation of research funding and publication restrictions (Brasil 2015).

A critical aspect of the Biodiversity Law is that it classifies access to biodiversity components

into three categories: (1) scientific research without economic purposes; (2) bioprospecting; and (3) sustainable use. Researchers in phylogenomics frequently work in the category of scientific research without economic purposes, but if the research results in a product of economic interest (for example, identification of a gene with biotechnological applications), the classification may change retroactively, triggering additional benefit-sharing obligations.

11.3.2 General Data Protection Law (LGPD - Lei nº 13.709/2018)

The General Data Protection Law (LGPD), promulgated in August 2018 and in full effect since August 2020, is the Brazilian legislation for personal data protection (Brasil 2018). Although often considered analogous to the General Data Protection Regulation (GDPR) of the European Union, the LGPD is specific Brazilian legislation that establishes rights and responsibilities for the processing of personal data in Brazil.

The LGPD is relevant to genomic research because genomic data, when associated with an identifiable or potentially identifiable individual, are classified as sensitive personal data. The law defines personal data as "information related to an identified or identifiable natural person" (Brasil 2018). Genomic data are often considered a special category of sensitive data because they can reveal information about health, genetic predisposition to disease, or ancestry.

Implications of the LGPD for genomic research include: (1) explicit consent required for collection and processing of personal genomic data; (2) right of individuals to access, correct, and delete their genomic data; (3) obligation to implement technical and administrative security measures for data protection; (4) obligation to report data breaches within a specified timeframe; and (5) right of individuals to object to processing of their genomic data for specific purposes (Brasil 2018).

For researchers, the LGPD means that any research project involving genomic data from Brazilian humans requires: written informed consent from the donor or their legal representatives; transparent privacy policy documenting how data will be collected, processed, stored, and shared; appropriate security measures; and a data retention plan documenting how long data will be maintained.

11.3.3 Brazilian Legislation on Research with Animals

For phylogenomic research on metazoans, Brazilian legislation on ethical use of animals in research is also relevant. The Arouca Law (Lei nº 11.794, of October 8, 2008) regulates the scientific use of animals in teaching and research in Brazil (Brasil 2008). The law establishes that: (1) every institution that conducts teaching or research activities with animals must establish an Animal Use Ethics Committee (CEUA); (2) every research proposal involving animals must be submitted to the CEUA before beginning; (3) the CEUA must evaluate whether the research is scientifically appropriate, ethically justified, and uses techniques that minimize animal suffering.

Researchers in phylogenomics who collect live animal specimens, who conduct experiments with animals, or who use animal tissues obtained through euthanasia for collection purposes must submit their protocols to the CEUA of their institution. The Arouca Law establishes principles of: (1) use of alternatives to animals when available; (2) reduction in the number of animals used (the 3Rs principle: Reduction, Replacement, Refinement); (3) refinement of techniques to minimize pain and suffering; and (4) humane euthanasia when necessary (Brasil 2008).

11.4 International Frameworks for Genomic Data Sharing

11.4.1 International Sequence Infrastructure: GenBank, ENA, and DDBJ

The international infrastructure for genomic sequence data rests on three central databases linked globally that share data daily: the National Center for Biotechnology Information (NCBI) GenBank maintained by the National Institutes of Health (NIH) of the United States, the European Nucleotide Archive (ENA) of the European Bioinformatics Institute, and the DNA Data Bank of Japan (DDBJ) (Coordinators et al. 2023; Harrison et al. 2023; Suzuki et al. 2023). Collectively, this infrastructure is referred to as the International Nucleotide Sequence Database Collaboration (INSDC).

The INSDC databases operate under the principle of immediate public access: sequences deposited are publicly accessible without restriction within 24 hours of submission, unless the submitter requests temporary retention of up to 4 months (called a hold period). This system was designed to facilitate rapid scientific discovery and ensure reproducibility of published analyses, allowing researchers to reanalyze underlying data for published studies.

However, for genomic data from humans that may be potentially identifying, the INSDC databases have implemented access restrictions. GenBank maintains a separate database for human genomic data that requires consent from the proposer for any public release, allowing indefinite retention of sensitive data while keeping the sequence available to researchers approved through a controlled access portal.

For researchers in phylogenomics working with metazoans, the implication is that sequences from non-human organisms are routinely deposited in INSDC with immediate public access. However, if the research involves genomic data from humans or specific human populations, the researcher must be aware of the Brazilian LGPD, the European GDPR, and other data protection laws that may impose restrictions on public data sharing.

11.4.2 Open Data Mandate Policies of Funding Agencies

International research funding agencies have increasingly adopted mandatory data sharing policies that require publicly funded researchers to make their data available in public repositories within a specified timeframe (often 6 months to 2 years after publication of results). The National Institutes of Health (NIH) of the United States, the European Research Council (ERC), and the Engineering and Physical Sciences Research Council (EPSRC) of the United Kingdom have all implemented open data policies.

In Brazil, funding agencies such as the São Paulo Research Foundation (FAPESP) and the National Council for Scientific and Technological Development (CNPq) have adopted similar policies, often in conformity with UNESCO recommendations for open data for science (UNESCO 2021). Funding for research by FAPESP or CNPq now typically requires that raw sequencing data be deposited in recognized public repositories and that analyses be reproducible through sharing of code and methods.

These policies have significant implications for phylogenomic research: genomic data must be planned to be shareable from the initial study design. For sensitive data (humans, data from indigenous populations), controlled access policies may be appropriate, but must be negotiated with funding agencies during proposal development.

11.5 Ethical Considerations in Genomic Research with Indigenous and Traditional Communities

11.5.1 Rights of Indigenous Communities and Biopiracy

Brazil is a signatory to the United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP), which establishes collective rights of indigenous peoples over their traditional knowledge, cultural and genetic resources (United Nations 2007). UNDRIP establishes that indigenous peoples have the right to: (1) consent or not consent to the collection and use of their biodiversity and associated knowledge; (2) control, protect, and preserve their traditional knowledge; and (3) maintain and strengthen their distinctive relationship with lands and resources.

Historically, genetic resources associated with traditional knowledge of indigenous communities were collected and used in research and commercial development without consent or benefit to indigenous communities, a process often termed "biopiracy." Notable cases include collection of medicinal plants and organisms from the Brazilian Amazon used in pharmaceutical development without participation or financial benefit to indigenous communities.

Brazilian Biodiversity Law establishes that access to traditional knowledge associated with a biodiversity component requires prior informed consent from the indigenous community or traditional population and results in benefit-sharing obligations. SisGen requires explicit documentation of consent when research accesses knowledge associated with biodiversity components.

For researchers in phylogenomics, this means that if the research involves collection of organisms on indigenous lands, or uses genomic data from organisms whose traditional knowledge was used in the research or development, the researcher must: (1) obtain prior informed consent from the indigenous community; (2) document consent and benefit-sharing in SisGen; (3) ensure that communities participate in decisions about data publication and sharing; and (4) implement benefit-sharing mechanisms, which may include financial participation or academic collaboration.

11.5.2 Principles of Respectful Research with Indigenous Communities

Beyond legal requirements, there is an ethical imperative to conduct research with indigenous communities in a respectful and participatory manner. Organizations such as the International Society of Ethnobiology have established ethical principles for research in ethnobiology, ethnobotany, and ethnogenomics (International Society of Ethnobiology 2006).

These principles include: (1) reciprocity — research should benefit collaborating communities, not just researchers; (2) transparency — communities must fully understand research objectives and possible uses; (3) sovereignty — indigenous communities must maintain control over their genomic data and traditional knowledge; (4) respect — researchers must respect indigenous cultural protocols and intellectual rights; and (5) benefit-sharing — researchers must share financial and non-financial benefits with communities.

In practice, this means that research involving genomics of organisms associated with indigenous traditional knowledge must include: (1) prior dialogue with communities to explain research objectives; (2) documented consent obtained freely without coercion; (3) participation of community members in research design when possible; (4) mechanisms for sharing results in the community's language and in accessible formats; and (5) pre-agreed benefit-sharing if research results in commercial developments.

11.6 European GDPR and International Genomic Data Protection

11.6.1 General Data Protection Regulation (GDPR) and Genomic Sequences

The General Data Protection Regulation (GDPR), adopted by the European Union in 2016 and in force since May 2018, is data protection legislation even more stringent than the Brazilian LGPD. The GDPR is particularly relevant to genomic research because: (1) it classifies genomic data as a special category of personal data requiring explicit consent; (2) it establishes individuals' right of access to their genomic data; (3) it establishes the right to delete data ("right to be forgotten"); and (4) it requires that any processing of genomic data be proportionate and justified (European Union 2018).

For researchers working with genomic data from European individuals, the GDPR imposes substantial obligations. Genomic data cannot be transferred outside the European Union without guarantees of data protection equivalent to GDPR. Researchers need a legal basis for processing genomic data (typically informed consent) and must document this legal basis in a Data Protection Impact Assessment (DPIA).

11.6.2 Implications for International Genomics Research

Researchers in phylogenomics working in an international context must be aware that different jurisdictions involved in research may have different requirements. A collaborative project involving collection of genomic data in Brazil, processing and analysis in Europe, and publication in an American journal may be simultaneously subject to: Brazilian Biodiversity Law and SisGen; Brazilian LGPD if data are from Brazilian humans; European GDPR if analysis occurs in a European institution; and United States data export regulations if data are transferred to American institutions.

Compliance with all these frameworks requires deliberate planning from the initial study design. Recommendations include: (1) consult regulatory compliance personnel and legal advisors during study design; (2) document the origin, consent, and use restrictions of all data from the point of collection; (3) implement appropriate technical data security measures; (4) establish international data transfer agreements if data will be shared between jurisdictions; and (5) maintain a record of all regulatory authorities involved and their specific requirements.

11.7 Open Access and Open Science in Genomics

11.7.1 Movement for Open Data and Open Science

There is a growing global movement toward what is often called "open science" or "open data" — the practice of making raw scientific data, computational code, and analyses publicly available in a manner that any researcher can access, review, and reuse this data and code without restriction. UNESCO published a Recommendation on Open Science in 2021, endorsed by Brazil, that establishes principles of openness in scientific data (UNESCO 2021).

For genomics, the open data movement has been particularly influential. The Human 1000 Genomes Project, completed in 2015, set a powerful precedent by making publicly accessible, without restriction, the genomic data of 2,500 individuals from diverse global populations. This open access to data allowed thousands of researchers to use the data for their own research, substantially

accelerating progress in population genomics and precision medicine (The 1000 Genomes Project Consortium 2015).

However, the open data model for genomics is in tension with personal data protection legislation such as GDPR and LGPD, which impose restrictions on public sharing of genomic data from identifiable humans. Resolution of this tension continues to evolve, with ongoing discussions about how to implement "responsible access" — an intermediate model in which genomic data are publicly accessible for scientific research through authorization processes, but with restrictions on commercial sharing or re-identification (Global Alliance for Genomics and Health 2018).

11.7.2 Controlled Access Repositories and Data Governance Models

To resolve the tension between open data and personal data protection, controlled access repository infrastructure and data governance models have been developed. The Global Alliance for Genomics and Health (GA4GH) has established international standards for federating controlled access genomic databases (Global Alliance for Genomics and Health 2018). Models such as NIH's dbGaP (database of Genotypes and Phenotypes) allow researchers to deposit genomic data with access restrictions applied, making data available for qualified academic research but not for the general public or commercial use.

In Brazil, discussions are ongoing about development of a similar model that balances scientific openness with compliance with the LGPD and Biodiversity Law. An emerging recommendation is that genomic data with identifying potential should be made available through a controlled access model in which: (1) researchers request access to the repository; (2) a data review committee evaluates whether the requester has scientific qualifications and legitimate purposes; (3) the researcher agrees to terms of use restricting use to non-commercial research purposes; and (4) data are provided in a secure computing environment where the researcher can perform analyses but cannot export raw data.

11.8 Practical Compliance: A Guide for Researchers

11.8.1 Research Planning with Legal and Ethical Considerations

Researchers in phylogenomics planning projects involving genomic data should implement procedures to ensure legal and ethical compliance. A recommended practical protocol includes:

1. **Identify data sources:** Document the origin of all genomic data. Are they public data from repositories like GenBank? Is data newly collected? If newly collected, in which country? From which organism?

2. **Identify applicable regulatory requirements:** Based on the origin of the data, identify which laws apply. If data are of Brazilian origin, the Biodiversity Law and SisGen apply. If data include humans, Brazilian LGPD or European GDPR apply. If data include indigenous traditional knowledge, UNDRIP and Biodiversity Law apply.

3. **Obtain consent and approval:** If research involves new collection of organisms or human data, obtain appropriate consent. Submit protocol for approval from relevant ethics committees (CEUA for animal research, ethics committee for human data).

4. **Register with SisGen (Brazil):** If research involves Brazilian biodiversity, register access with SisGen before beginning collection or analysis.

5. **Document compliance:** Keep detailed records of all compliance actions — dates of consent, committee approvals, SisGen registrations.

6. **Plan data sharing:** Document how data will be shared. If data can be public, choose an appropriate repository. If data are sensitive, establish controlled access. Document reuse restrictions in metadata.

7. **Report compliance in publications:** Publications should include a "Data Availability" section documenting: origin of data, where data can be accessed, any access restrictions, compliance with consent requirements and regulations.

11.8.2 Documentation of Methods for Transparency

In addition to legal compliance, methodological transparency practices are essential. Genomic studies should clearly document:

- Origin of all data, including information about sequencing (which capture protocol, which coverage depth, which sequencing technology)
- Inclusion and exclusion criteria for organisms or samples
- Information about data quality (amount of missing data, sequence coverage)
- Analysis methodology, including software used and parameters
- Computational code used, deposited in a public repository (GitHub, GitLab) for reproducibility
- Informed consent and ethics approval if applicable
- Author contributions in CASRAI CRediT format

11.9 Future Perspectives: International Harmonization and Genomic Data as a Common Good

11.9.1 Efforts to Harmonize International Regulations

Recognizing that divergent regulations between jurisdictions hinder international scientific research, there are ongoing efforts to harmonize legal and ethical requirements around genomic data. GA4GH works with global regulatory authorities to develop shared standards. UNESCO has established recommendations on open data for science. Discussions are underway in international forums about how to balance scientific openness with personal data protection.

A critical emerging question is how to treat genomic data from indigenous communities and specific populations as a "global common good" — a resource that belongs to humanity as a whole for research purposes — while simultaneously respecting the sovereign rights of nation states and the collective rights of indigenous communities over their genetic resources. This tension remains largely unresolved and should be central in future discussions of international science policy.

11.10 Conclusion

The contemporary researcher in phylogenomics and bioinformatics operates in a complex normative context marked by multiple legal frameworks, often in tension with one another. Brazilian Biodiversity Law and SisGen establish requirements for registration of access to biodiversity. Brazilian LGPD protects personal data, including genomic data from humans. Internationally, the Nagoya Protocol establishes sovereign rights of States over genetic resources. European GDPR imposes even more stringent restrictions on genomic data of Europeans. Indigenous communities possess rights under UNDRIP.

Simultaneously, funding agencies require open data sharing, international repositories such as GenBank facilitate public access to sequences, and scientific norms value reproducibility and transparency that demand broad data and code sharing.

Compliance with this complex normative apparatus requires: (1) knowledge of the multiple applicable legal frameworks; (2) careful planning from the initial project design; (3) detailed documentation of data origin, consent, and compliance; and (4) respectful and participatory engagement with indigenous and traditional communities.

Researchers who successfully navigate this complex normative landscape contribute not only to scientific progress but also to responsible science practice that respects individual rights, community rights, national sovereignty, and the global imperative for scientific sharing for human well-being. The path forward requires continued international collaboration to harmonize regulations, development of responsible access infrastructure, and commitment from the scientific community to ethical practices.

Chapter Review Outline

1. CBD (Rio, 1992, in force 1993): state sovereignty over genetic resources; access requires prior informed consent (PIC); fair sharing of benefits (United Nations 1992).
2. Nagoya Protocol (2010, in force 2014): operationalizes CBD; requires disclosure of origin in IP; mutually agreed terms; monitoring along the chain (United Nations Environment Programme 2014).
3. Biodiversity Law (Lei 13.123/2015) + Decree 8.772/2016: mandatory registration in **SisGen** before accessing Brazilian biodiversity; fine up to R$200 thousand; retroactive reclassification if research generates economic product.
4. LGPD (Lei 13.709/2018, in force 2020): human genomic data = **sensitive** personal data; explicit consent; right of access, correction, and deletion; mandatory breach reporting.
5. GDPR (EU, in force 2018): more stringent than LGPD; prohibits transfer outside EU without equivalent safeguards; mandatory DPIA; "right to be forgotten".
6. Arouca Law (Lei 11.794/2008): ethical use of animals; mandatory CEUA in every institution; 3Rs principles (Reduction, Replacement, Refinement); humane euthanasia.
7. UNDRIP: collective rights of indigenous peoples over traditional knowledge and genetic resources; free prior consent; biopiracy = access without benefit.
8. INSDC (GenBank/ENA/DDBJ): public access in 24 h; hold of up to 4 months; controlled access for identifying human data (dbGaP).
9. Open data mandates: NIH, ERC, EPSRC, FAPESP, CNPq; UNESCO 2021 endorsed by Brazil.
10. GA4GH: standards for federating controlled access genomic databases; "responsible access" model (openness with restrictions on re-identification/commercial use).
11. Practical compliance (7 steps): identify sources → regulatory requirements → consent/approval → SisGen registration → document compliance → plan sharing → report in publications (Data Availability).

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